AI6126 Advanced Computer Vision

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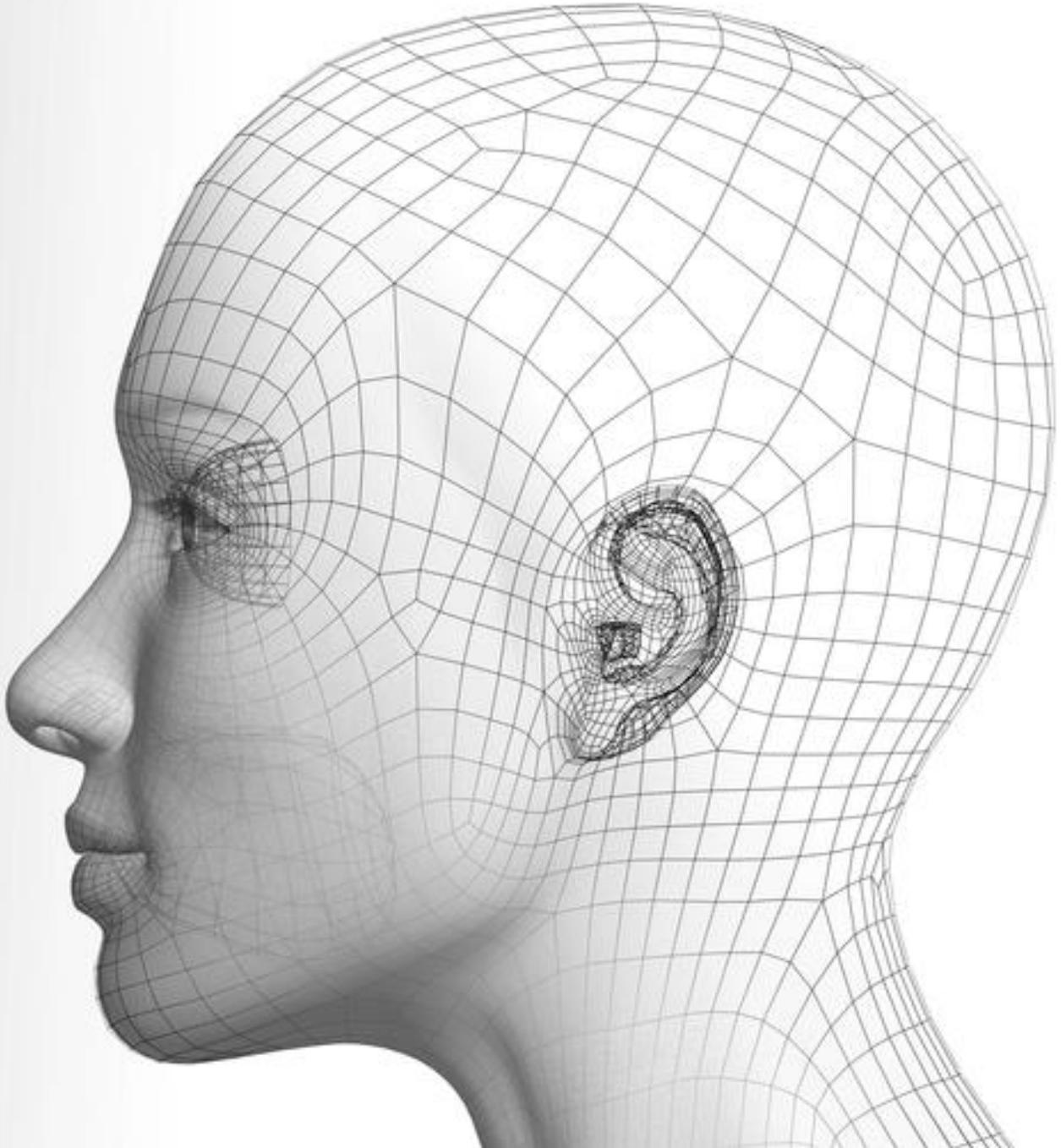
Image Segmentation

Ziwei Liu

刘子纬

https://liuziwei7.github.io/



Slide Credits

- Justin Johnson, EECS 498/598
- David Fouhey, EECS 442
- Paper Authors

THE PERSON IN THE DRIVER'S SEAT
IS ONLY THERE FOR LEGAL REASONS.

HE IS NOT DOING ANYTHING.
THE CAR IS DRIVING ITSELF.

Outline

Semantic Segmentation:

- Fully Convolutional Network
- Skip Connections
- Spatial Contexts

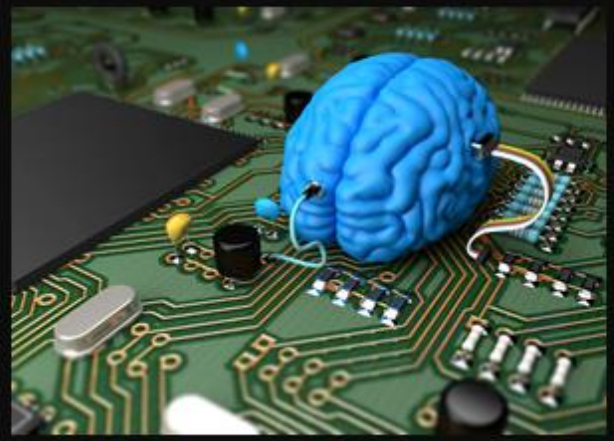
Instance Segmentation:

- Object Detection
- Mask R-CNN
- Joint Mask+X Prediction

Open Problems:

Deep Learning

What society thinks I do



What my friends think I do

What other computer scientists think I do

What mathematicians think I do

What I think I do

```
from theano import *
```

What I actually do

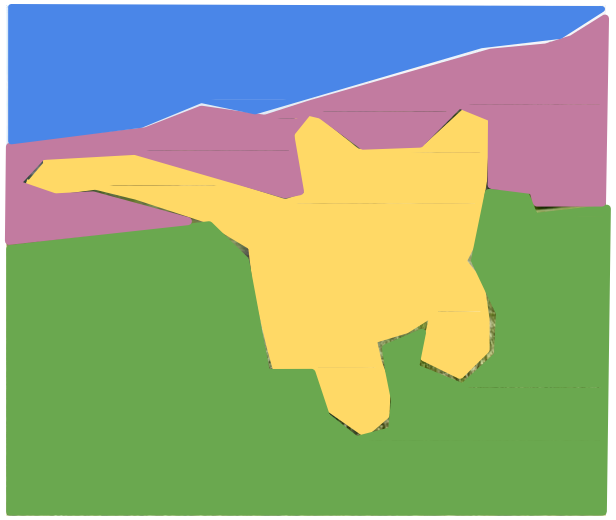
Many Structured Prediction Tasks

Classification

CAT

No spatial extent

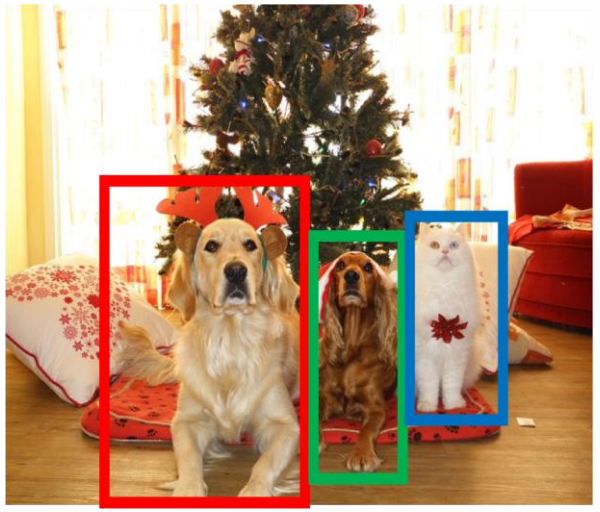
Semantic Segmentation



GRASS, **CAT**, **TREE**, **SKY**

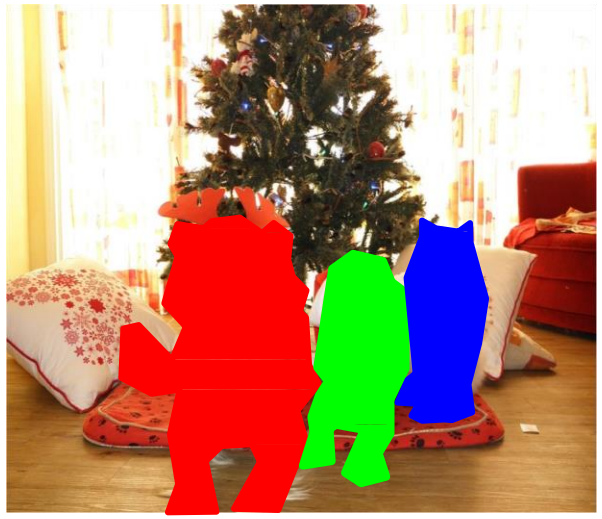
No objects, just pixels

Object Detection



DOG, **DOG**, **CAT**

Instance Segmentation



DOG, **DOG**, **CAT**

Multiple Objects

Part I:
Semantic Segmentation

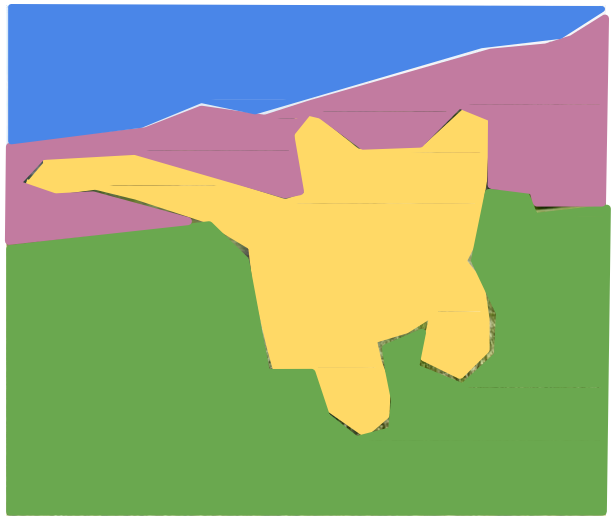
Structured Prediction Tasks: Semantic Segmentation

Classification

CAT

No spatial extent

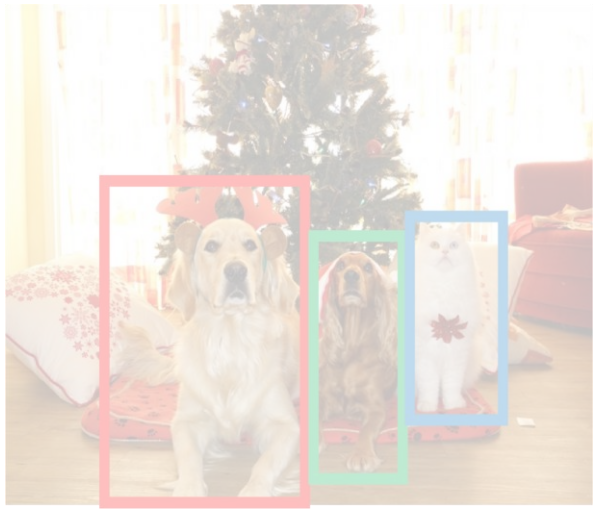
Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation

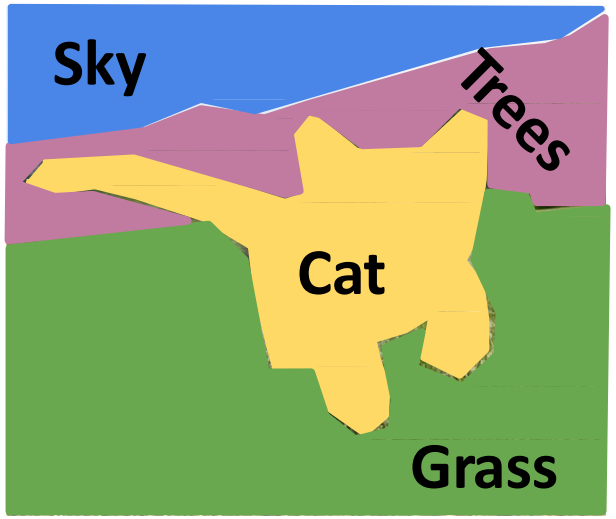
DOG, DOG, CAT

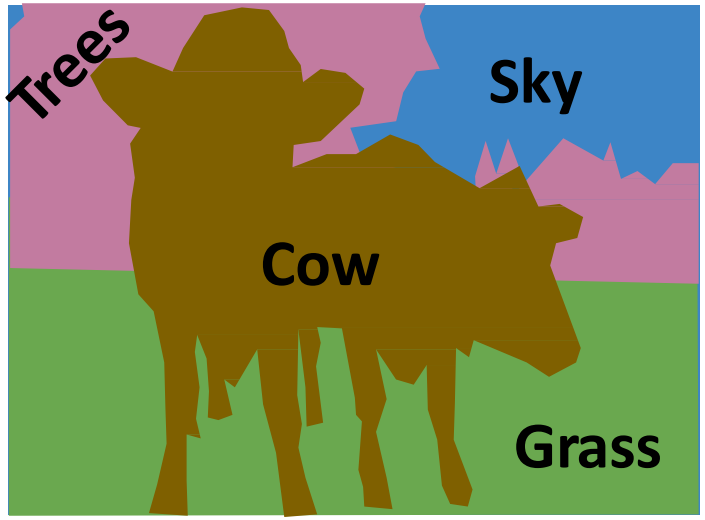
Multiple Objects

Semantic Segmentation

Label each pixel in the image with a category label

Don’t differentiate instances, only care about pixels

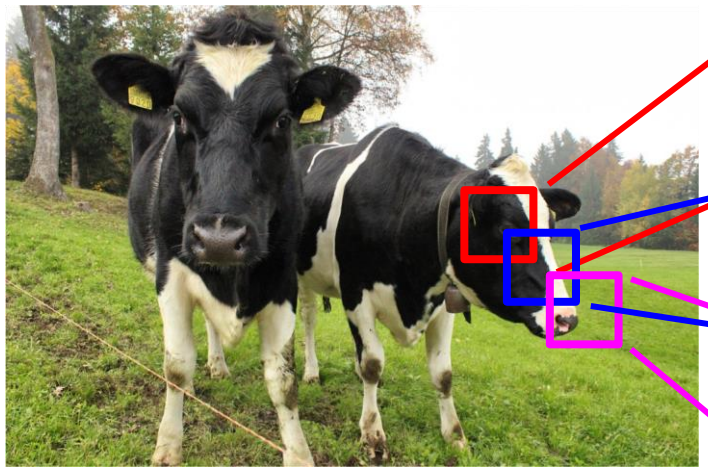




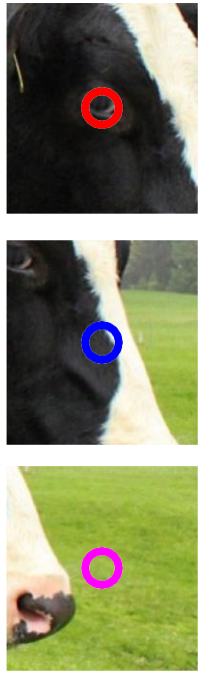
Fully Convolutional Network

Semantic Segmentation Idea: Sliding Window

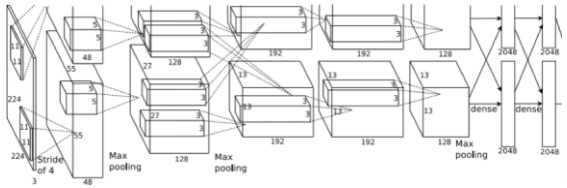
Full image



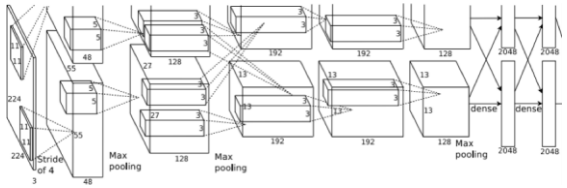
Extract patch



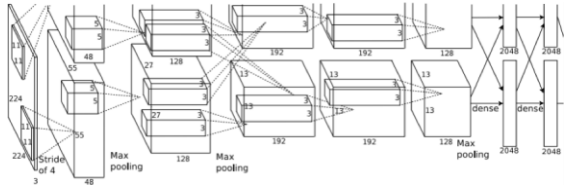
Classify center pixel with CNN



Cow



Cow



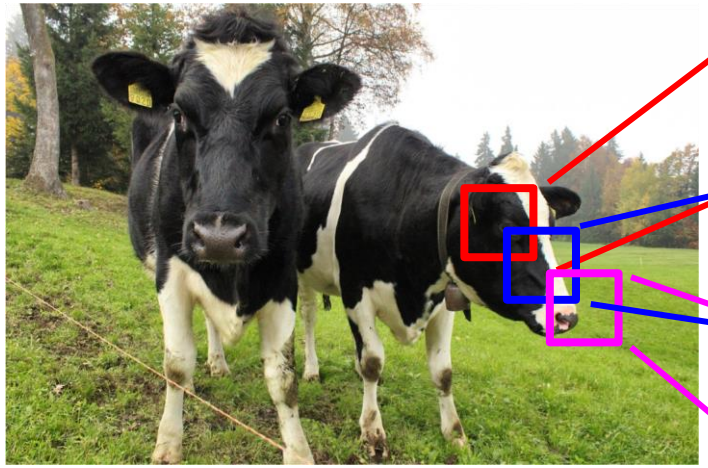
Grass

Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013

Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

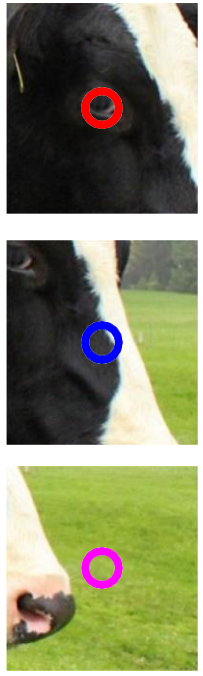
Semantic Segmentation Idea: Sliding Window

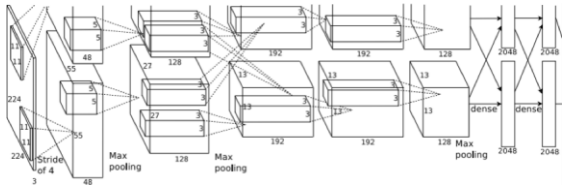
Full image



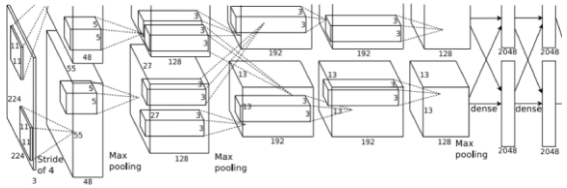
Extract patch

Classify center pixel with CNN

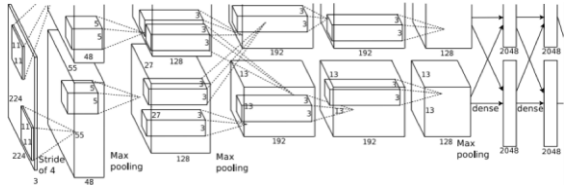




Cow



Cow



Grass

Problem: Very inefficient! Not reusing shared features between overlapping patches

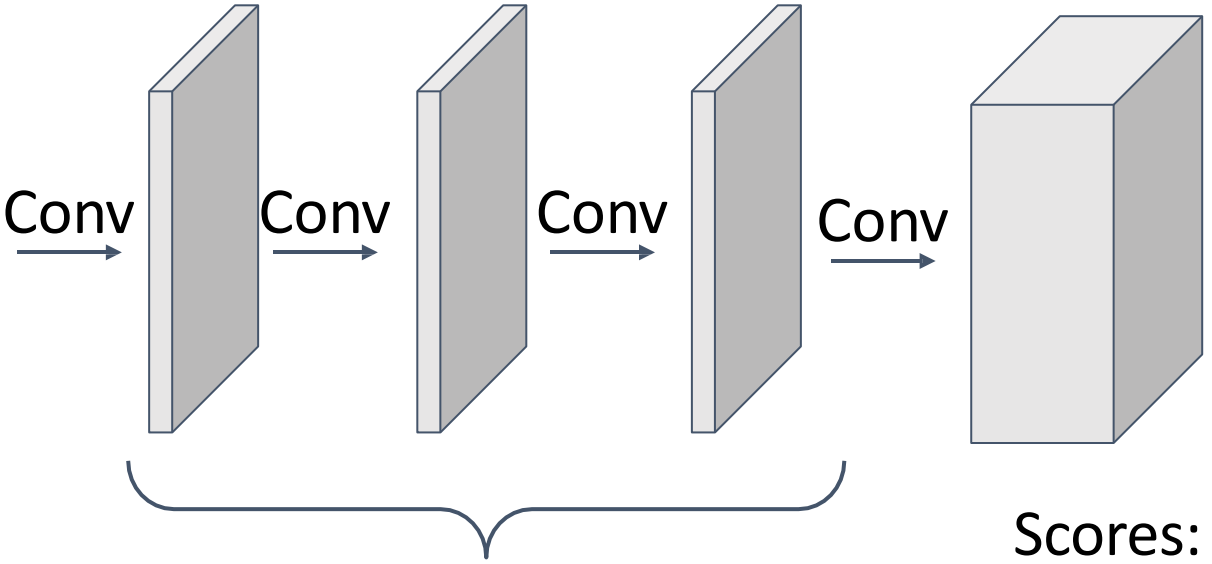
Farabet et al, “Learning Hierarchical Features for Scene Labeling,” TPAMI 2013
Pinheiro and Collobert, “Recurrent Convolutional Neural Networks for Scene Labeling”, ICML 2014

“Fully” Convolution

Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!

Input:
3 x H x W



Convolutions:
D x H x W

Scores:
C x H x W

argmax

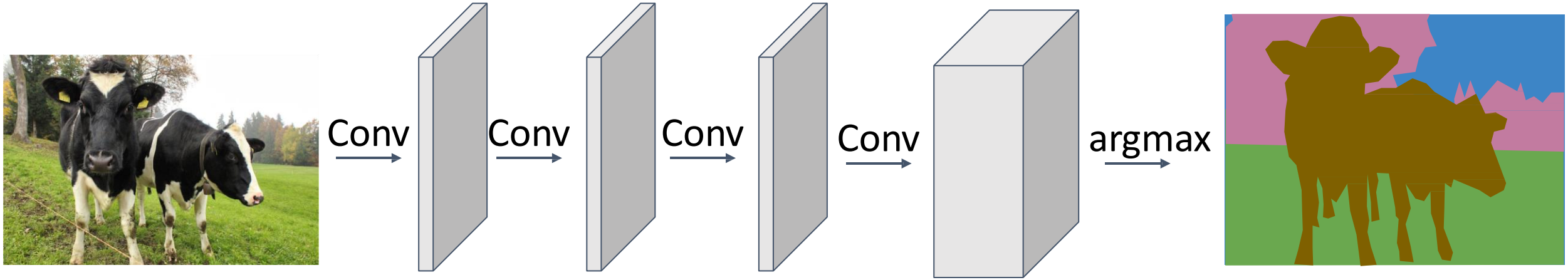
Predictions:
H x W

Loss function: Per-Pixel cross-entropy

Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015

Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



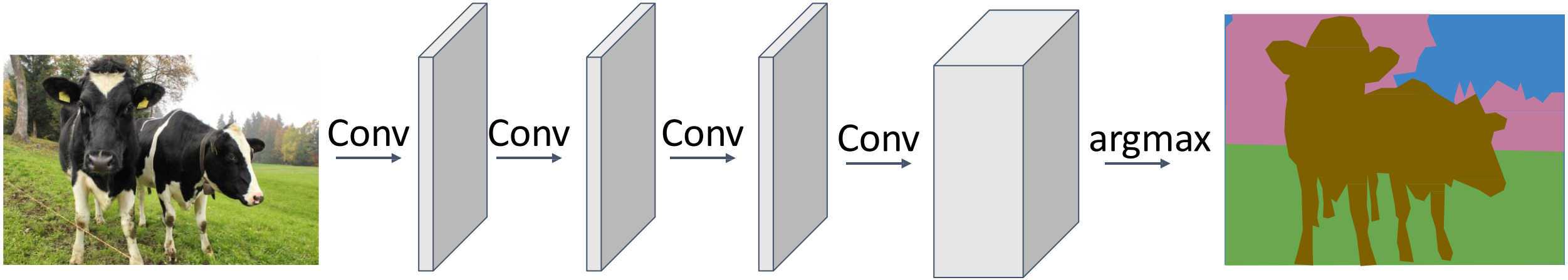
Input:
3 x H x W

Problem #1: Effective receptive field size is linear in number of conv layers: With L 3x3 conv layers, receptive field is 1+2L

Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015

Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



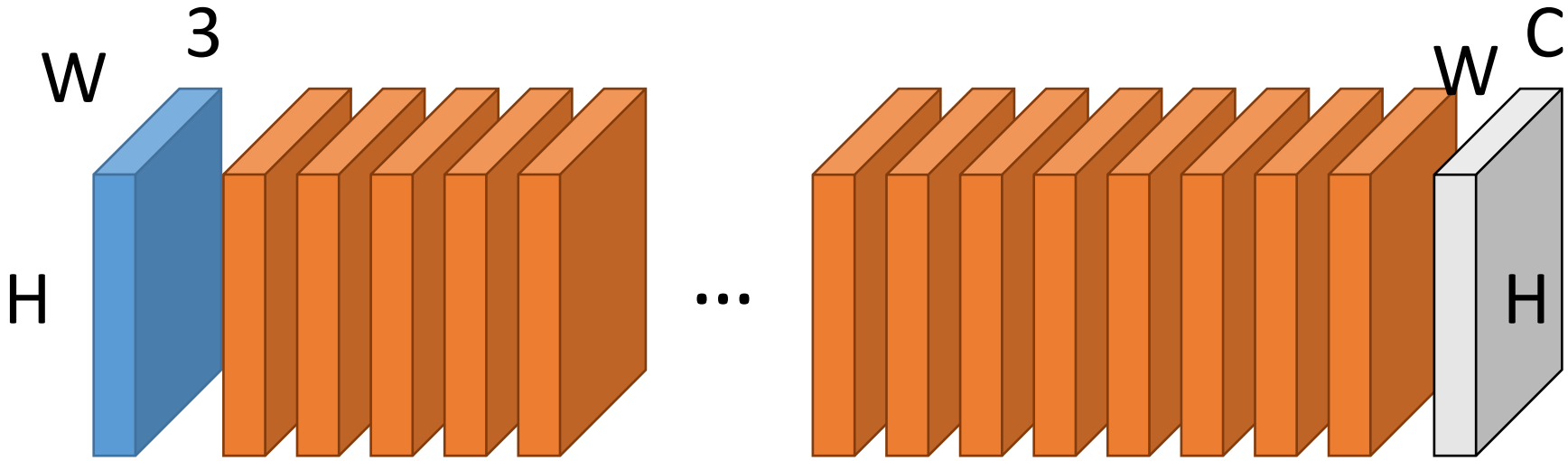
Input:
3 x H x W

Problem #1: Effective receptive field size is linear in number of conv layers: With L 3x3 conv layers, receptive field is 1+2L

Problem #2: Convolution on high res images is expensive! Recall ResNet stem aggressively downsamples

Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015

Why Not Stack Convolutions?

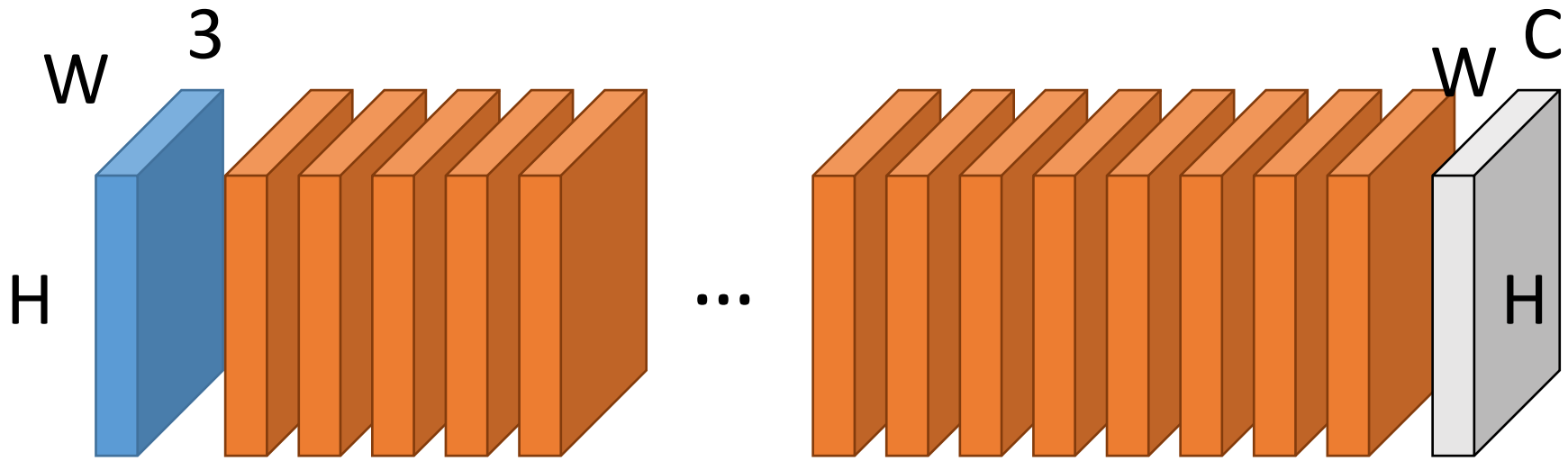


n 3x3 convs have a receptive field of 2n+1 pixels

How many convolutions until >=200 pixels?

100

Why Not Stack Convolutions?



Suppose 200 3x3 filters/layer, H=W=400

Storage/layer/image: 200 * 400 * 400 * 4 bytes = 122MB

Uh oh!*

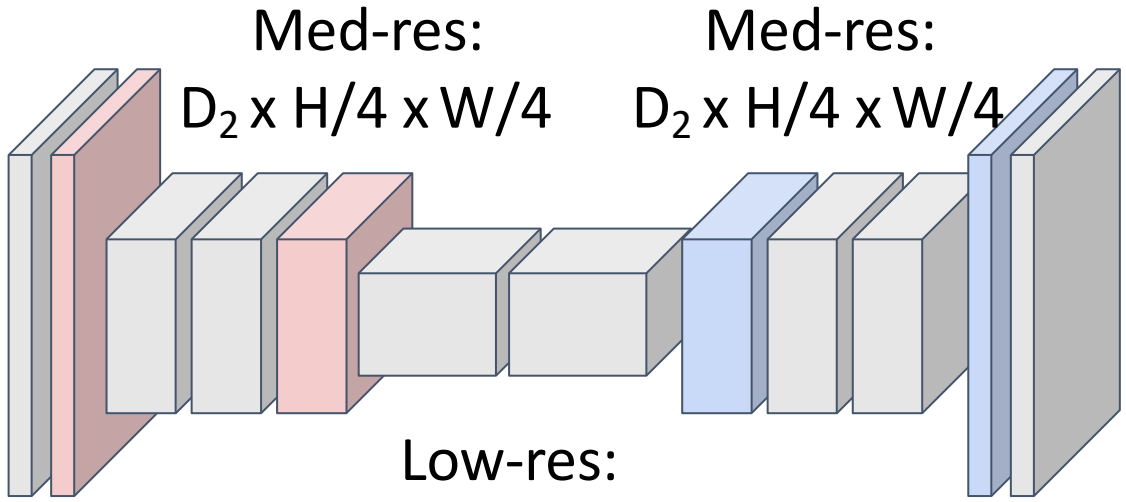
*100 layers, batch size of 20 = 238GB of memory!

Downsampling and Upsampling

Fully Convolutional Network

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Input:
3 x H x W



Predictions:
H x W

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

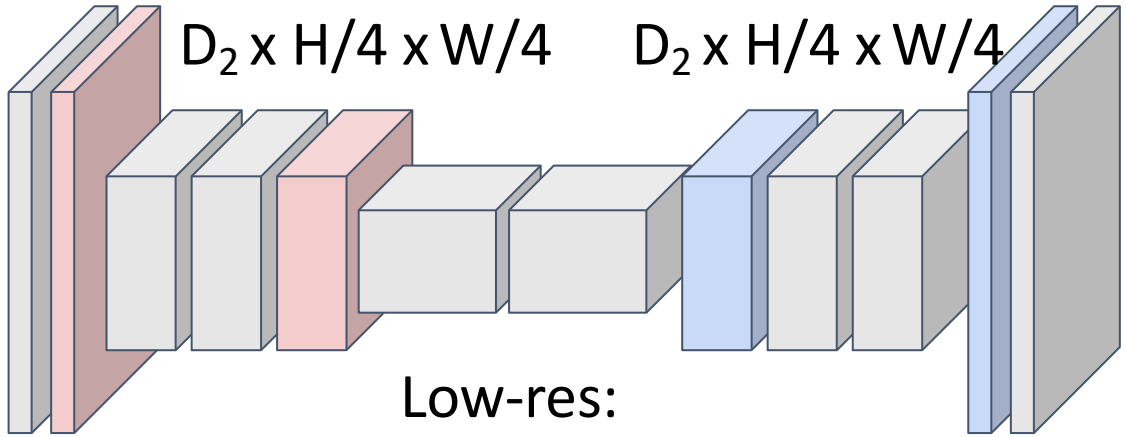
Fully Convolutional Network

Downsampling: Pooling, strided convolution

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Upsampling: ???

Input:
3 x H x W

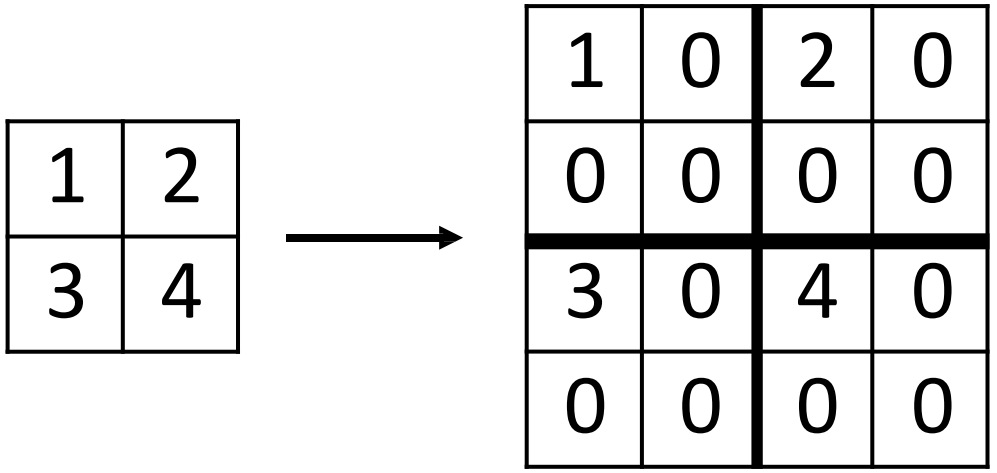


Predictions:
H x W

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

In-Network Upsampling: “Unpooling”

Bed of Nails

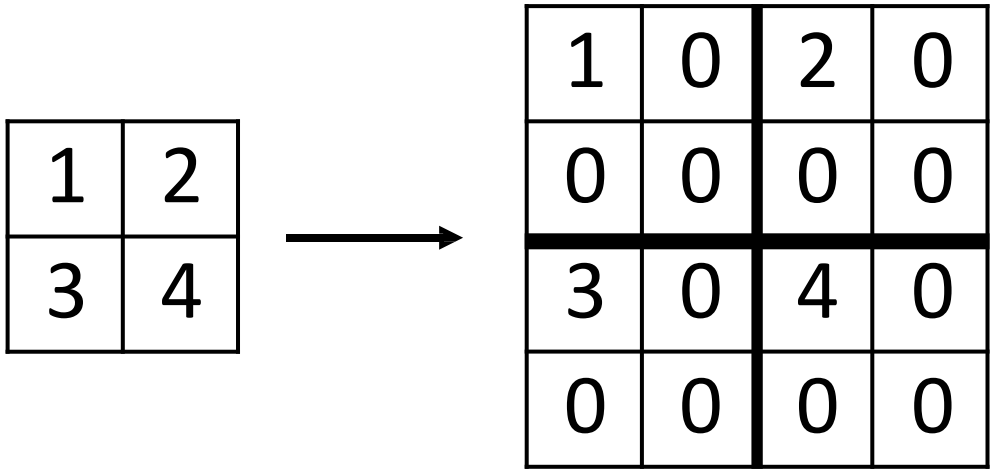


Input
C x 2 x 2

Output
C x 4 x 4

In-Network Upsampling: “Unpooling”

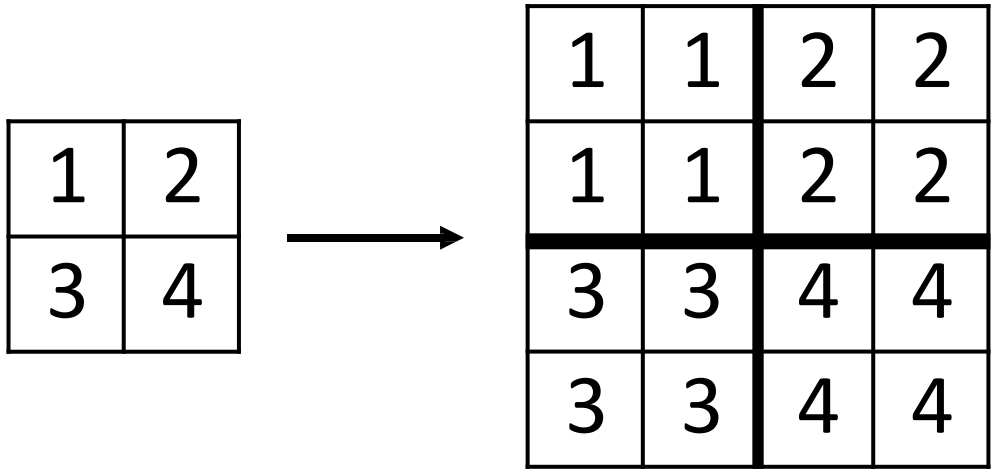
Bed of Nails



Input
C x 2 x 2

Output
C x 4 x 4

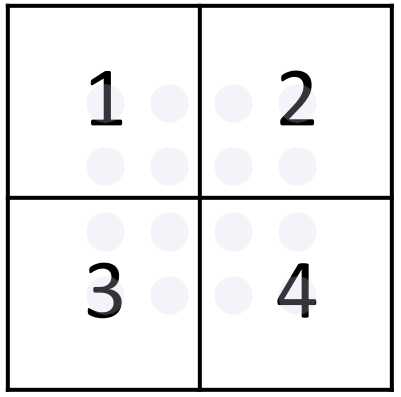
Nearest Neighbor



Input
C x 2 x 2

Output
C x 4 x 4

In-Network Upsampling: Bilinear Interpolation



1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

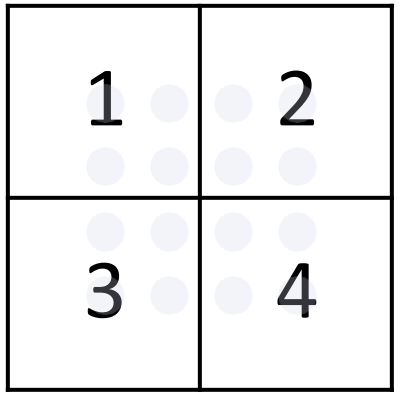
Input: C x 2 x 2

Output: C x 4 x 4

$$f_{x,y} = \sum_{i,j} f_{i,j} \max(0, 1 - |x - i|) \max(0, 1 - |y - j|) \quad i \in \{\lfloor x \rfloor - 1, \ldots, \lceil x \rceil + 1\} \quad j \in \{\lfloor y \rfloor - 1, \ldots, \lceil y \rceil + 1\}$$

Use two closest neighbors in x and y to construct linear approximations

In-Network Upsampling: Bicubic Interpolation



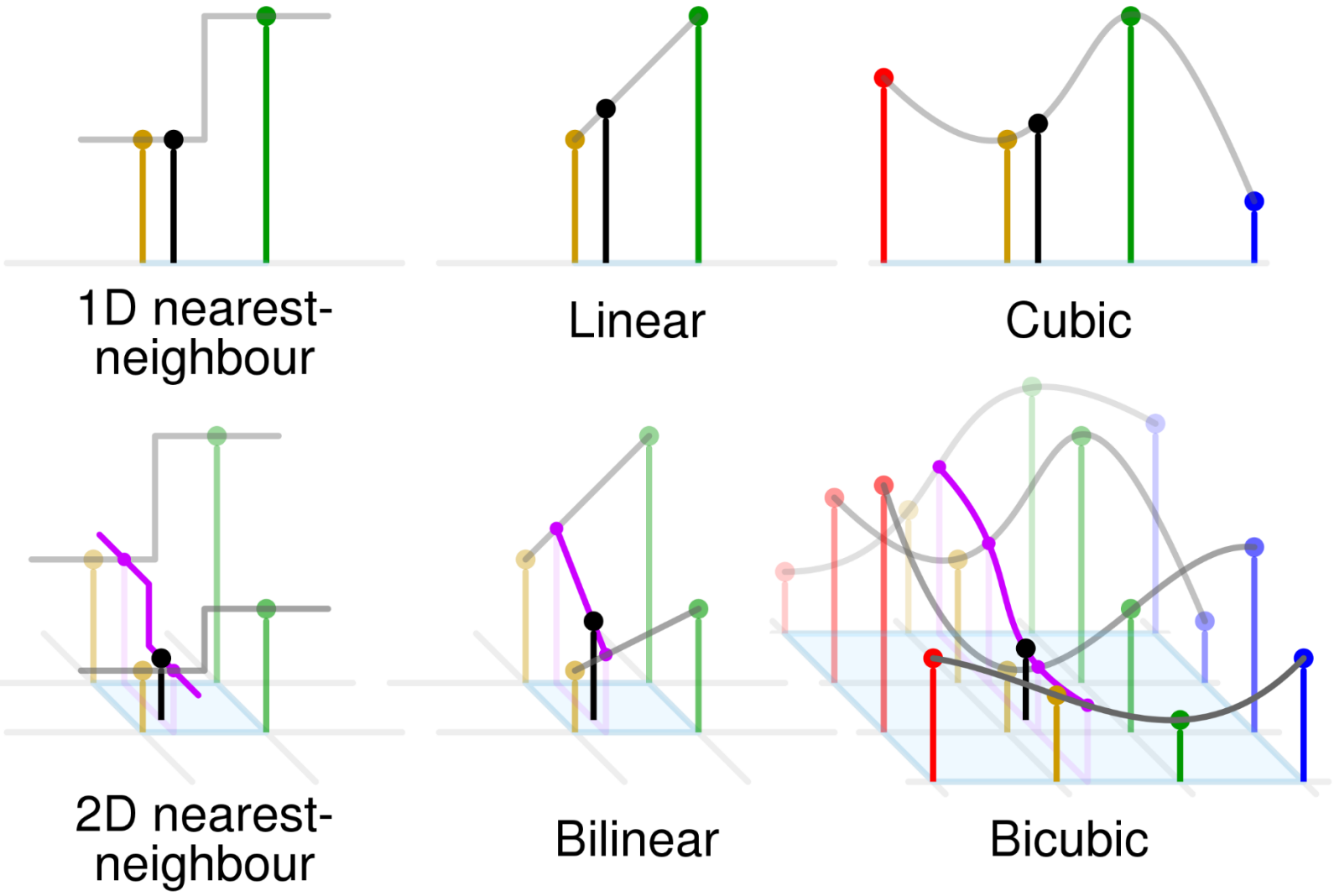
0.68	1.02	1.56	1.89
1.35	1.68	2.23	2.56
2.44	2.77	3.32	3.65
3.11	3.44	3.98	4.32

Input: C x 2 x 2

Output: C x 4 x 4

Use **three** closest neighbors in x and y to construct **cubic** approximations
(This is how we normally resize images!)

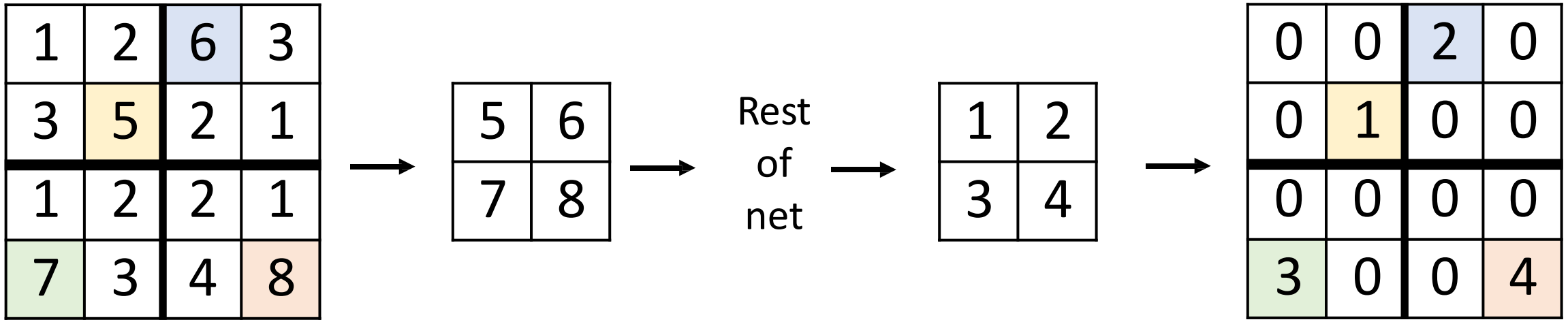
In-Network Upsampling

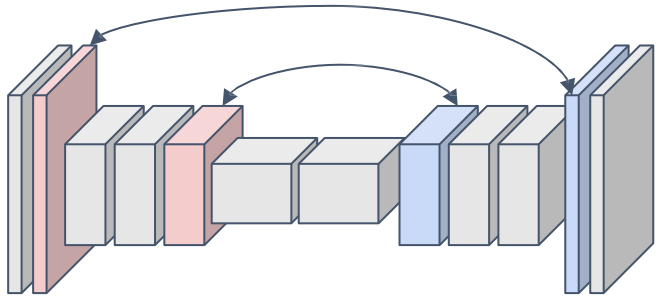


In-Network Upsampling: “Max Unpooling”

Max Pooling: Remember which position had the max

Max Unpooling: Place into remembered positions



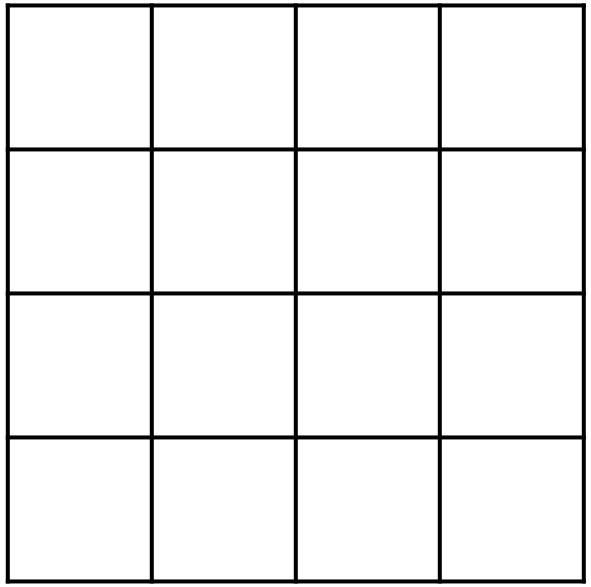


Pair each downsampling layer with an upsampling layer

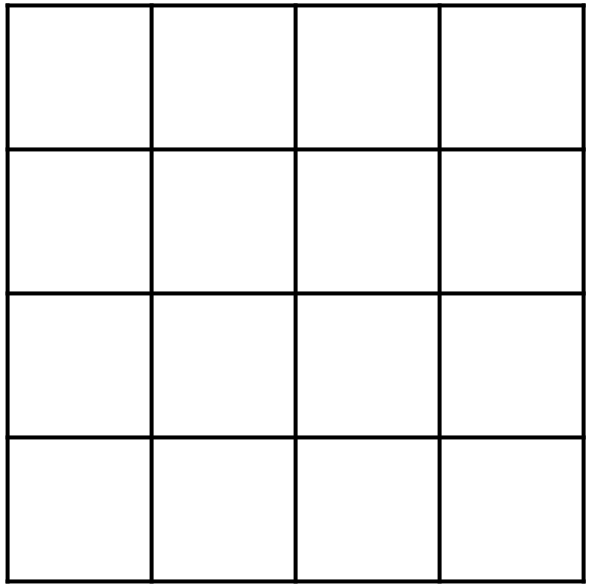
Noh et al, “Learning Deconvolution Network for Semantic Segmentation”, ICCV 2015

Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 1, pad 1



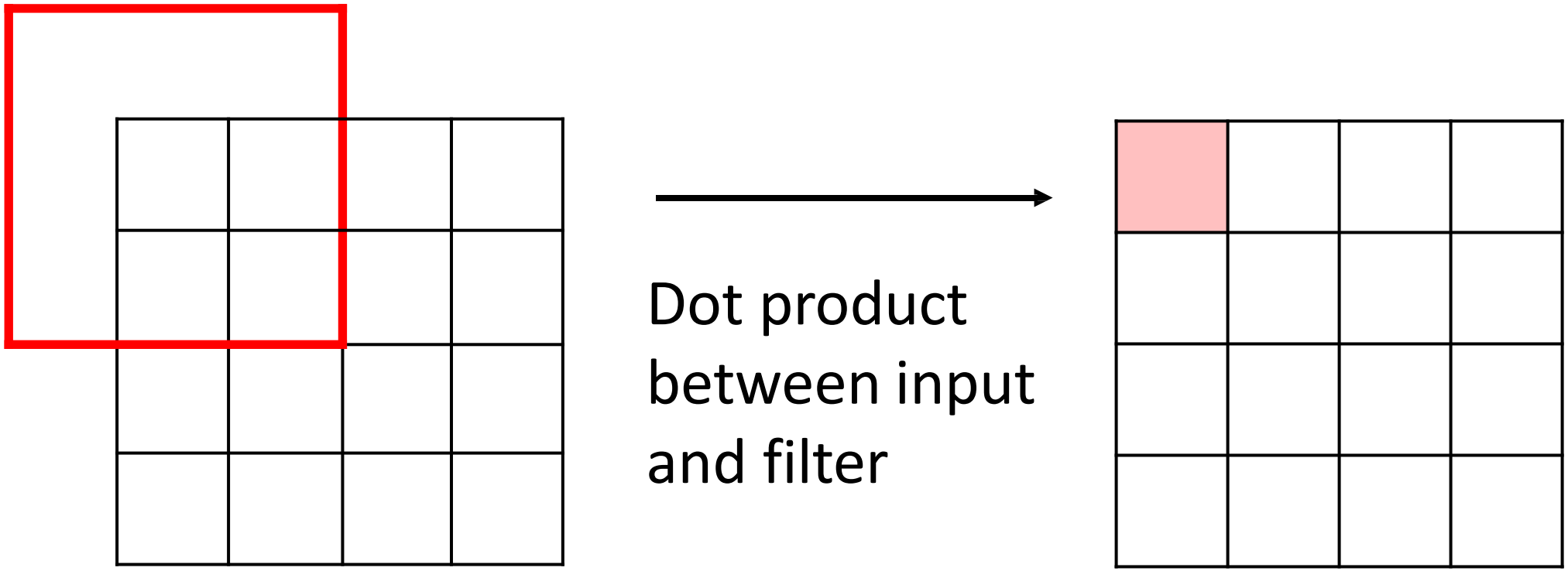
Input: 4 x 4



Output: 4 x 4

Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 1, pad 1

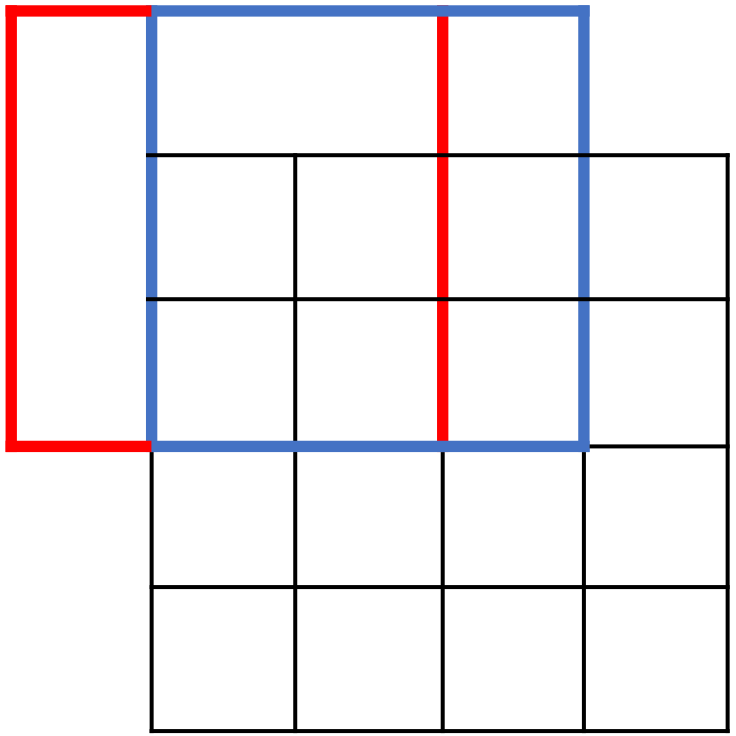


Input: 4 x 4

Output: 4 x 4

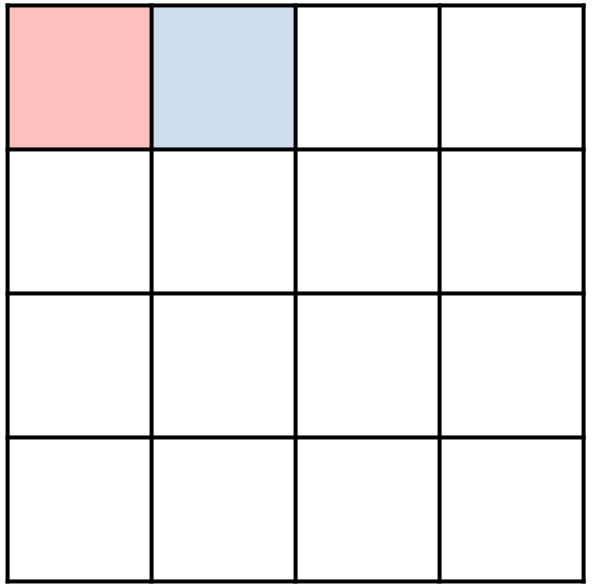
Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 1, pad 1



Input: 4 x 4

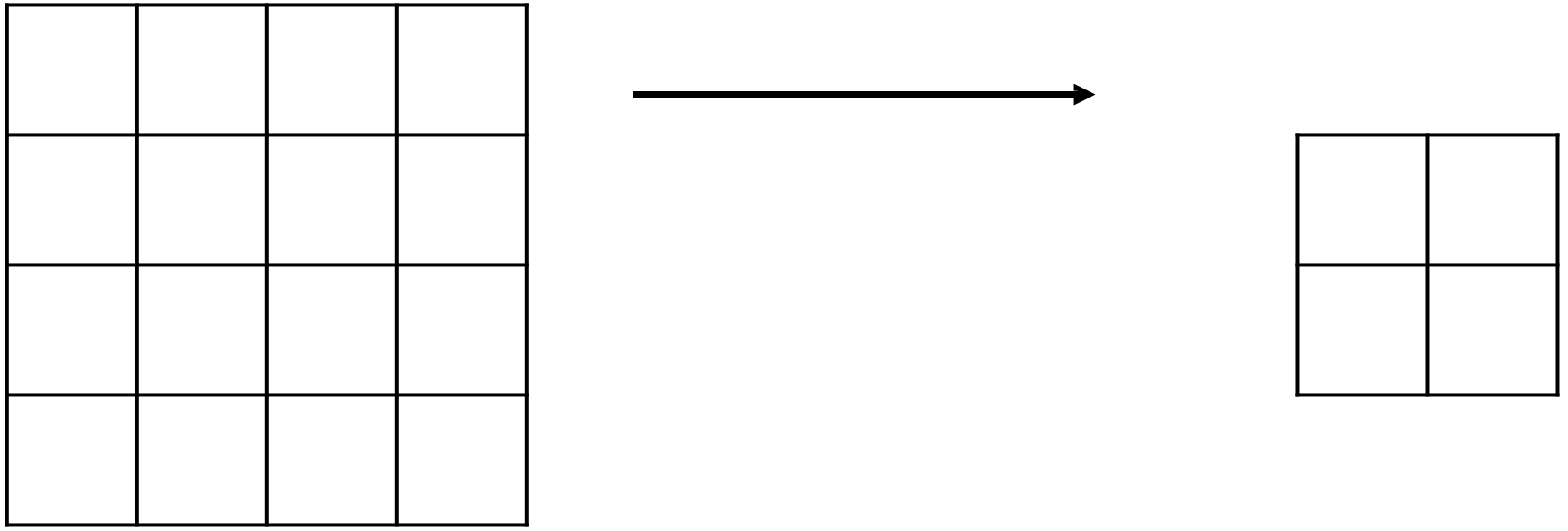
Dot product between input and filter



Output: 4 x 4

Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 2, pad 1

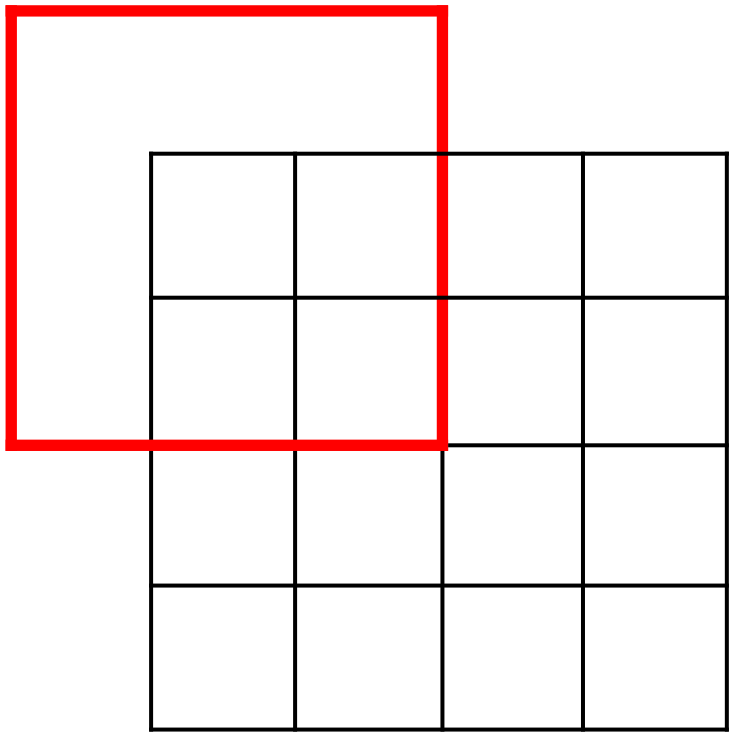


Input: 4 x 4

Output: 2 x 2

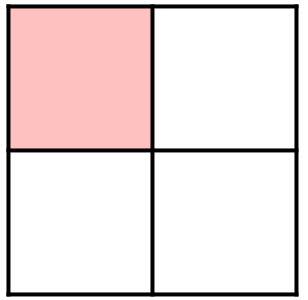
Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 2, pad 1



Input: 4 x 4

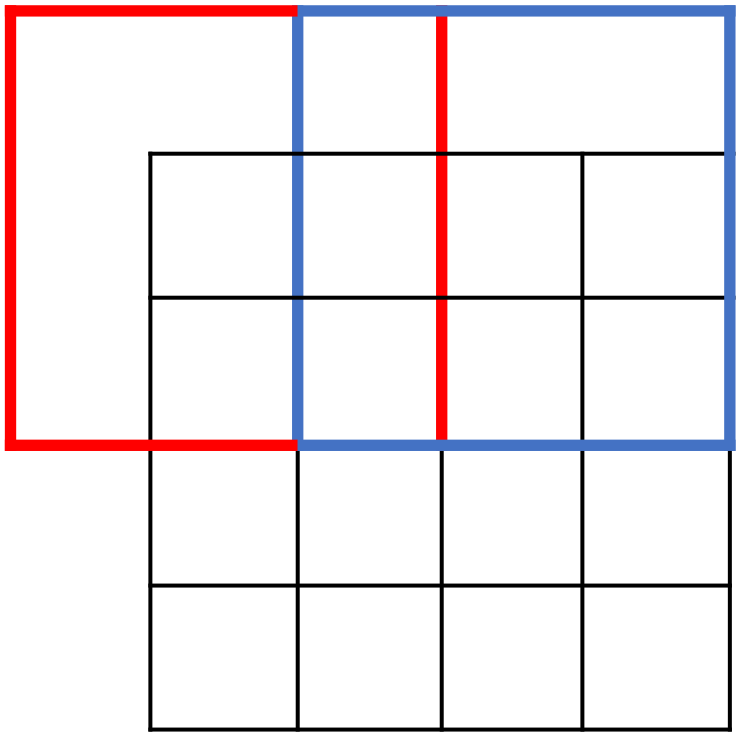
Dot product between input and filter



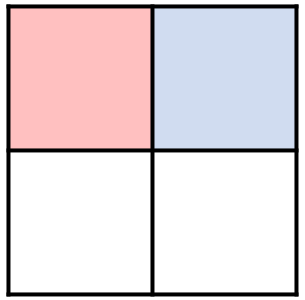
Output: 2 x 2

Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 2, pad 1



Dot product between input and filter

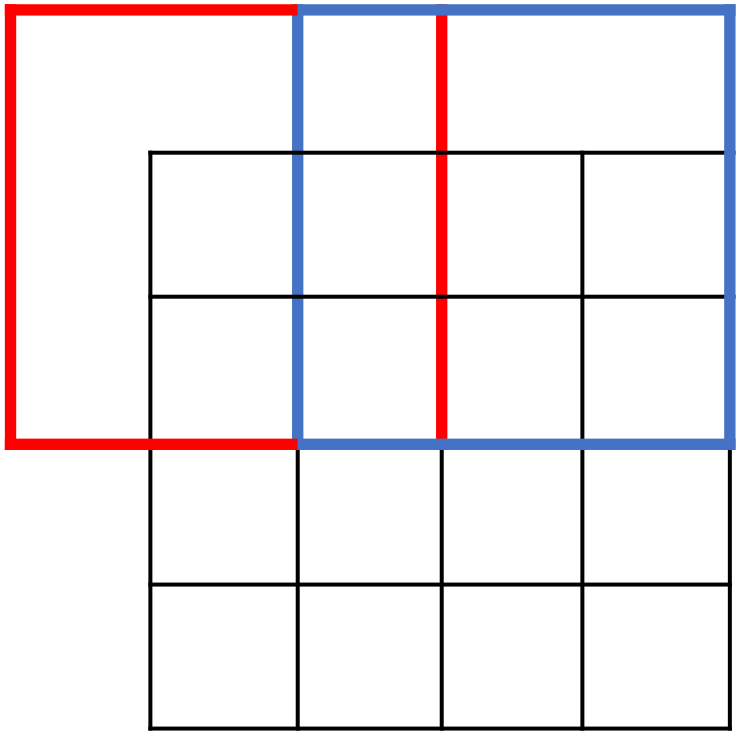


Input: 4 x 4

Output: 2 x 2

Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 2, pad 1

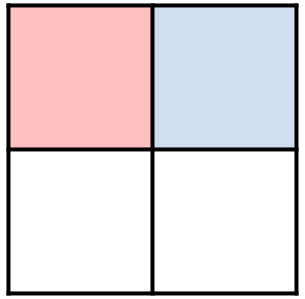


Input: 4 x 4

Convolution with stride > 1 is “Learnable Downsampling”

Can we use stride < 1 for “Learnable Upsampling”?

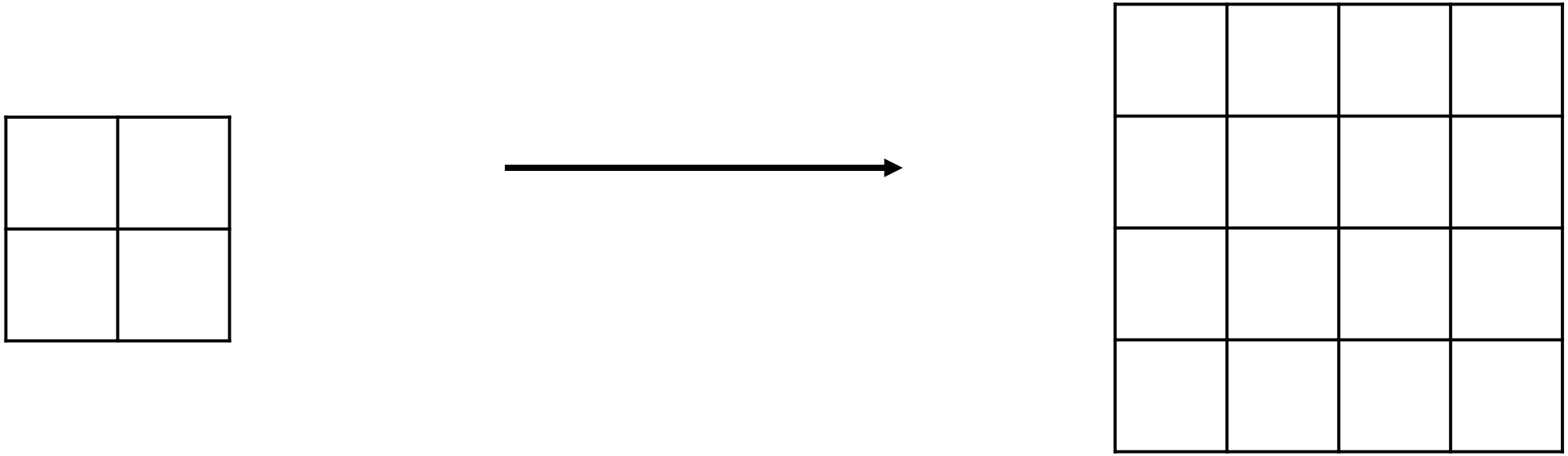
Dot product between input and filter



Output: 2 x 2

Learnable Upsampling: Transposed Convolution

3 x 3 **convolution transpose**, stride 2

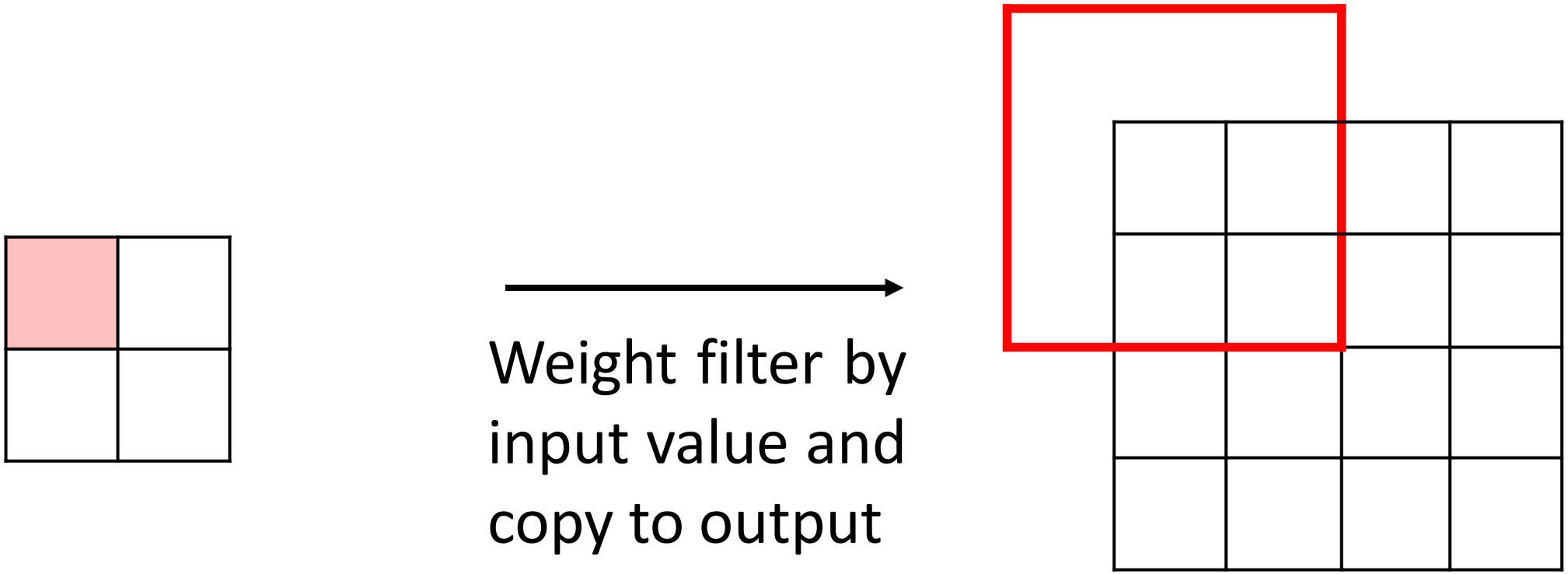


Input: 2 x 2

Output: 4 x 4

Learnable Upsampling: Transposed Convolution

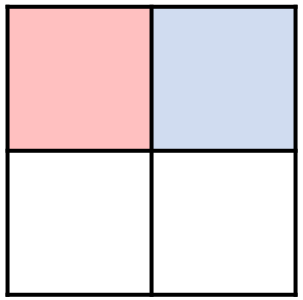
3 x 3 **convolution transpose**, stride 2



Learnable Upsampling: Transposed Convolution

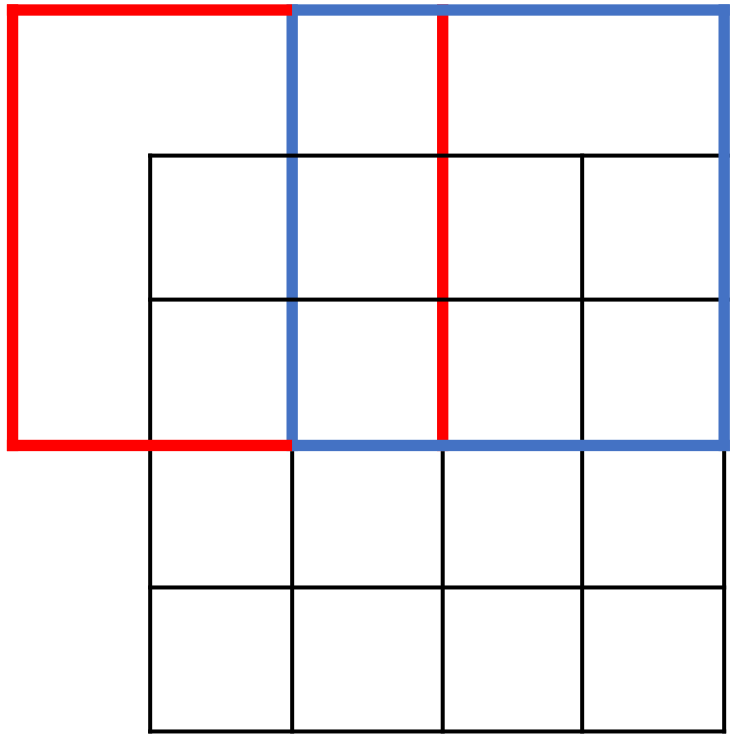
3 x 3 **convolution transpose**, stride 2

Filter moves 2 pixels in output
for every 1 pixel in input



Input: 2 x 2

Weight filter by input value and copy to output

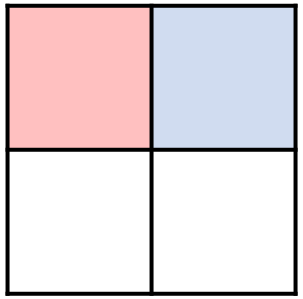


Output: 4 x 4

Learnable Upsampling: Transposed Convolution

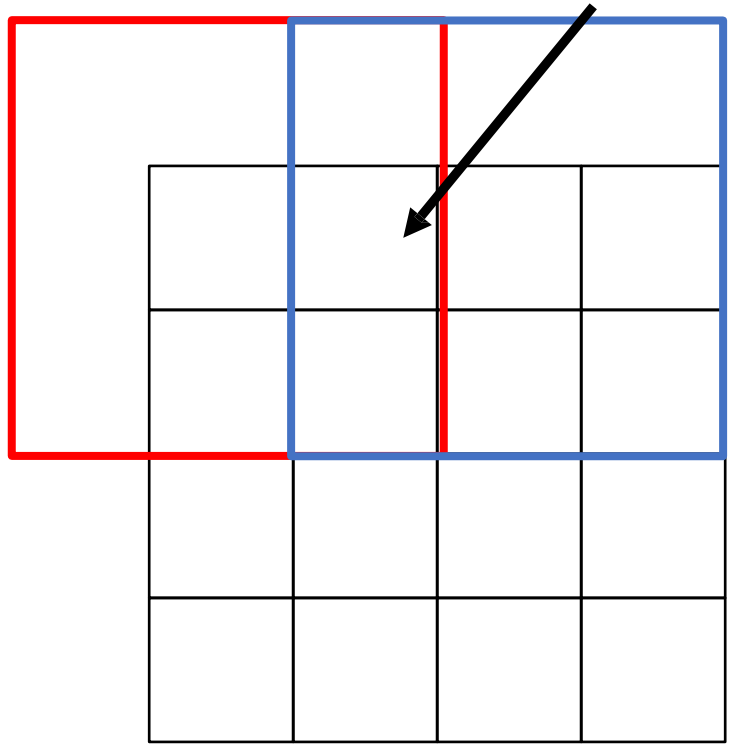
3 x 3 **convolution transpose**, stride 2

Filter moves 2 pixels in <u>output</u>
for every 1 pixel in <u>input</u>



Input: 2 x 2

Weight filter by input value and copy to output



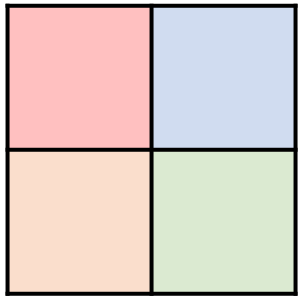
Output: 4 x 4

Learnable Upsampling: Transposed Convolution

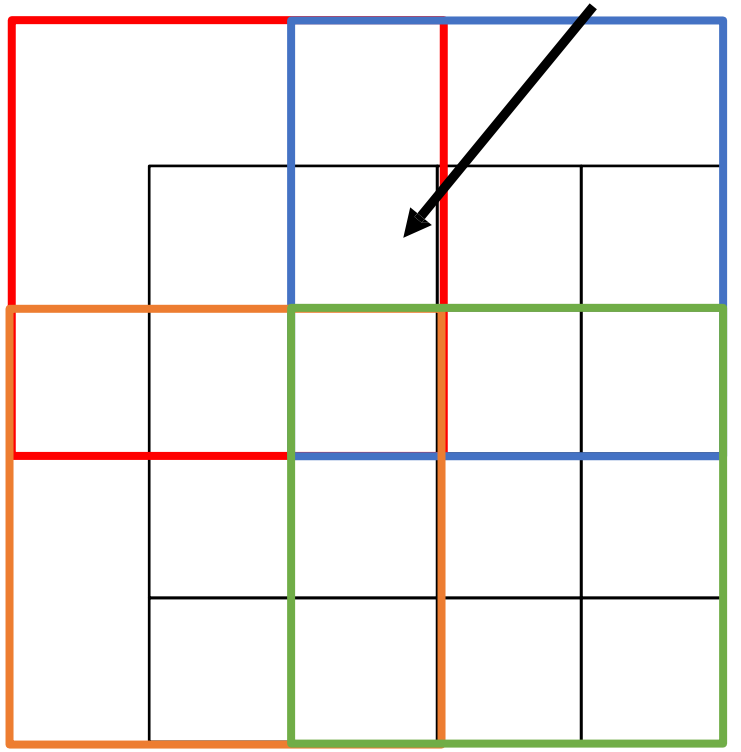
3 x 3 **convolution transpose**, stride 2

Sum where output overlaps

This gives 5x5 output – need to trim one pixel from top and left to give 4x4 output



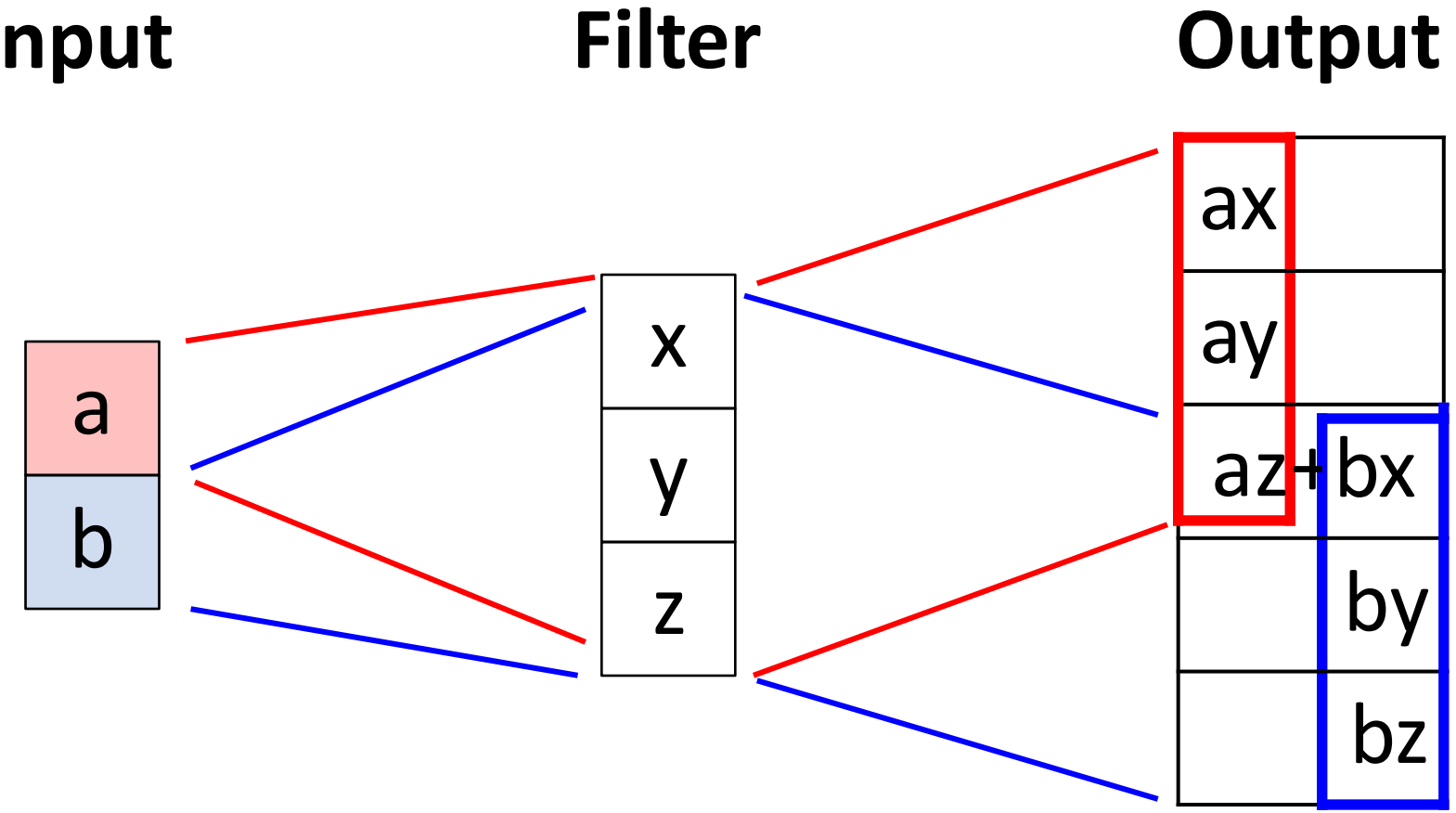
Weight filter by input value and copy to output



Input: 2 x 2

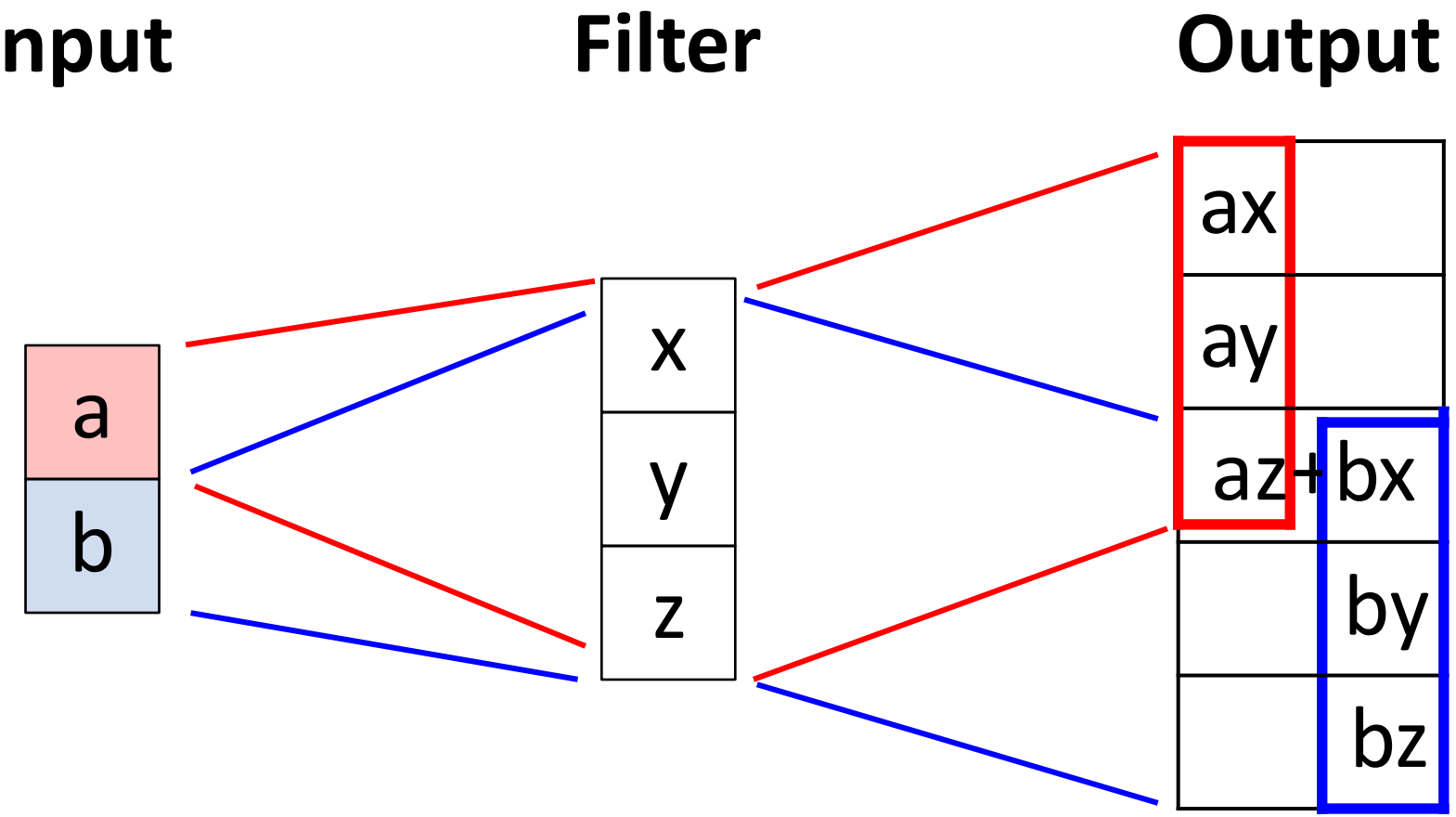
Output: 4 x 4

Transposed Convolution: 1D example



- Output has copies of filter weighted by input
- Stride 2: Move 2 pixels output for each pixel in input
- Sum at overlaps

Transposed Convolution: 1D example



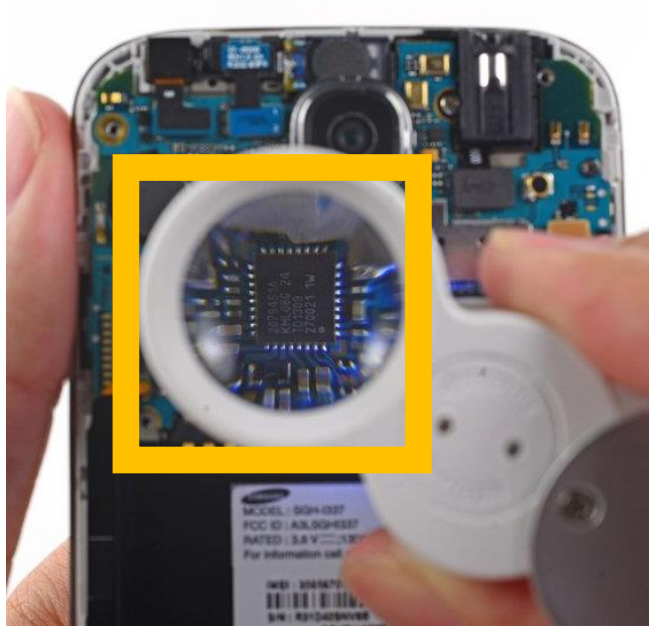
This has many names:

- Deconvolution (bad)!
- Upconvolution
- Fractionally strided convolution
- Backward strided convolution
- Transposed Convolution (best name)

Transposed Convolution

Convolution

Filter: little lens that looks at a pixel.



Transposed Conv.

Filter: tiles used to make image

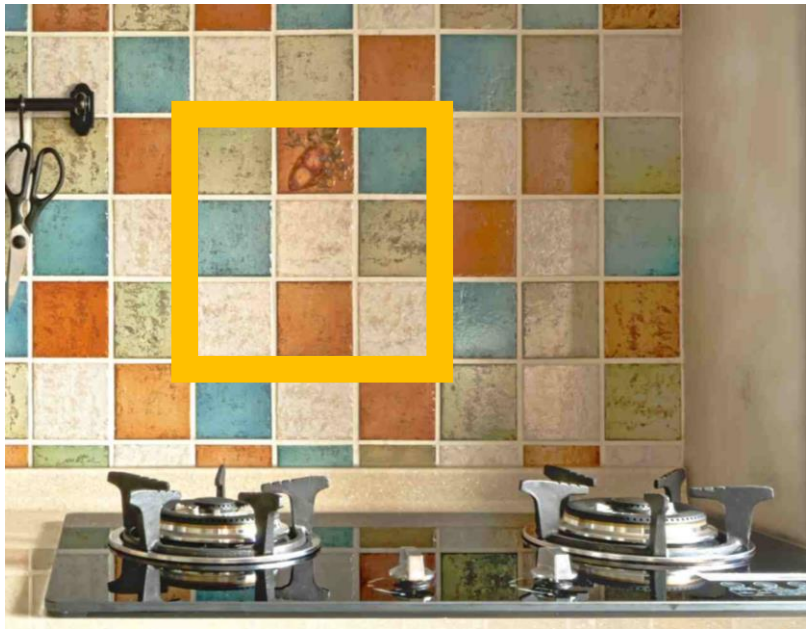


Image credit: ifixit.com, thespruce.com

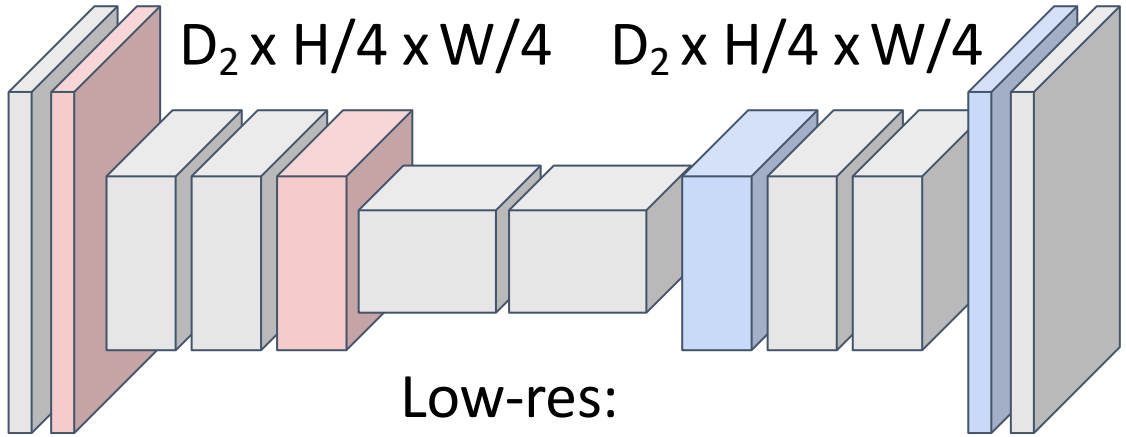
Fully Convolutional Network

Downsampling: Pooling, strided convolution

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Upsampling: interpolation, transposed conv

Input:
3 x H x W



Predictions:
H x W

Loss function: Per-Pixel cross-entropy

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

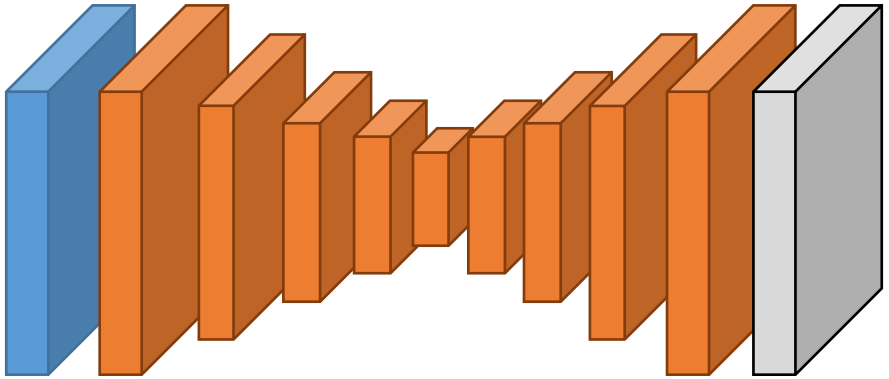
Skip Connections

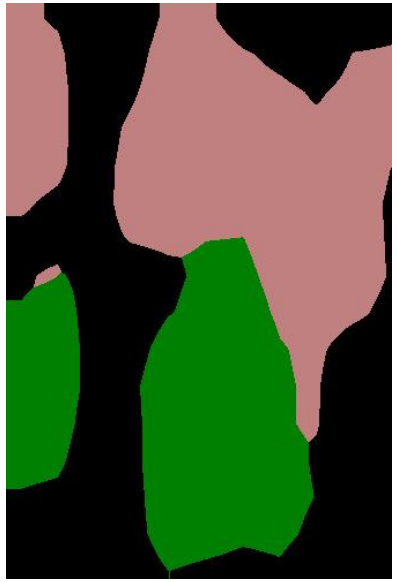
Retaining Spatial Details

While the output *is* HxW, just upsampling often produces results without details/not aligned with the image.

Why?

Information about details lost when downsampling!

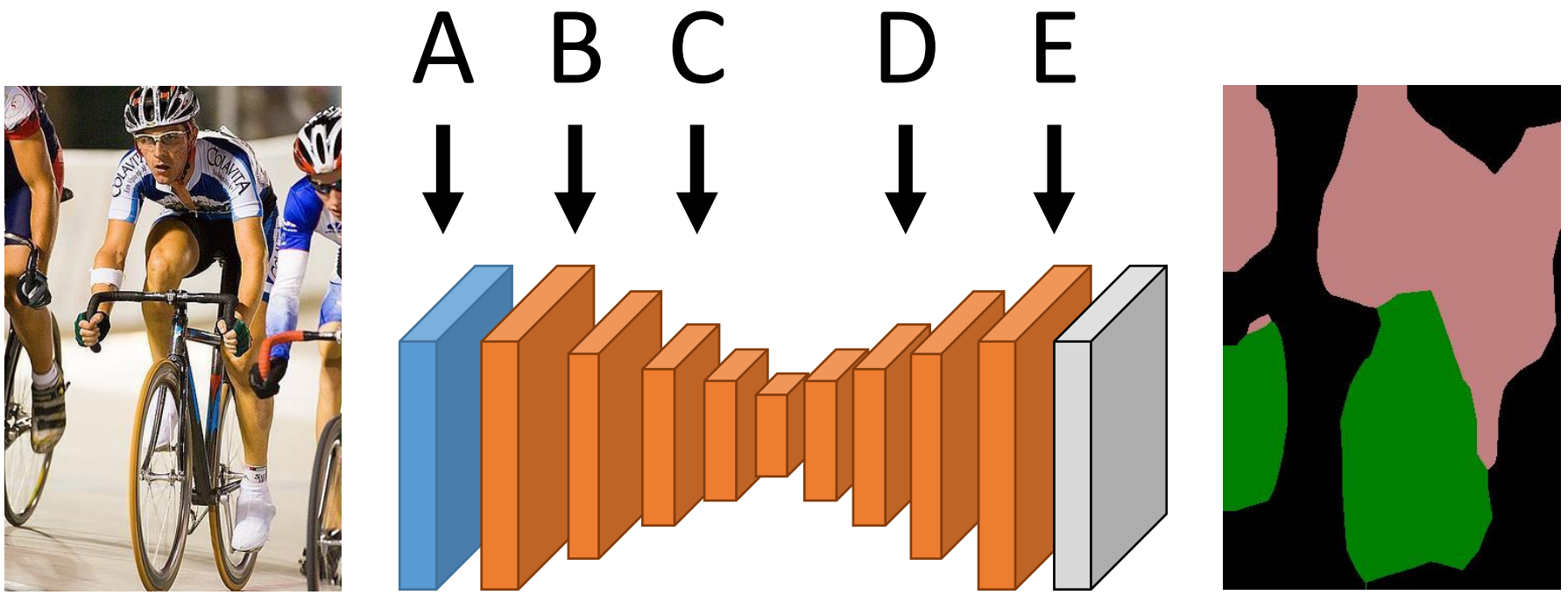




Result from Long et al. *Fully Convolutional Networks For Semantic Segmentation*. CVPR 2014

Retaining Spatial Details

Where is the useful information about the high-frequency details of the image?



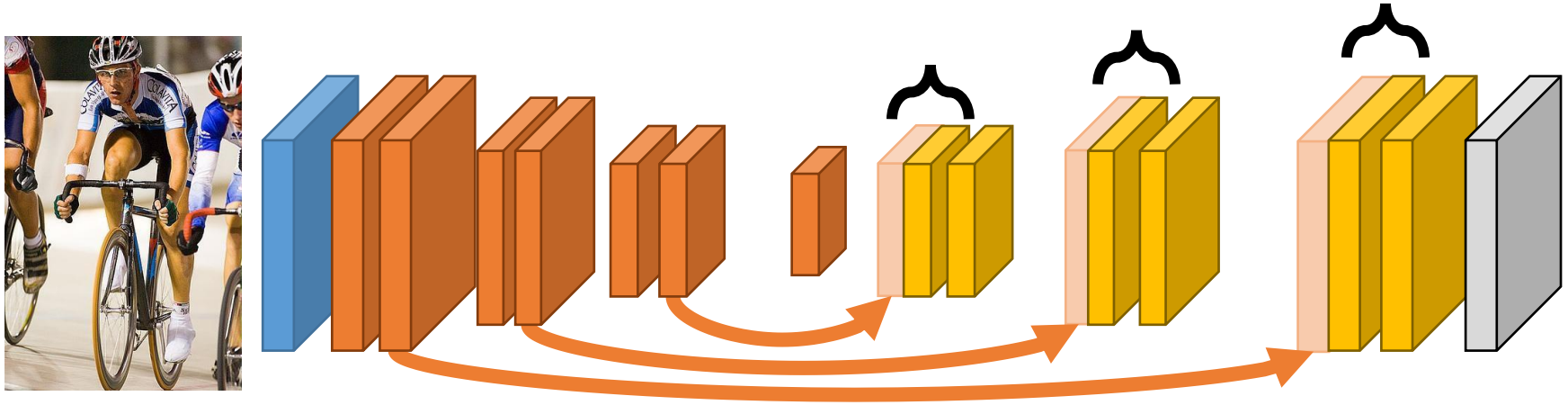
Result from Long et al. *Fully Convolutional Networks For Semantic Segmentation*. CVPR 2014

Retaining Spatial Details

How do you send details forward in the network?

You copy the activations forward.

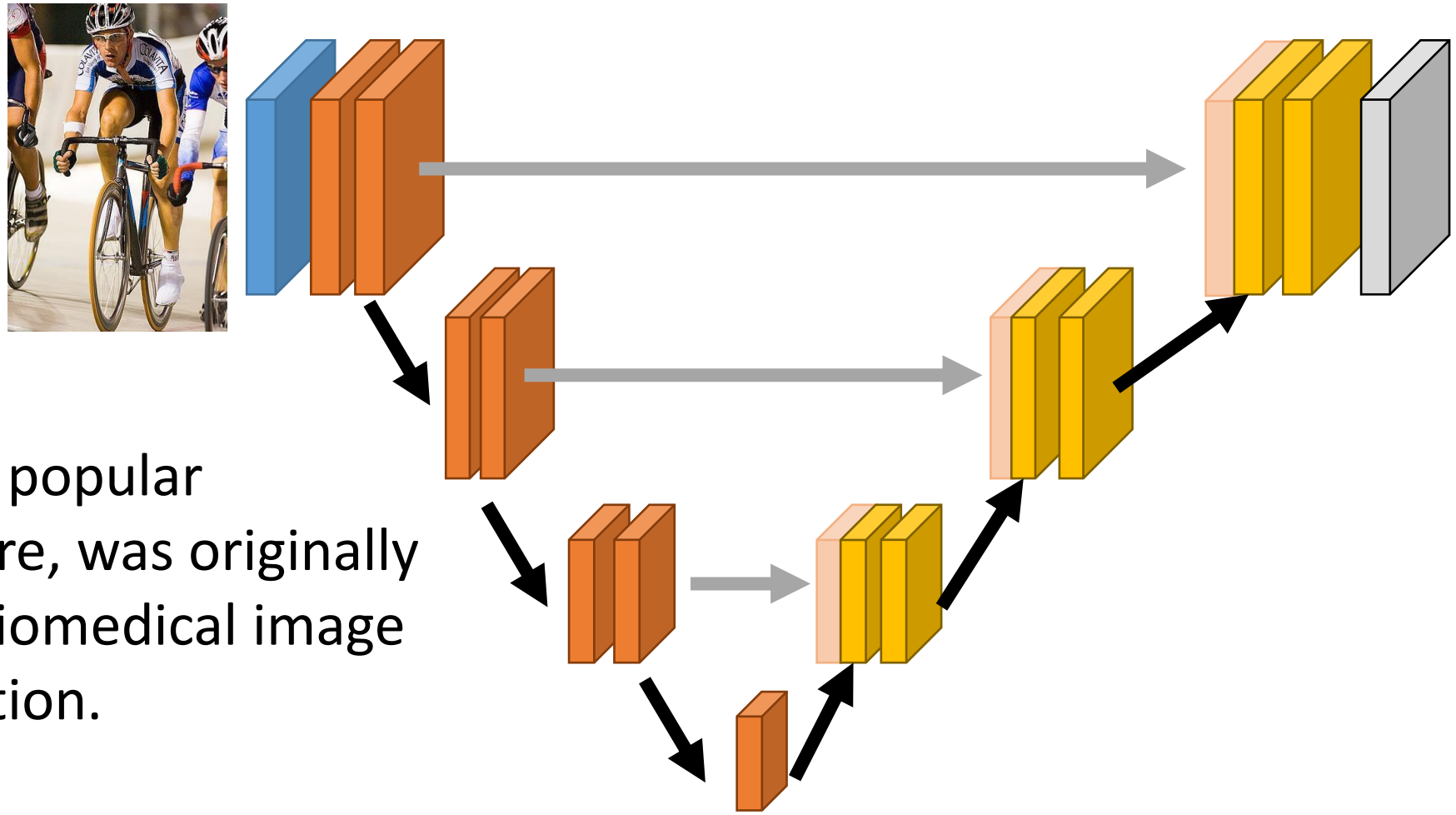
Subsequent layers at the same resolution figure out how to fuse things.



Copy

Result from Long et al. *Fully Convolutional Networks For Semantic Segmentation*. CVPR 2014

U-Net



Extremely popular architecture, was originally used for biomedical image segmentation.

Ronneberger et al, “U-Net: Convolutional Networks for Biomedical Image Segmentation”, MICCAI 2015

Spatial Contexts

The Importance of Spatial Contexts

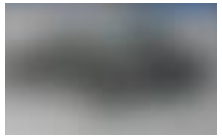
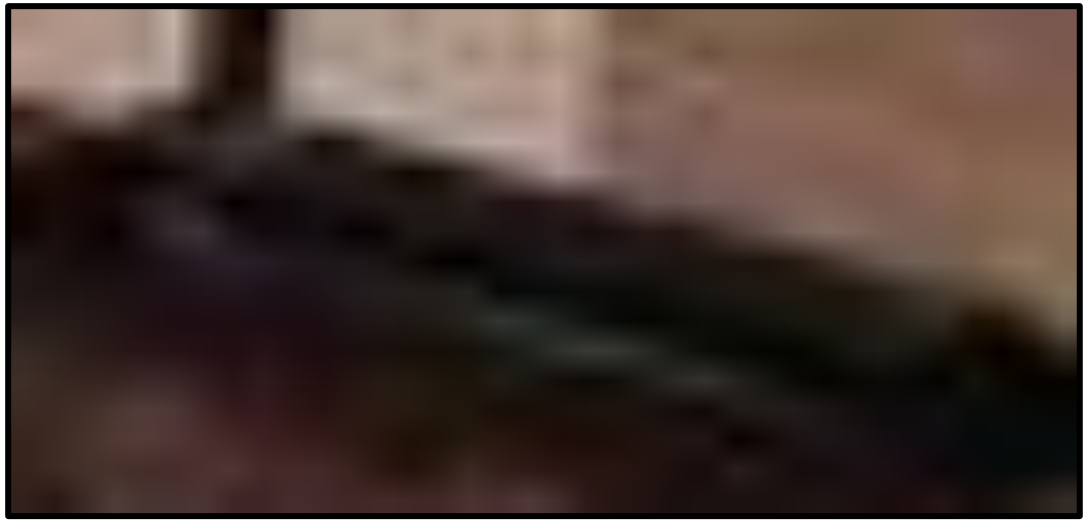


Image credit: A. Torralba

The Importance of Spatial Contexts

What's this? (No Cheating!)



(a) Keyboard?

(b) Hammer?

(c) Old cell phone?

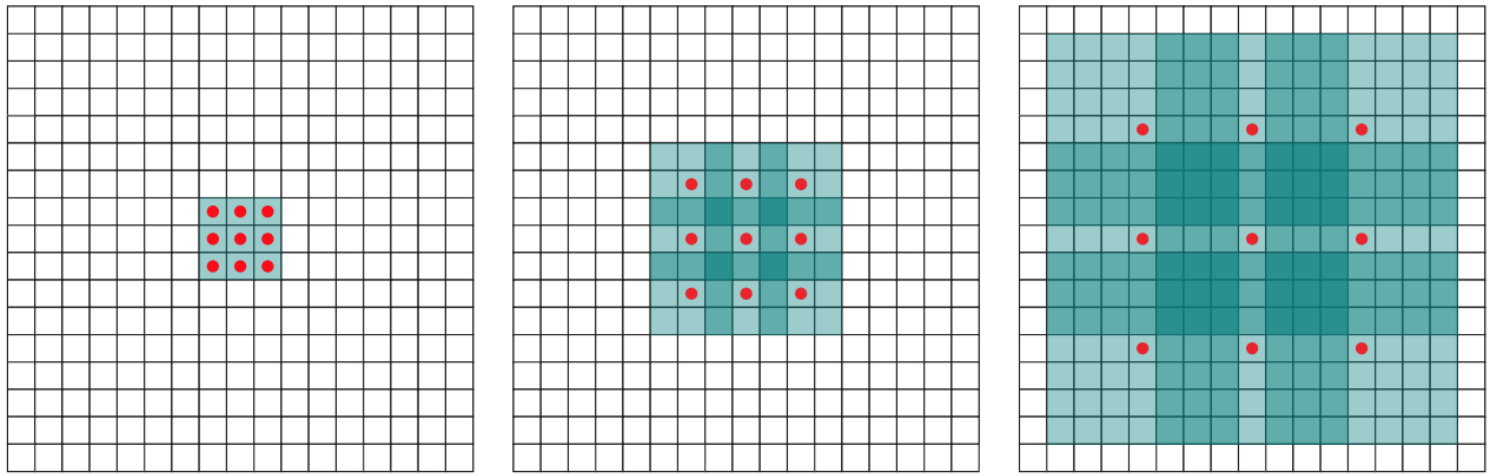
(d) Xbox controller?

Image credit: COCO dataset

The Importance of Spatial Contexts

Image credit: COCO dataset

Spatial Contexts: Dilated Convolution

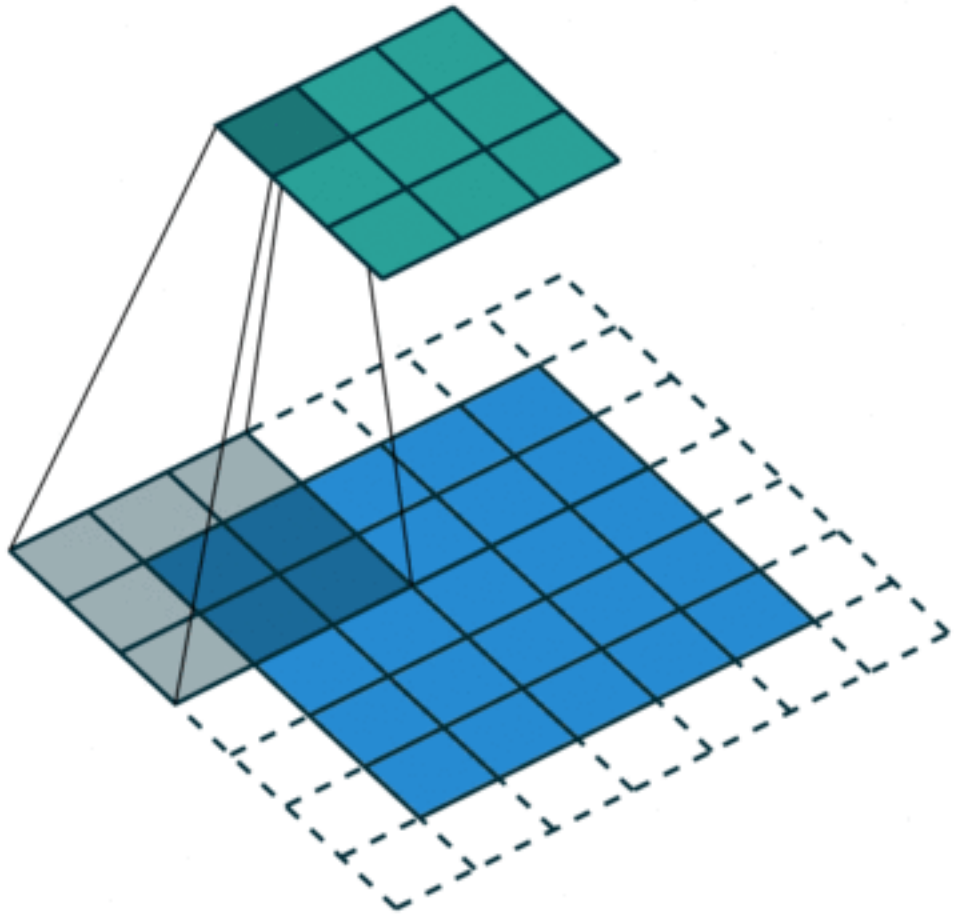


Consecutive 1-Dilated (left), 2-Dilated (middle), 4-Dilated (right) 3×3 Convolution

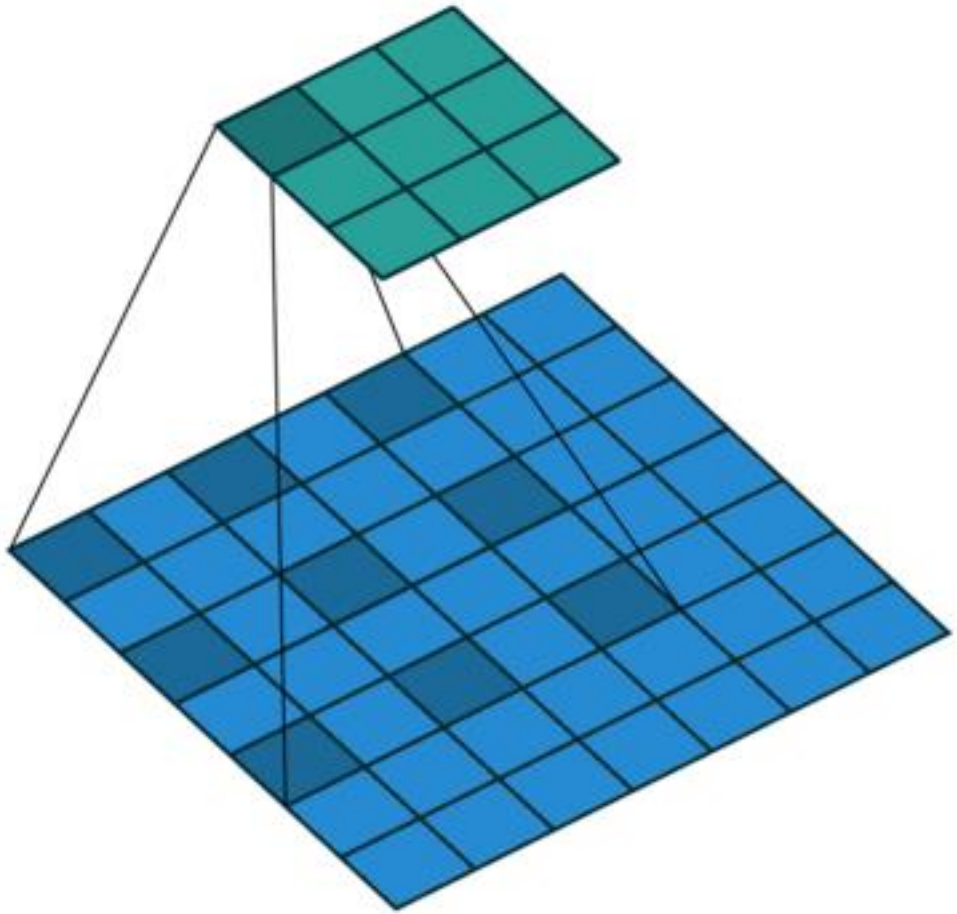
- Receptive field of an element x in layer k + 1 is the set of elements in layer k that influence it
- Resulting receptive field of 2^i -Dilated feature map is size $(2^{i+2} - 1)^2$
- Receptive field grows exponentially while number of parameters is constant

Yu et al, "Multi-Scale Context Aggregation By Dilated Convolutions", ICLR 2016

Spatial Contexts: Dilated Convolution

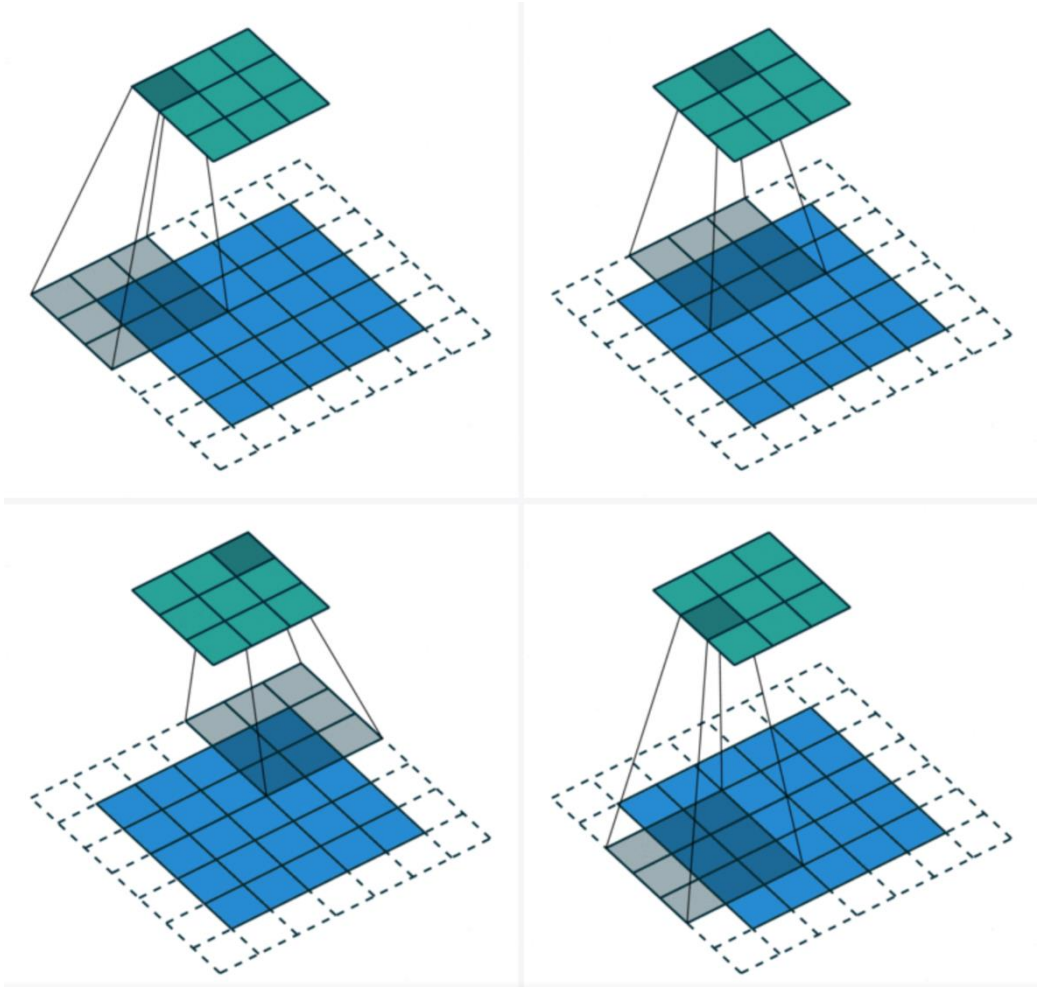


standard convolution

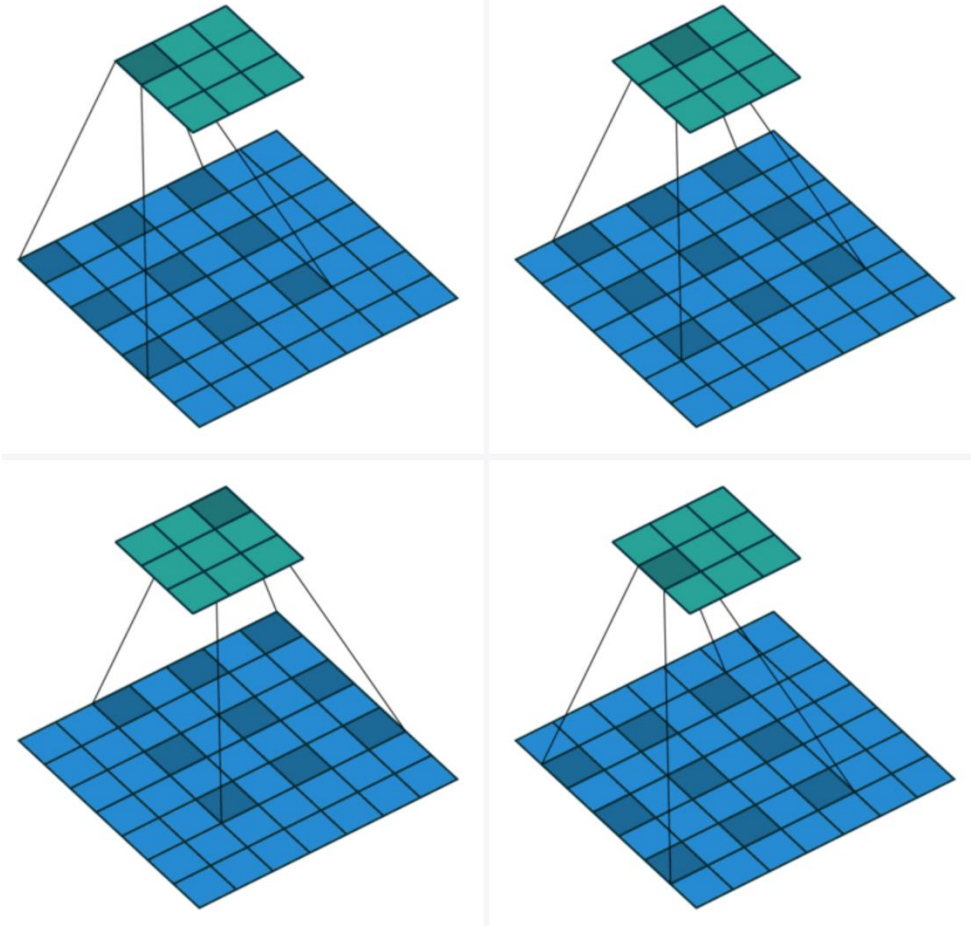


dilated convolution

Spatial Contexts: Dilated Convolution

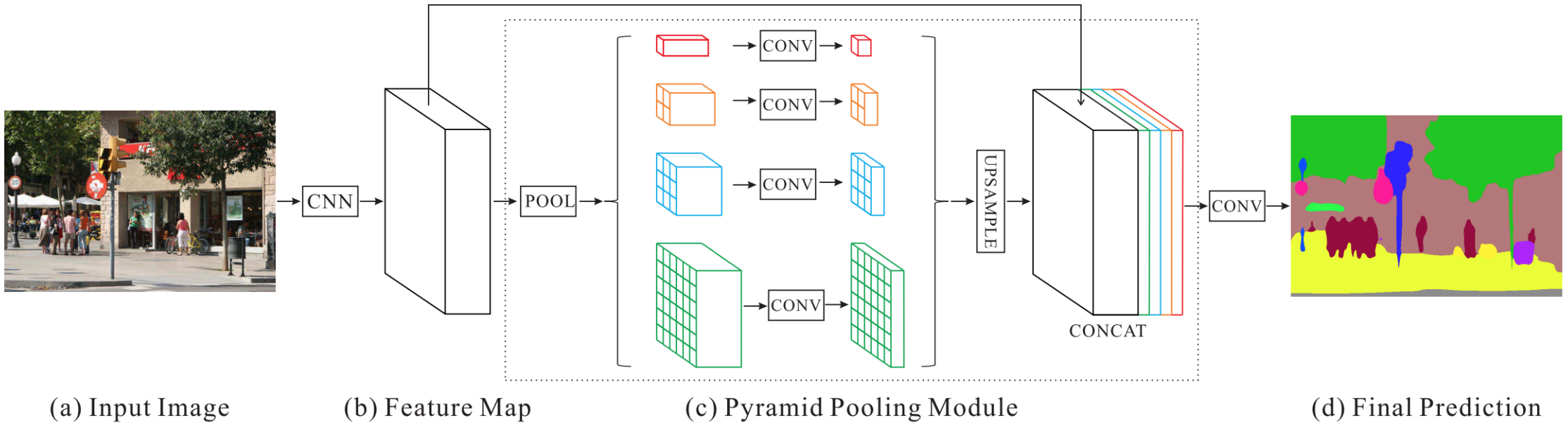


standard convolution



dilated convolution

Spatial Contexts: Pyramid Scene Parsing Net



- Hierarchical global prior, containing information with different scales and varying among different sub-regions
- Pyramid pooling module for global scene prior on the final feature map
- Use 1x1 convolution to reduce the number of channels

Zhao et al, “Pyramid Scene Parsing Network”, CVPR 2017

Spatial Contexts: Markov Random Field

$p_i(label =' table') = 0.8$

i

Energy Function

$$\min \quad E = Unary + Pair$$

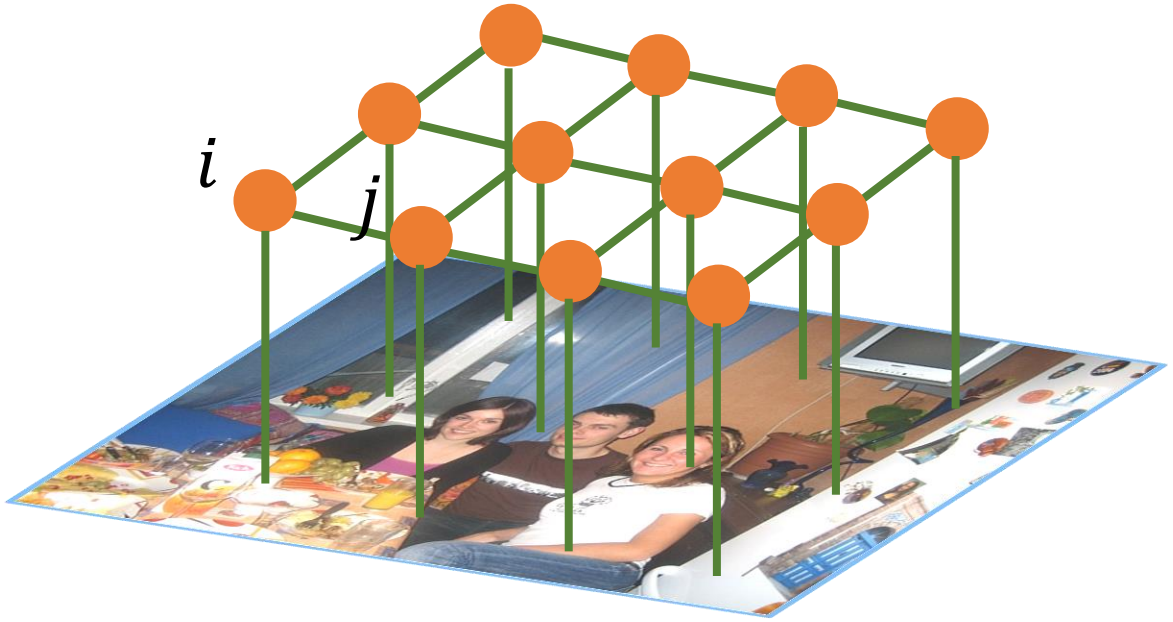
Unary Term

$$Unary = -\sum_i \ln p_i(label)$$

Liu et al, “Semantic Image Segmentation via Deep Parsing Network”, ICCV 2015

Spatial Contexts: Markov Random Field

$diss(i,j) = ($  ,  $) = 0.8$



Appearance Consistency

Energy Function

$$\min \quad E = Unary + Pair$$

Unary Term

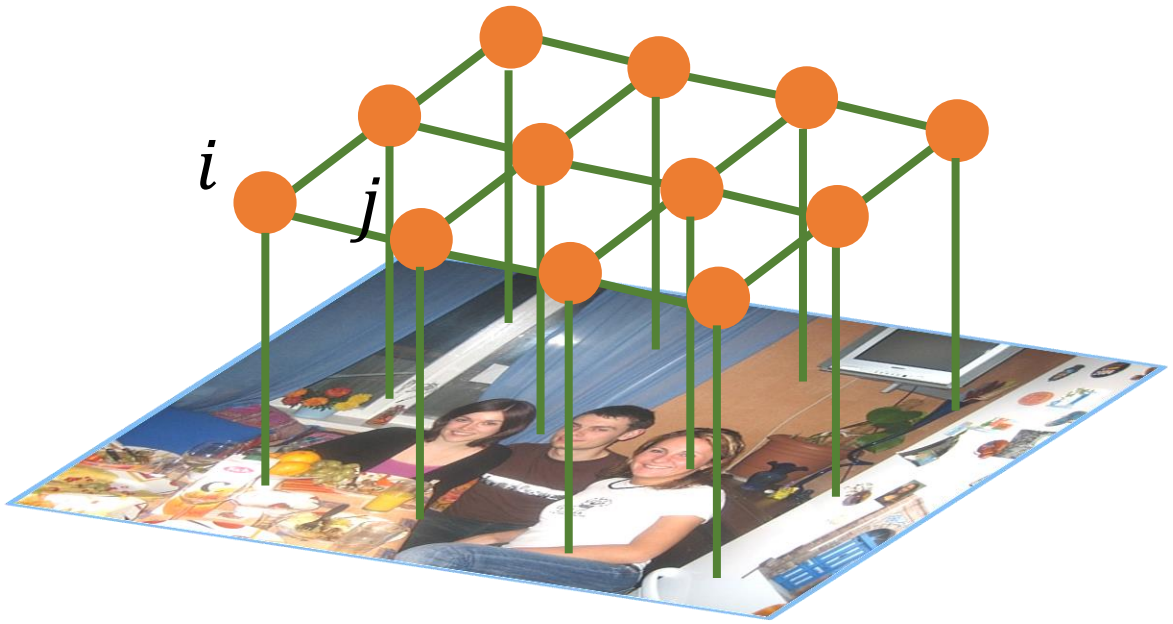
$$Unary = -\sum_{i} \ln p_i(label)$$

Pairwise Term

$$Pair = \sum_{i,j} cost(i) * diss(i,j)$$

Spatial Contexts: Markov Random Field

$cost(i; label =' table') = 0.1$



Label Consistency

Energy Function

$$\min \quad E = Unary + Pair$$

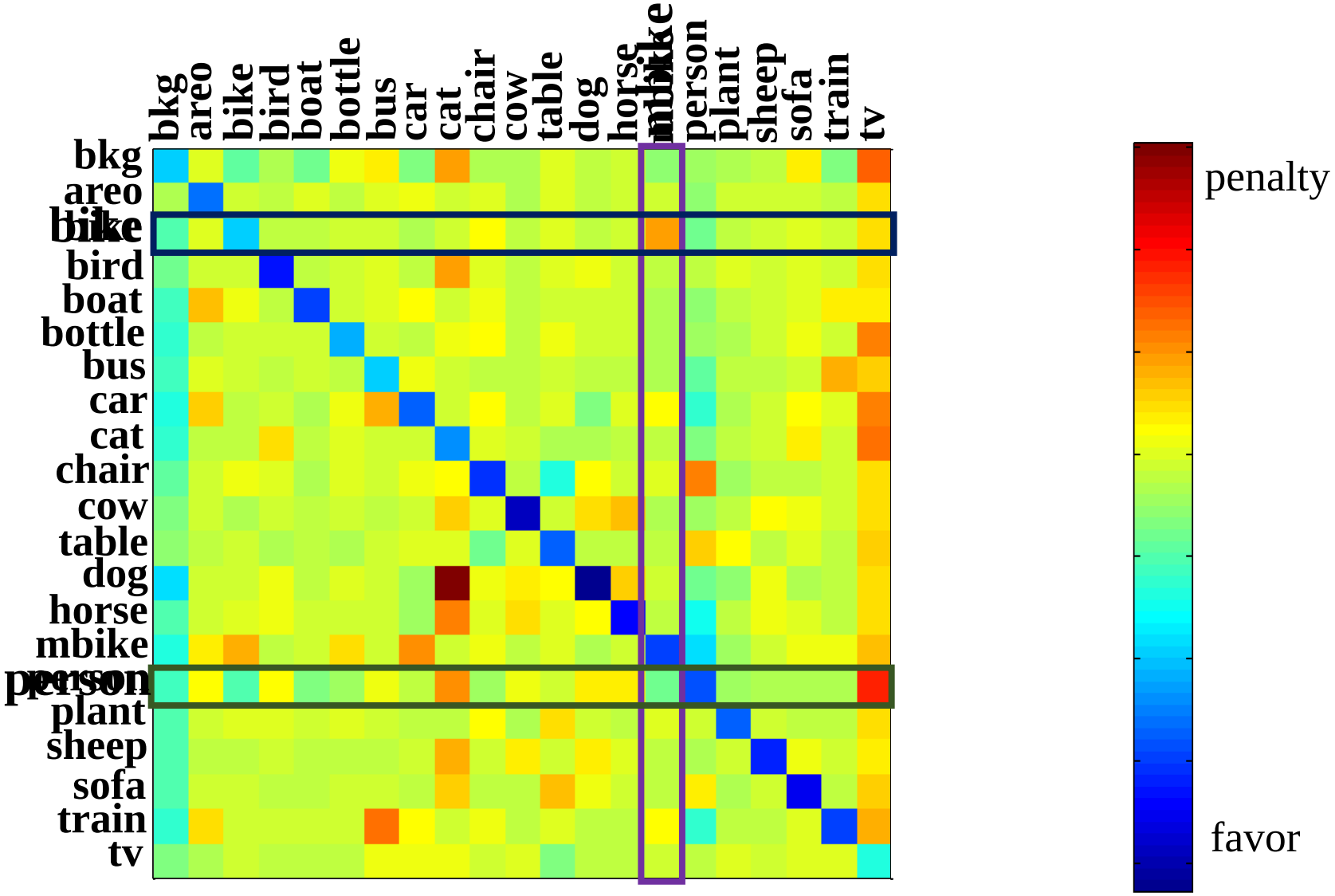
Unary Term

$$Unary = -\sum_i \ln p_i(label)$$

Pairwise Term

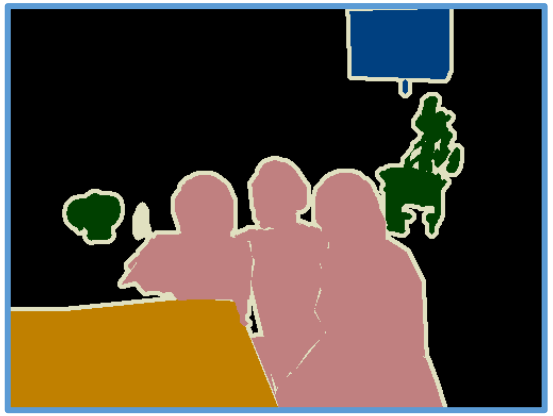
$$Pair = \sum_{i,j} cost(i) * diss(i,j)$$

Spatial Contexts: Markov Random Field



Spatial Contexts: Markov Random Field

Original Image



Ground Truth

Unary Term

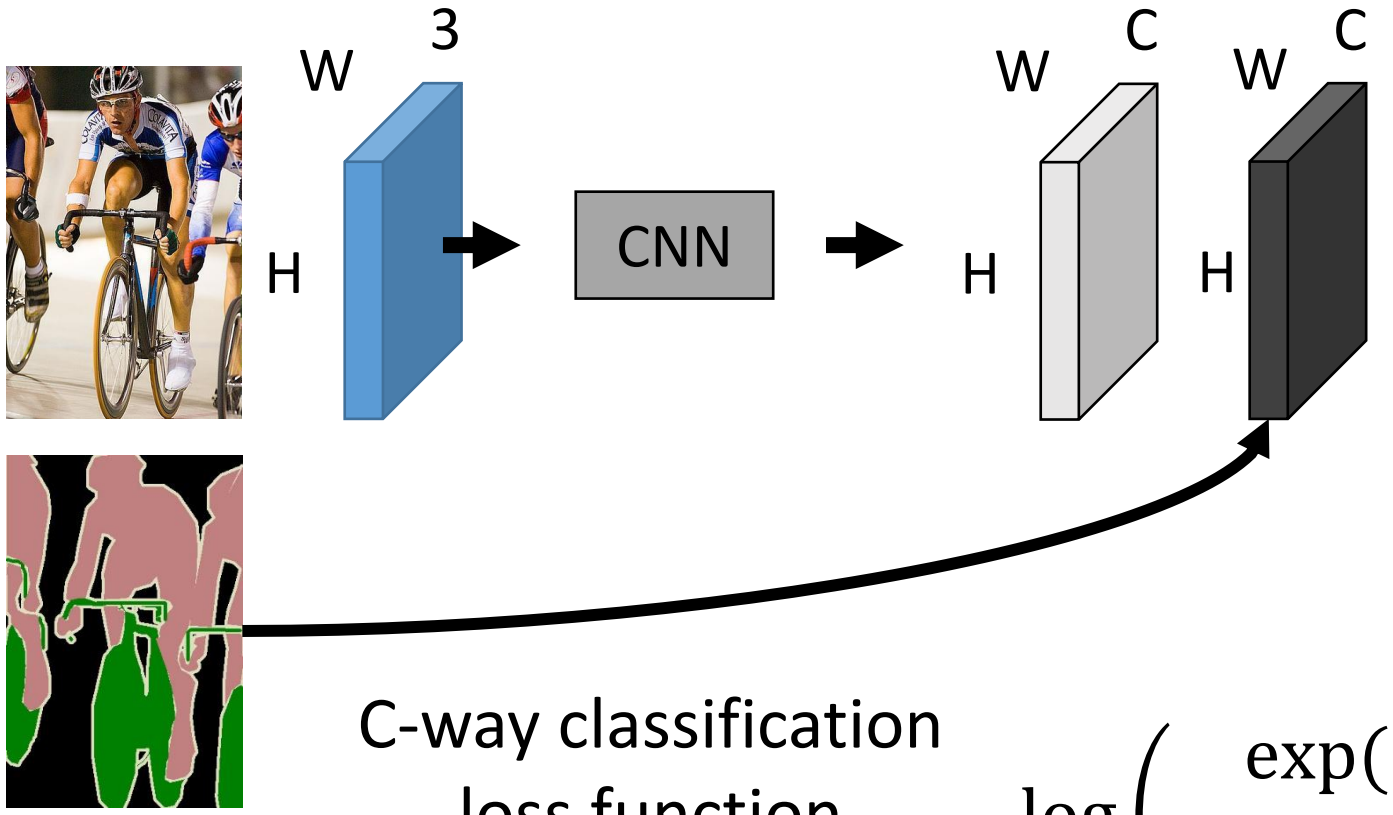
Triple Penalty

Label Contexts

Joint Tuning

Train and Evaluate FCN

Train Fully Convolutional Network



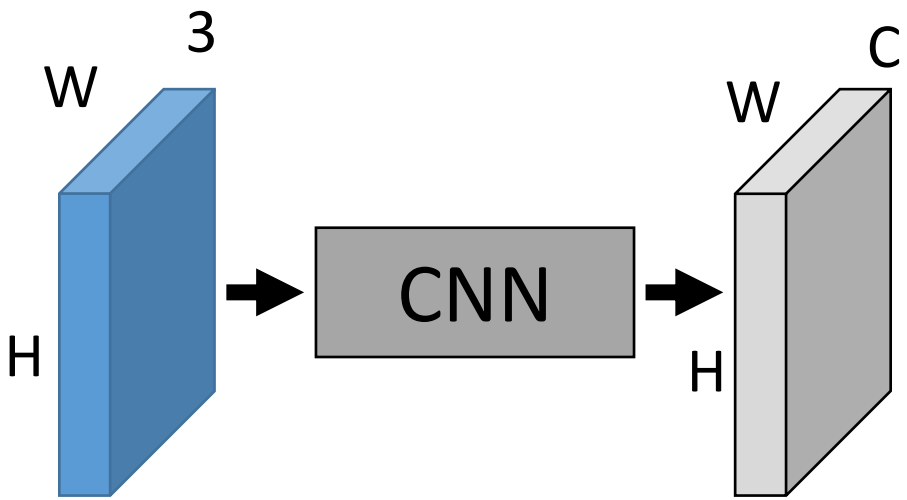
C-way classification loss function at every pixel:

$$-\log\left(\frac{\exp((Wx)_{y_i}}{\sum_k \exp((Wx)_k))}\right)$$

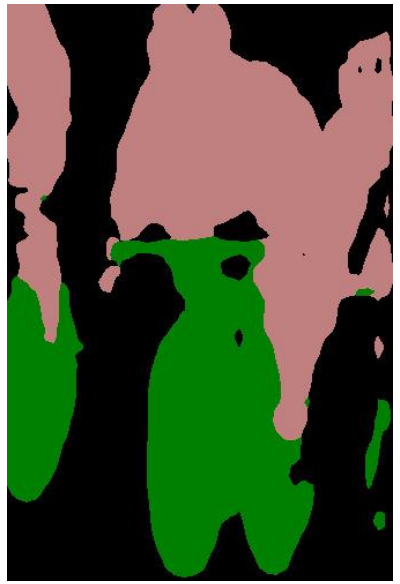
Image Credit: Everingham et al. Pascal VOC 2012.

Evaluate Fully Convolutional Network

Input Image



Predicted Classes



How do we convert final HxWxC into labels?

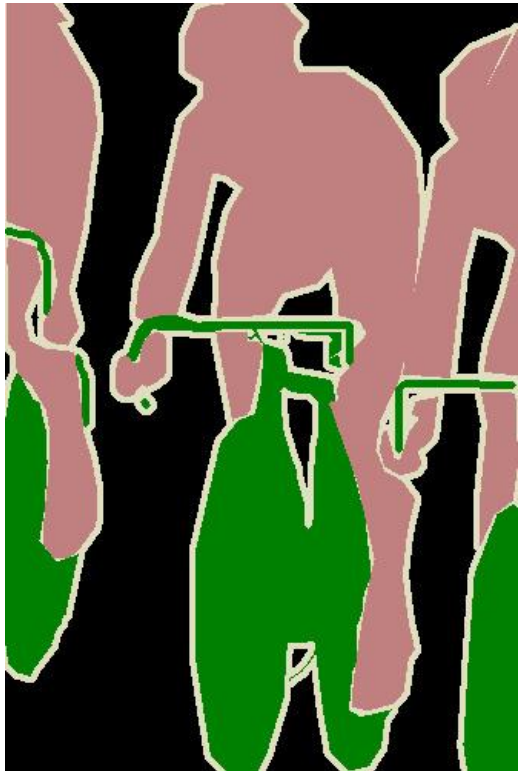
argmax over labels

Evaluate Fully Convolutional Network

Input

Prediction ($\hat{y}$)

Ground-Truth (y)

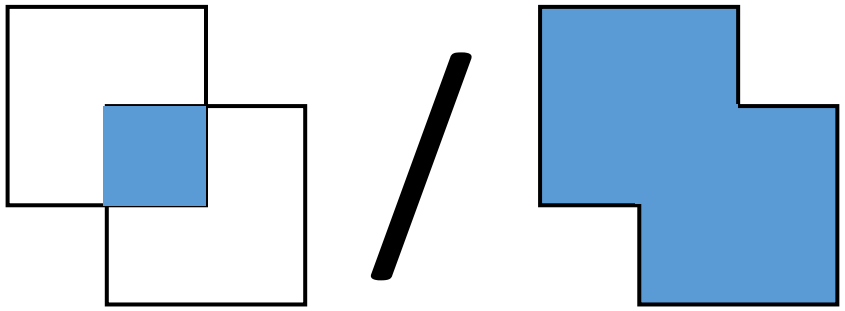


Evaluate Fully Convolutional Network

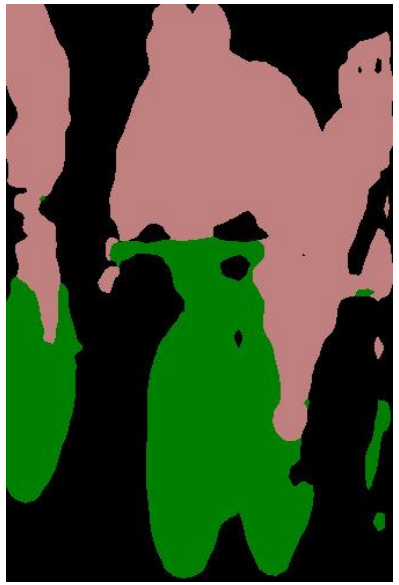
Prediction and ground-truth are images where each pixel is one of C classes.

Accuracy: mean($\hat{y} = y$)

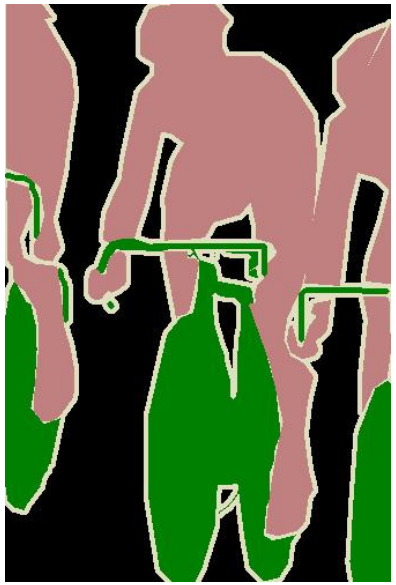
Intersection over union, averaged over classes



Prediction ($\hat{y}$)

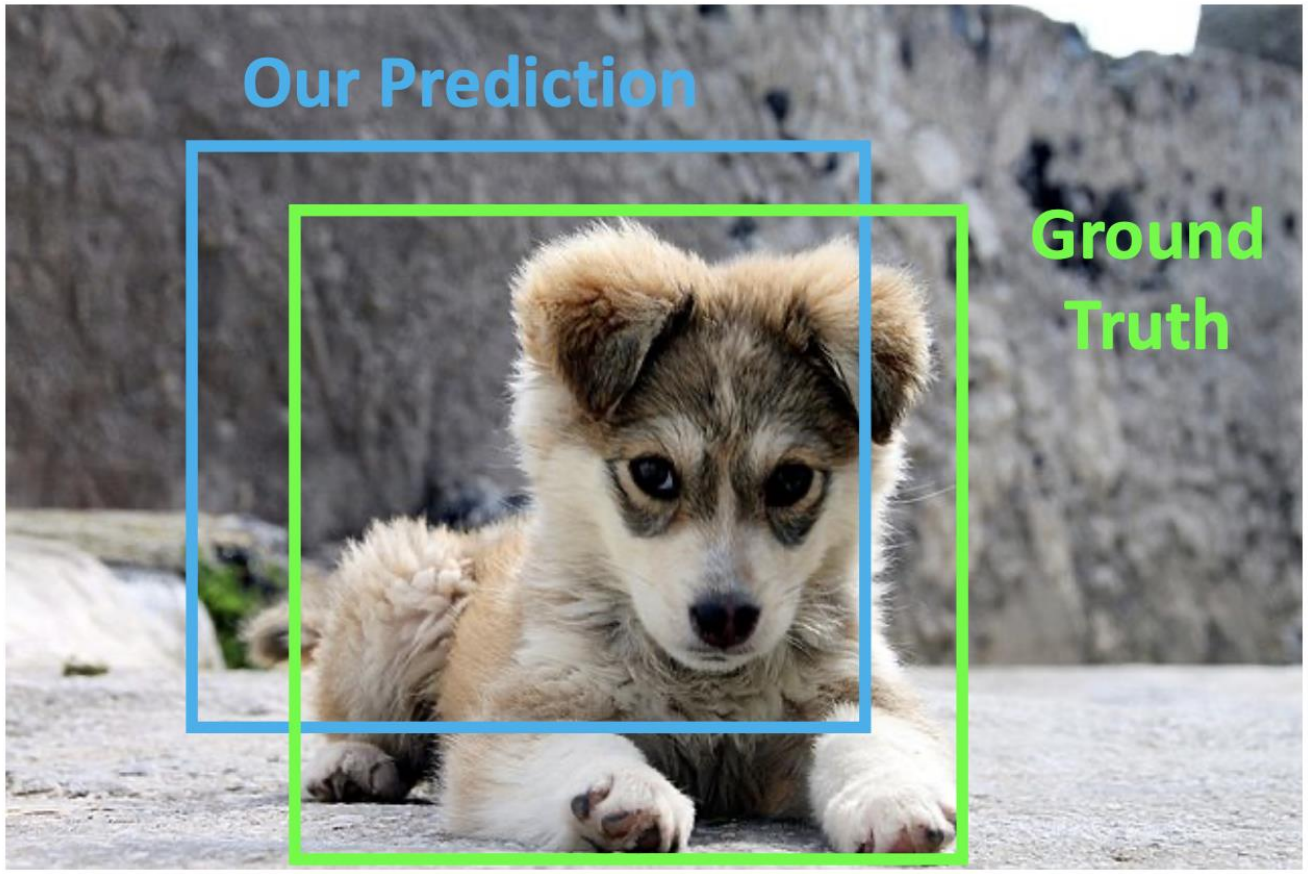


Ground-Truth (y)



Intersection over Union (IoU)

How can we compare our prediction to the ground-truth bounding box or mask?

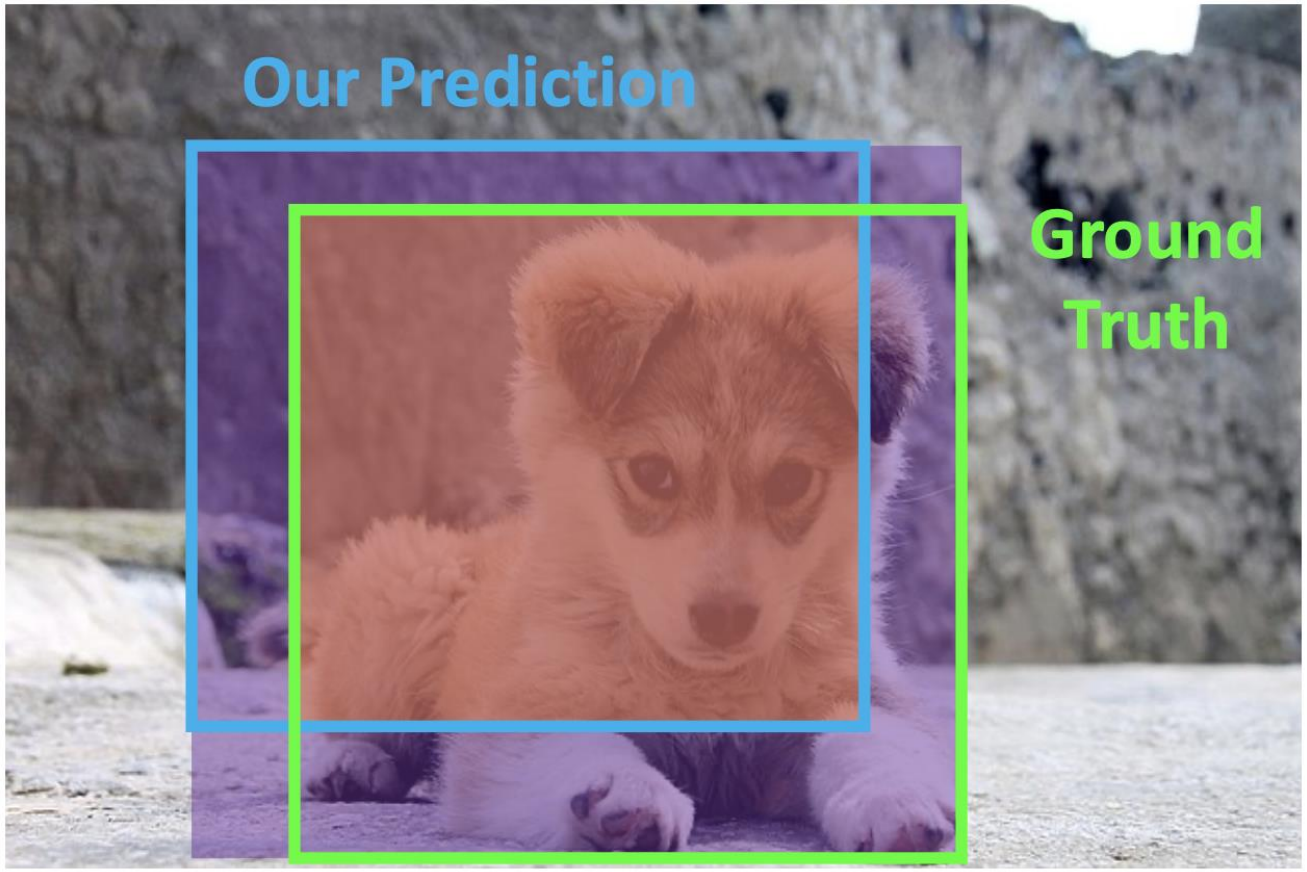


Intersection over Union (IoU)

How can we compare our prediction to the ground-truth bounding box or mask?

Intersection over Union (IoU) (Also called “Jaccard similarity” or “Jaccard index”):

$$\frac{Area\ of\ Intersection}{Area\ of\ Union}$$



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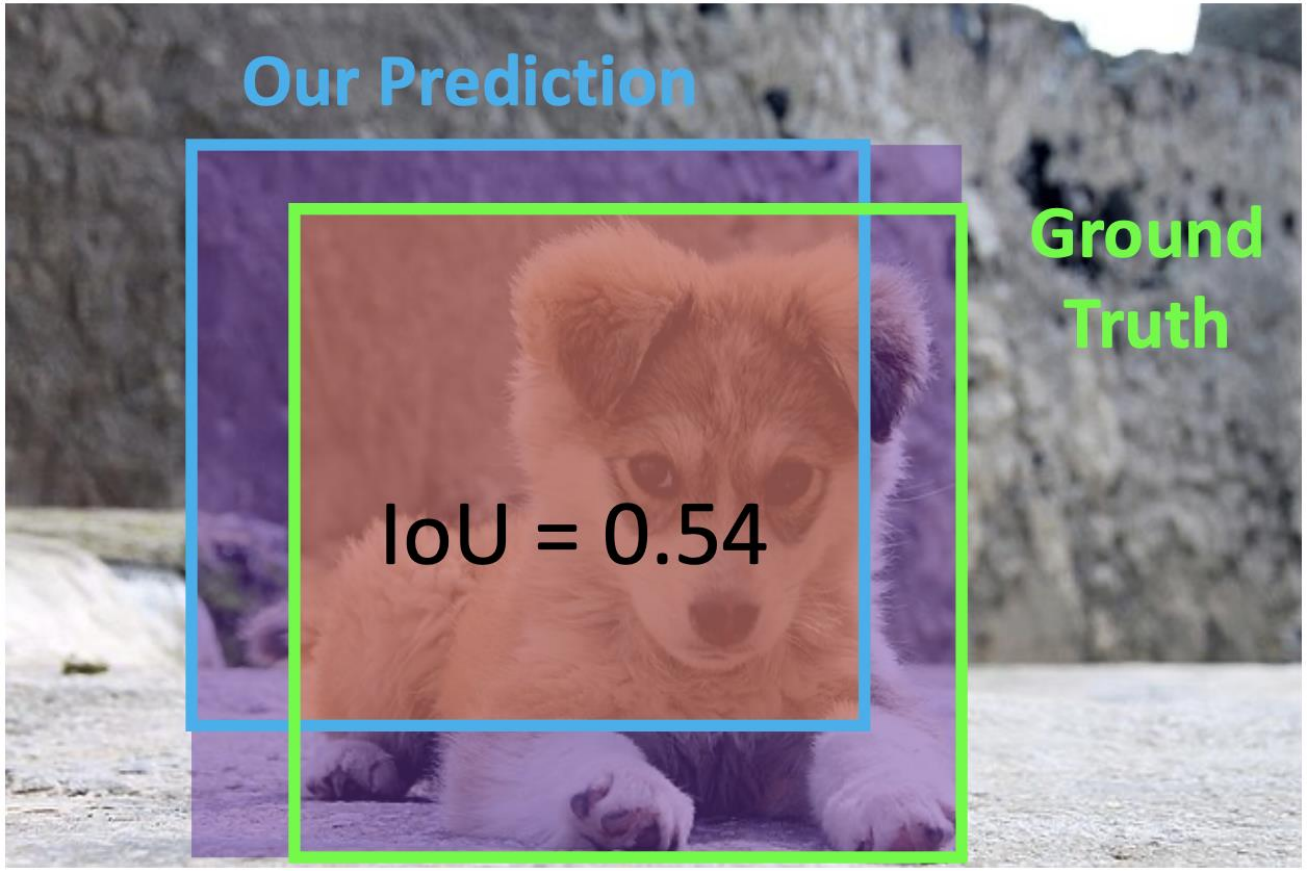
$$\frac{\textit{Area of Intersection}}{\textit{Area of Union}}$$

IoU > 0.5 is “decent”



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How can we compare our prediction to the ground-truth bounding box or mask?

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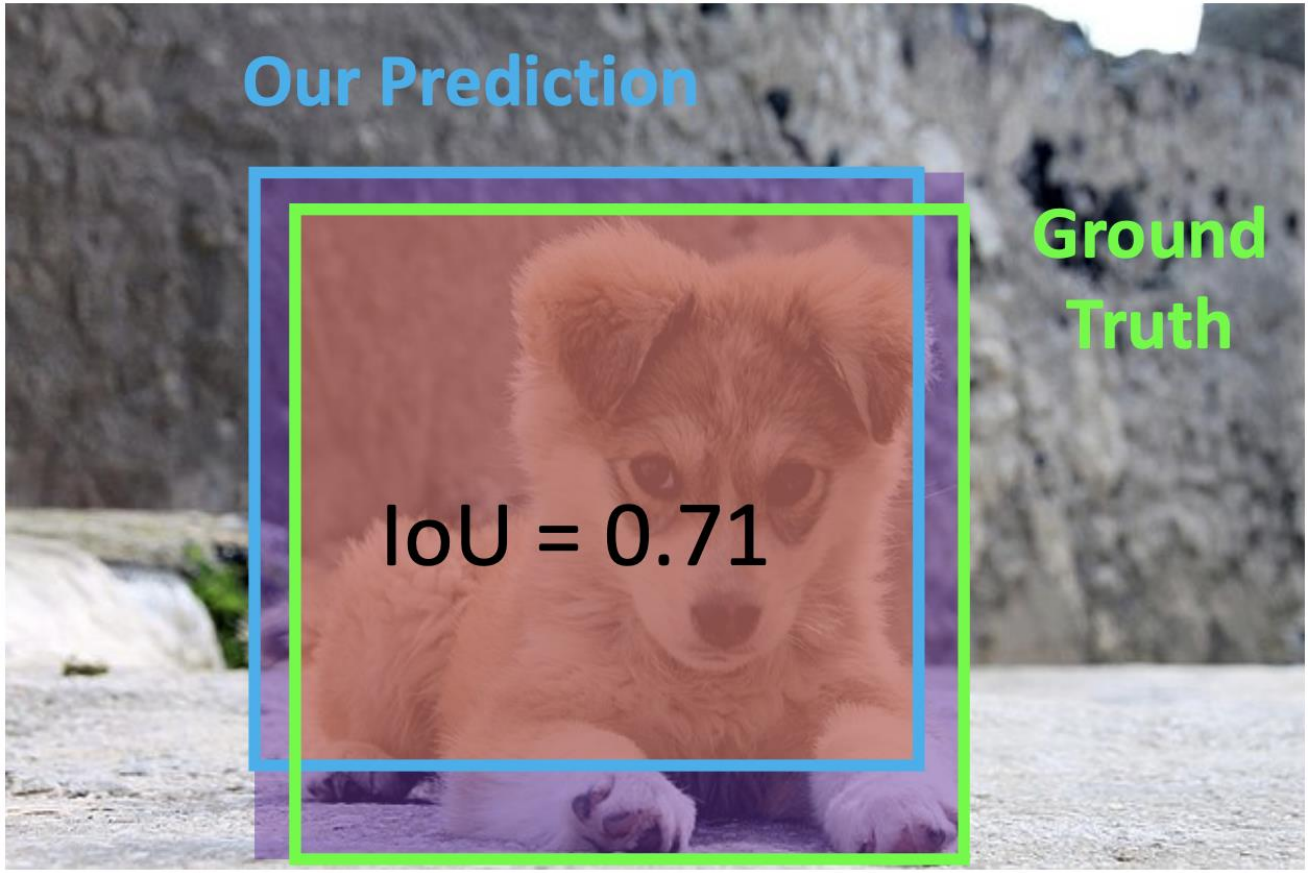
IoU > 0.5 is “decent”,
IoU > 0.7 is “pretty good”



Intersection over Union (IoU)

How can we compare our prediction to the ground-truth bounding box or mask?

Intersection over Union (IoU) (Also called “Jaccard similarity” or “Jaccard index”):

$$\frac{\textit{Area of Intersection}}{\textit{Area of Union}}$$

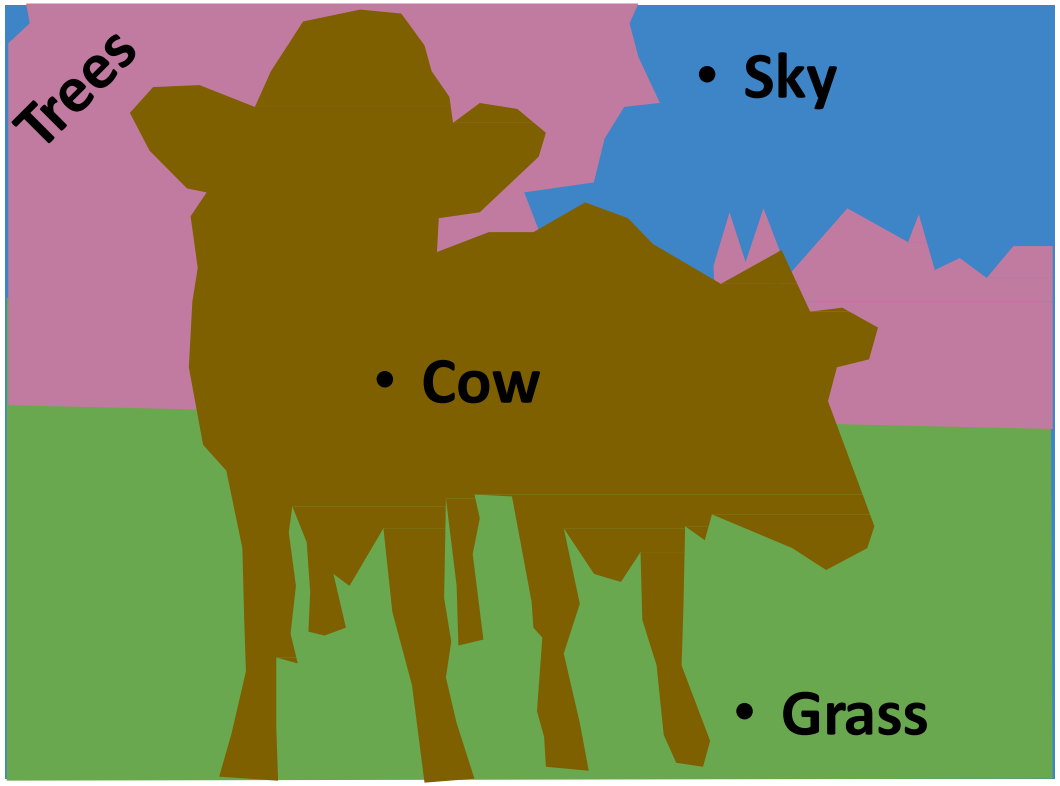
IoU > 0.5 is “decent”,
IoU > 0.7 is “pretty good”,
IoU > 0.9 is “almost perfect”

Part II:
Instance Segmentation

Structured Prediction Tasks

Object Detection: Detects individual object instances, but only gives box

Semantic Segmentation: Gives per-pixel labels, but merges instances

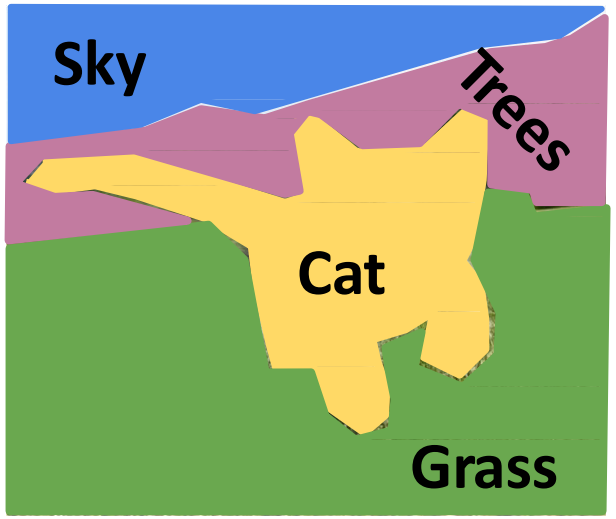


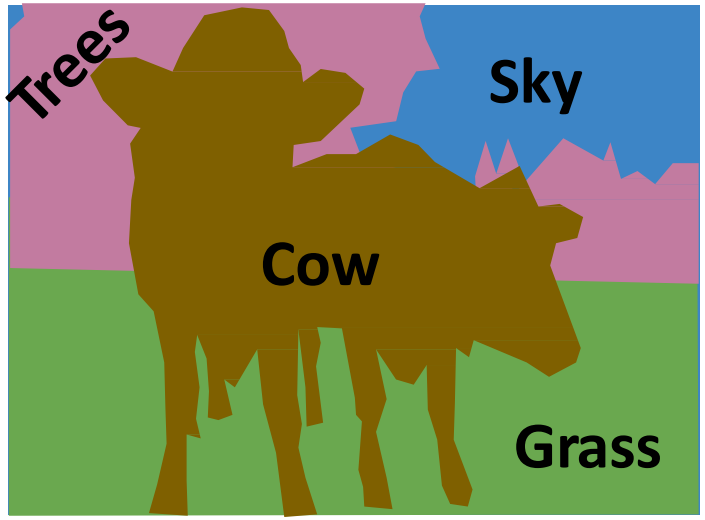
Things and Stuff

Things and Stuff

Things: Object categories that can be separated into object instances
(e.g. cats, cars, person)

Stuff: Object categories that cannot be separated into instances
(e.g. sky, grass, water, trees)





Things and Stuff

- Things



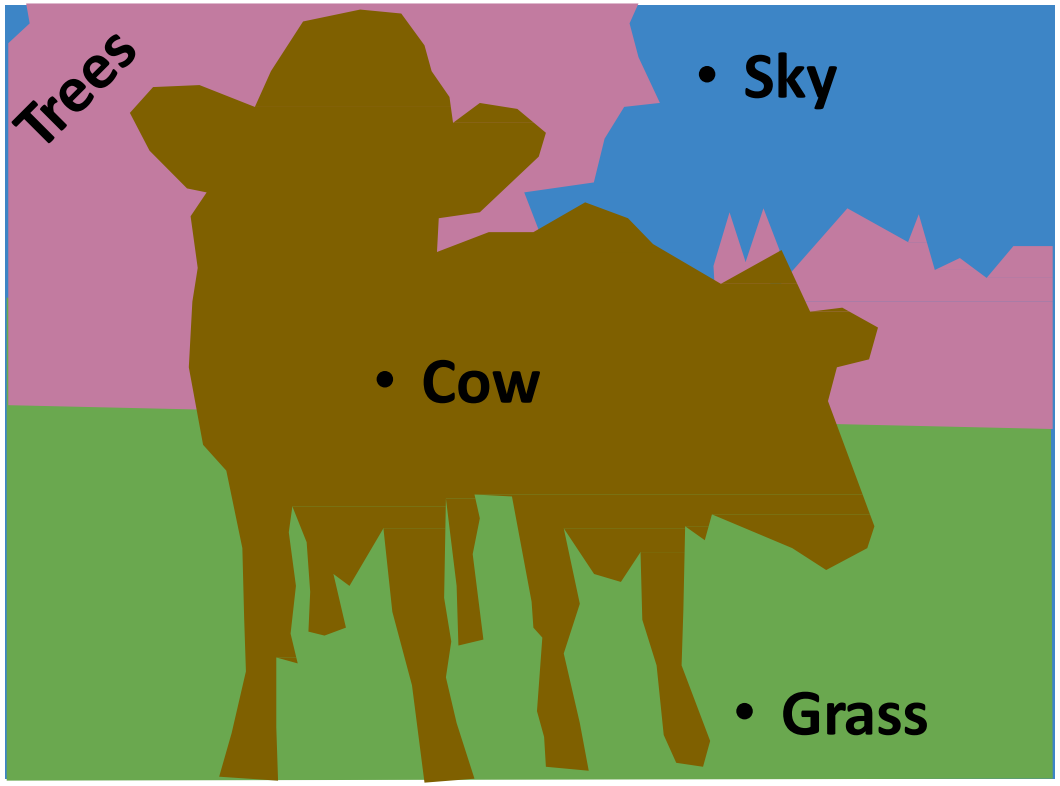

- Stuff




Structured Prediction Tasks

Object Detection: Detects individual object instances, but only gives box (Only things!)

Semantic Segmentation: Gives per-pixel labels, but merges instances (Both things and stuff)



Structured Prediction Tasks: Instance Segmentation

Classification

CAT

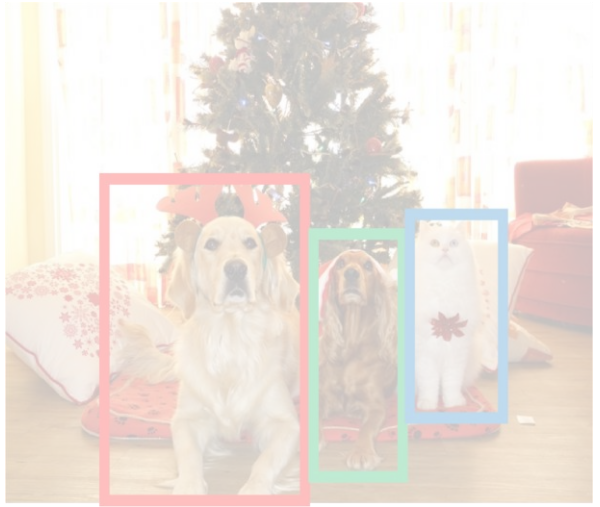
No spatial extent

Semantic Segmentation

GRASS, CAT, TREE, SKY

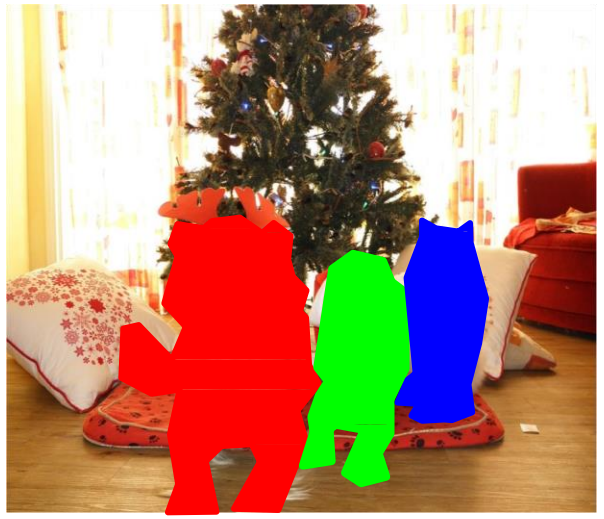
No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

Multiple Objects

Semantic or Instance Segmentation?

Liu et al, “Multi-Objective Convolutional Learning for Face Labeling”, CVPR 2015

Face Parsing

Eyeglasses

Earring

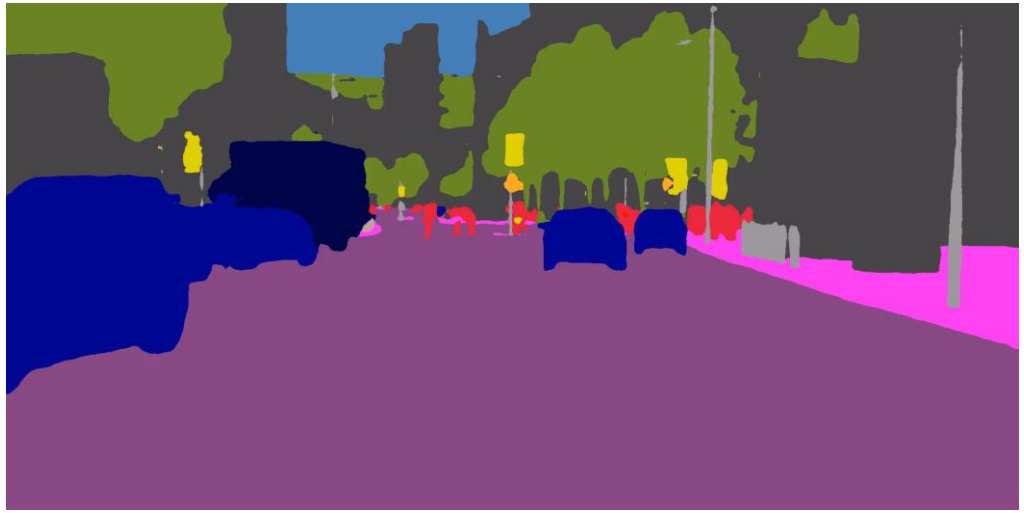
Cloth

Hat

Necklace

Hair

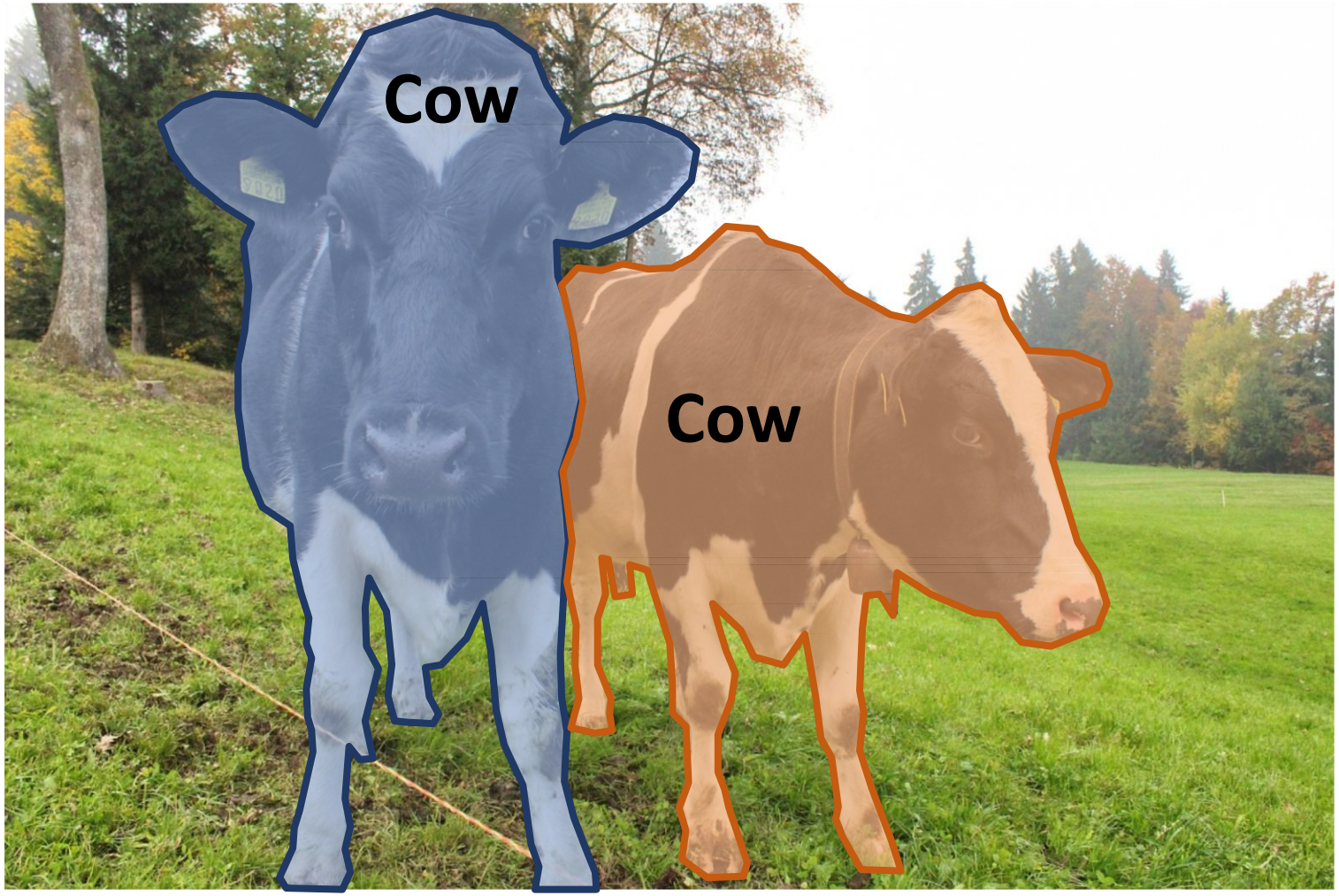
Semantic Segmentation



Instance Segmentation

Instance Segmentation

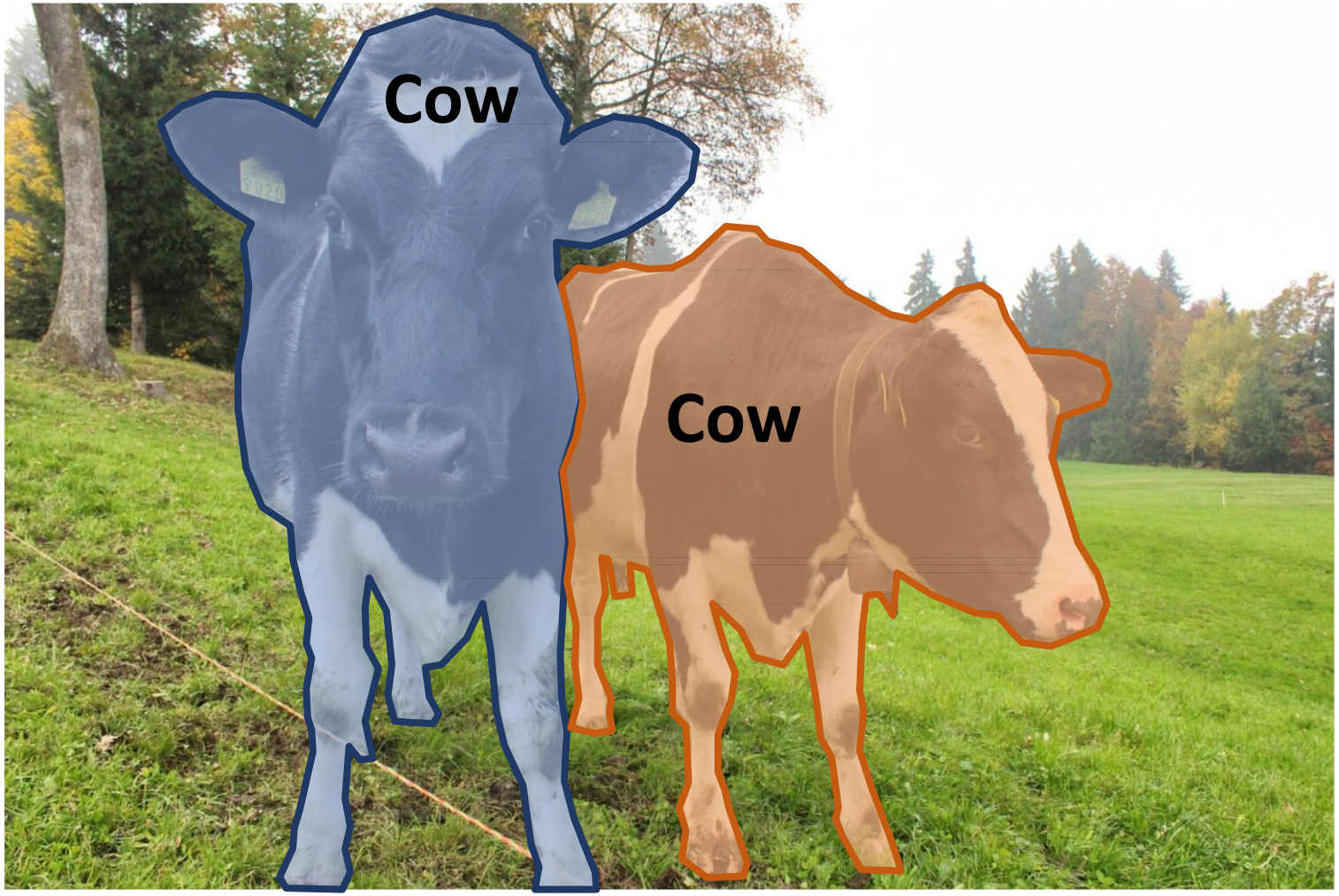
Instance Segmentation: Detect all objects in the image, and identify the pixels that belong to each object (Only things!)



Instance Segmentation

Instance Segmentation: Detect all objects in the image, and identify the pixels that belong to each object (Only things!)

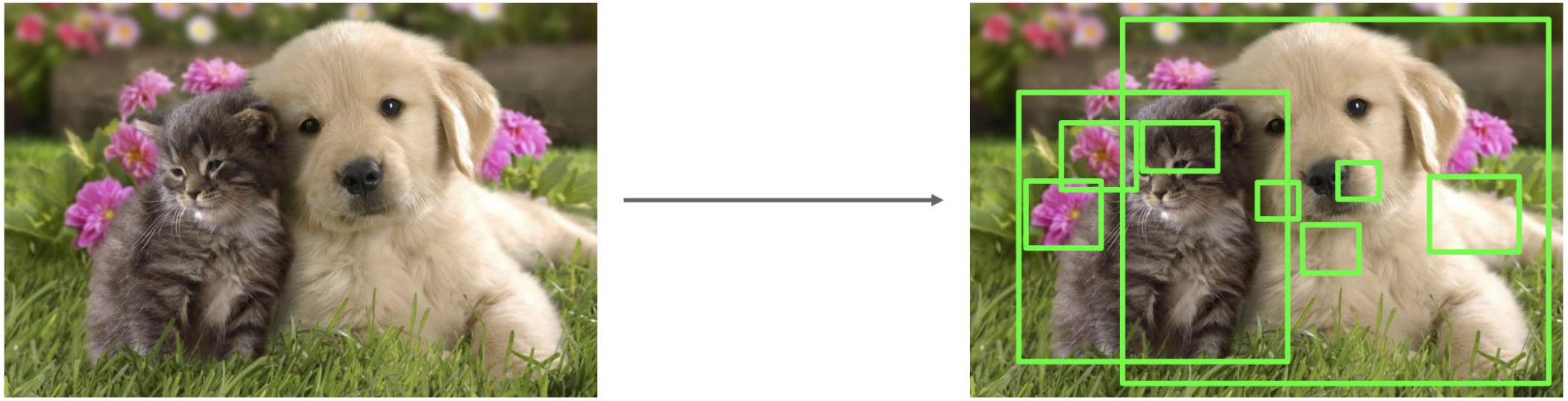
Approach: Perform object detection, then predict a segmentation mask for each object!



Object Detection

Region proposals

- Find a small set of boxes that are likely to cover all objects
- Often based on heuristics: e.g. look for “blob-like” image regions
- Relatively fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU



Alexe et al, “Measuring the objectness of image windows”, TPAMI 2012
Uijlings et al, “Selective Search for Object Recognition”, IJCV 2013
Cheng et al, “BING: Binarized normed gradients for objectness estimation at 300fps”, CVPR 2014 Zitnick and Dollar, “Edge boxes: Locating object proposals from edges”, ECCV 2014

Region-based CNN

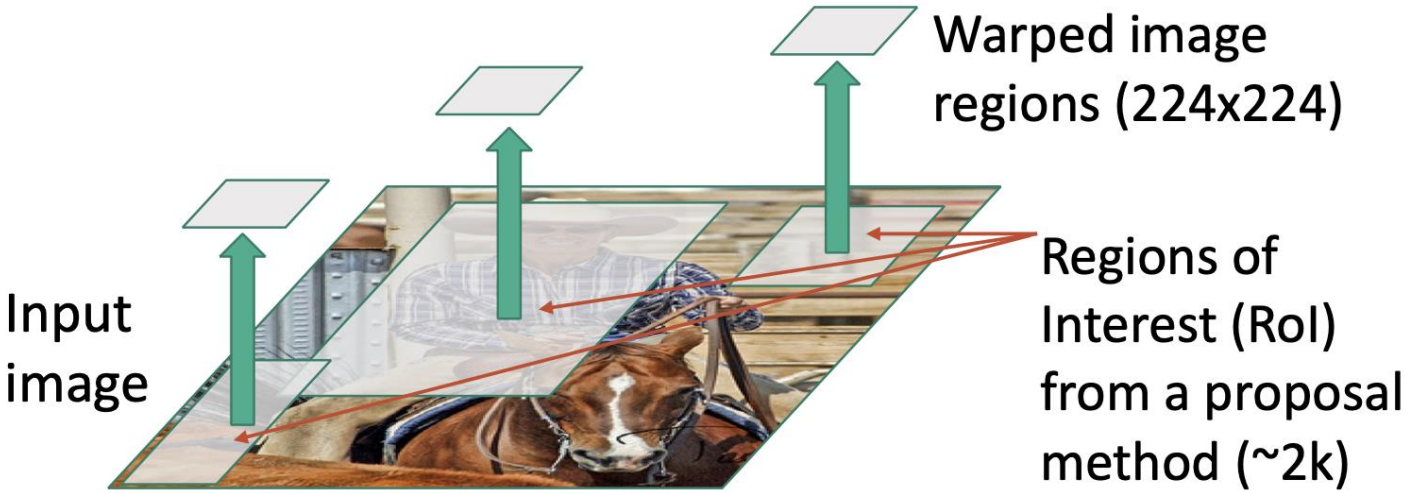
Input image

Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. Figure copyright Ross Girshick, 2015

Region-based CNN

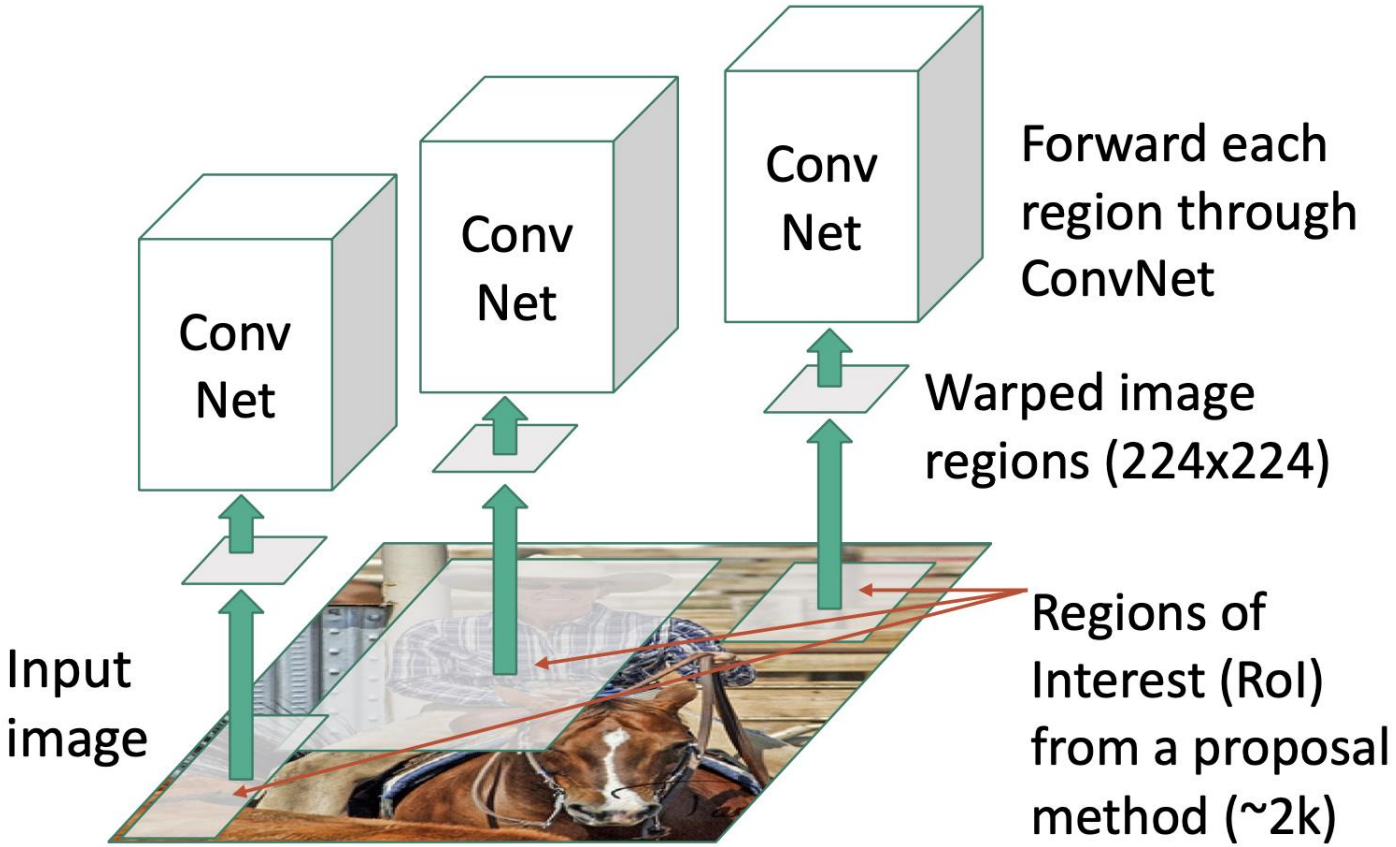
Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. Figure copyright Ross Girshick, 2015

Region-based CNN



Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. Figure copyright Ross Girshick, 2015

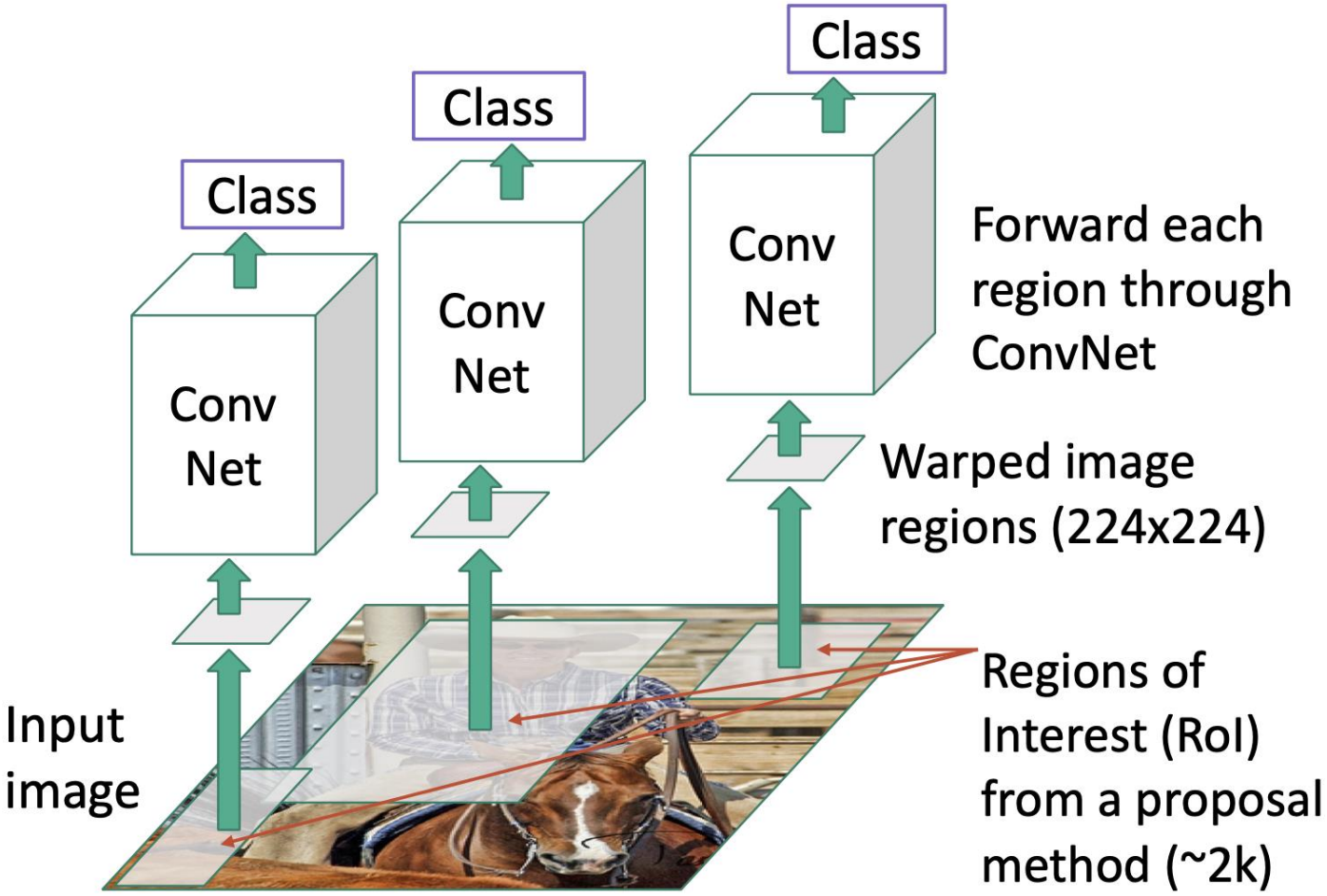
Region-based CNN



Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. Figure copyright Ross Girshick, 2015

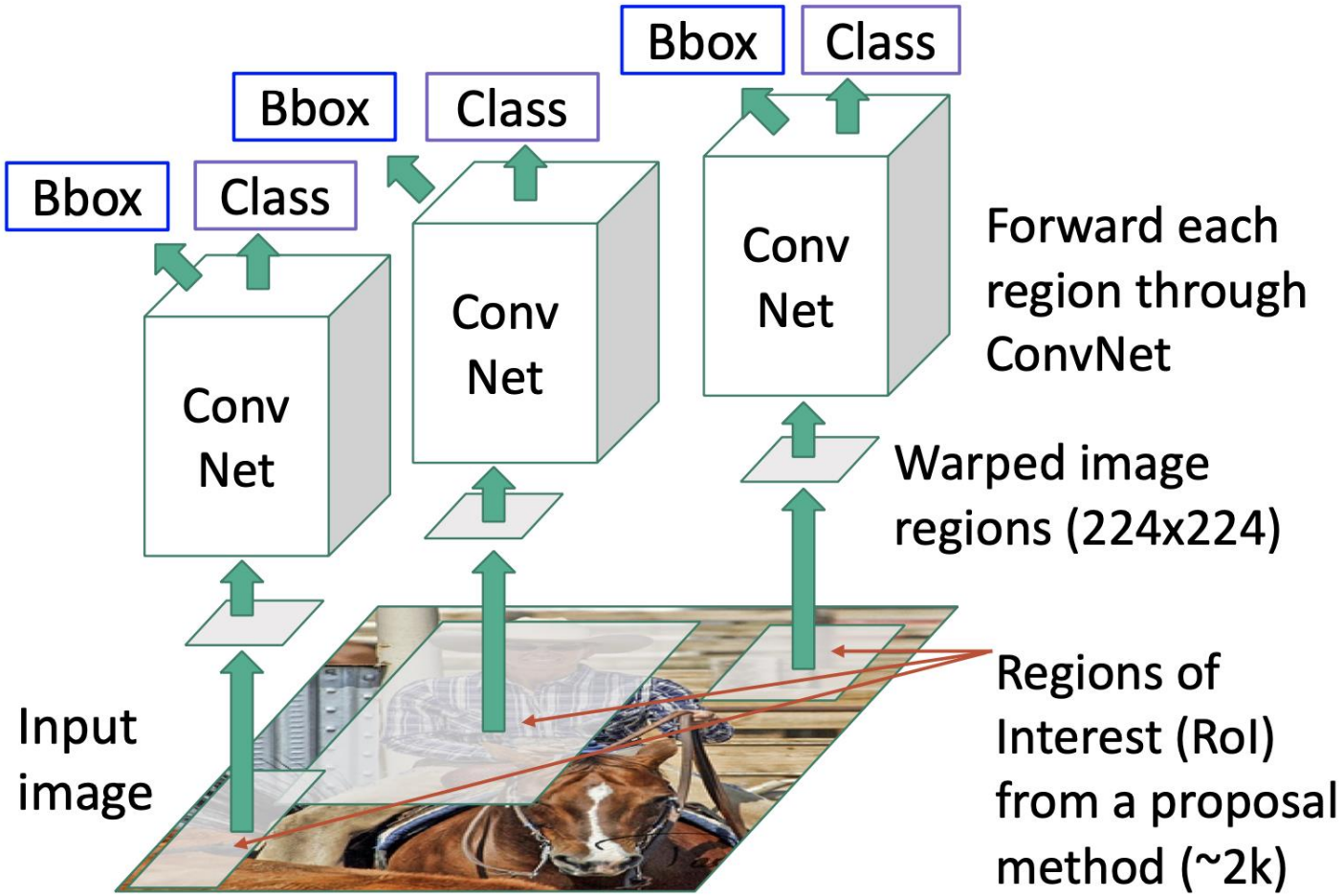
Region-based CNN

Classify each region



Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. Figure copyright Ross Girshick, 2015

Region-based CNN

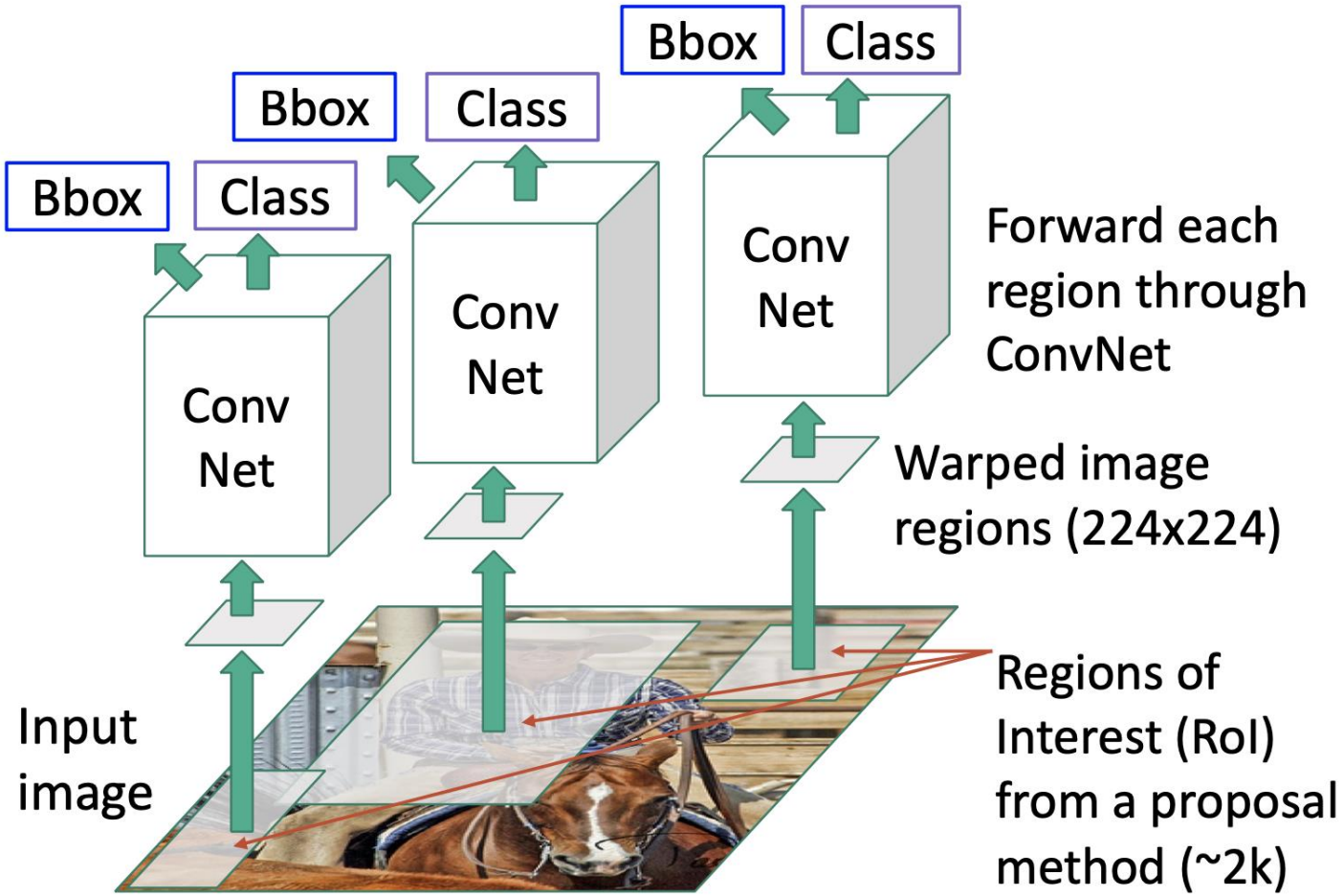


Classify each region

Bounding box regression:
Predict "transform" to correct the RoI: 4 numbers (t_x, t_y, t_h, t_w)

Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014. Figure copyright Ross Girshick, 2015

Region-based CNN (R-CNN)



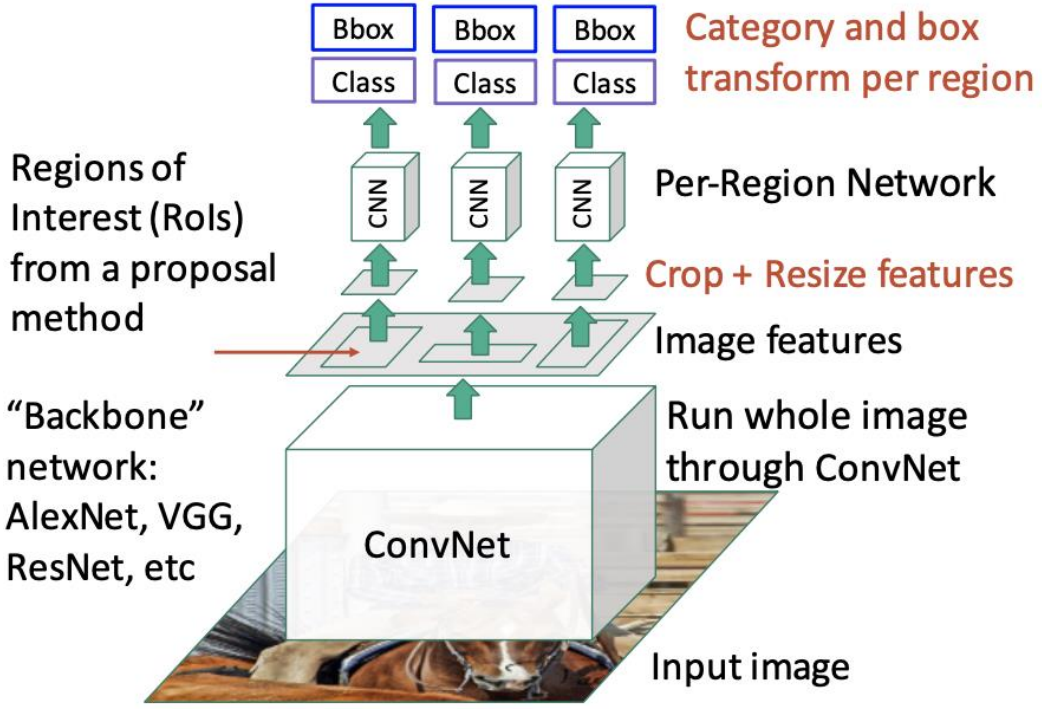
Problem: Very slow! Need to do ~2k forward passes for each image!

Solution: Run CNN *before* warping!

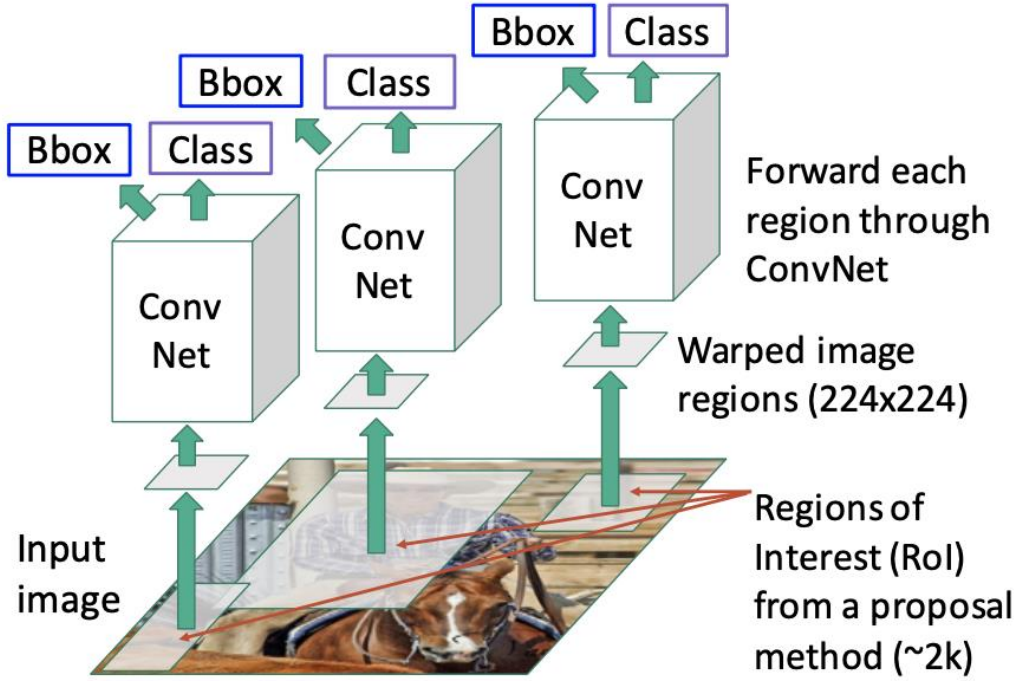
Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014. Figure copyright Ross Girshick, 2015

Fast R-CNN vs “Slow” R-CNN

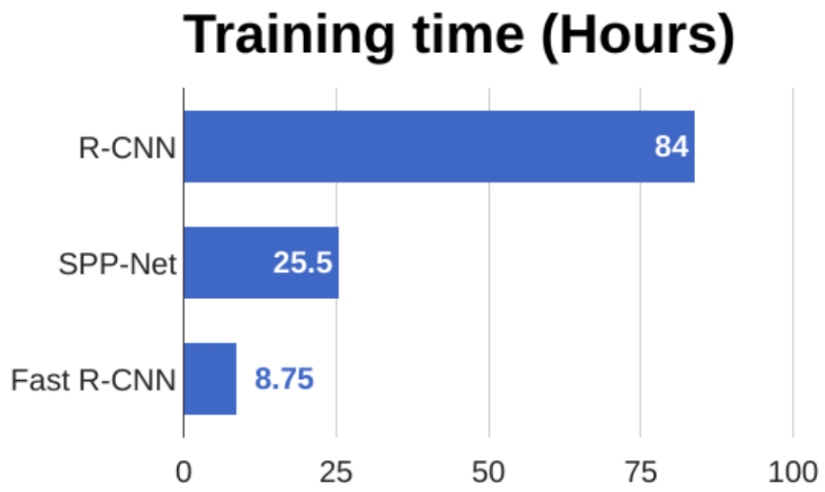
Fast R-CNN: Apply differentiable cropping to shared image features

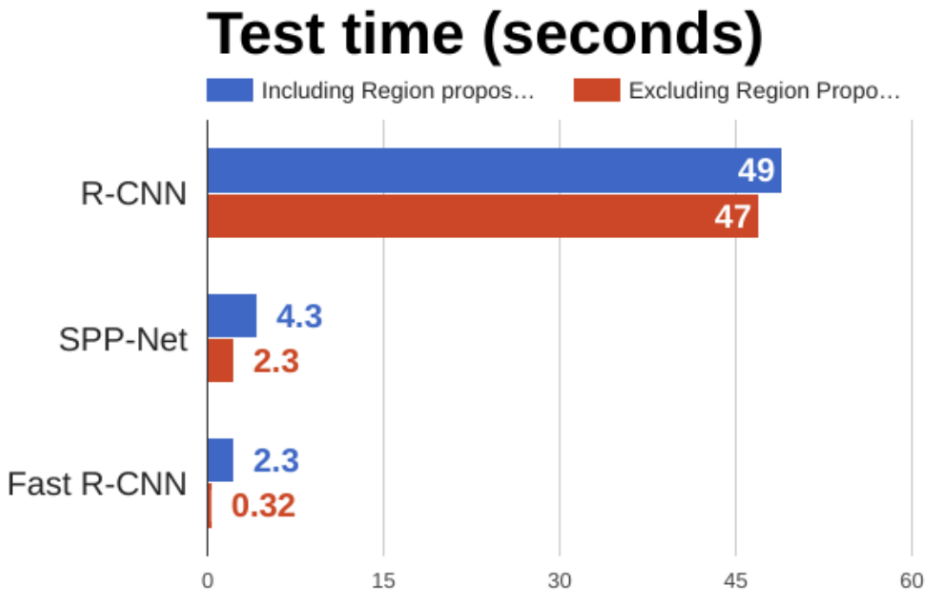


“Slow” R-CNN: Run CNN independently for each region



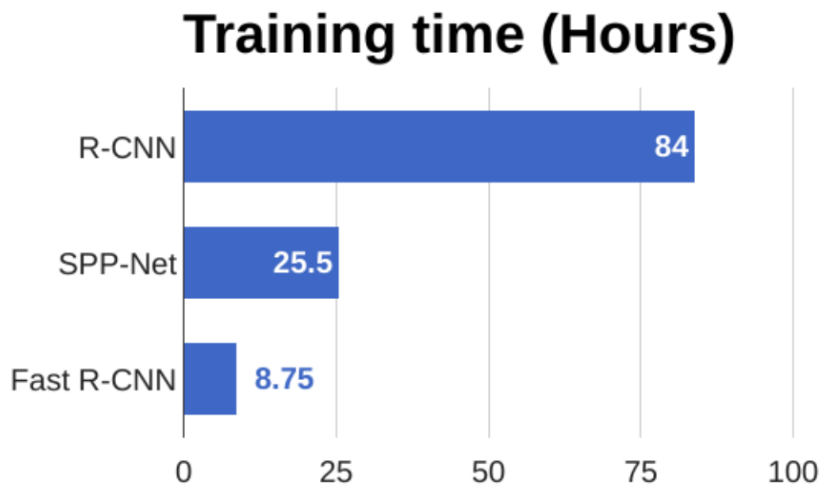
Fast R-CNN vs “Slow” R-CNN

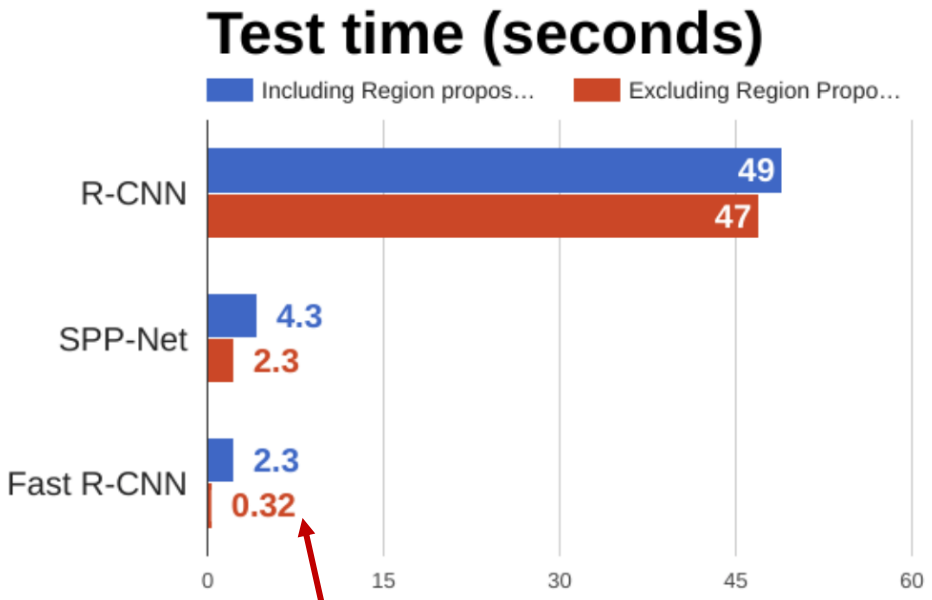




Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. He et al, “Spatial pyramid pooling in deep convolutional networks for visual recognition”, ECCV 2014
Girshick, “Fast R-CNN”, ICCV 2015

Fast R-CNN vs “Slow” R-CNN

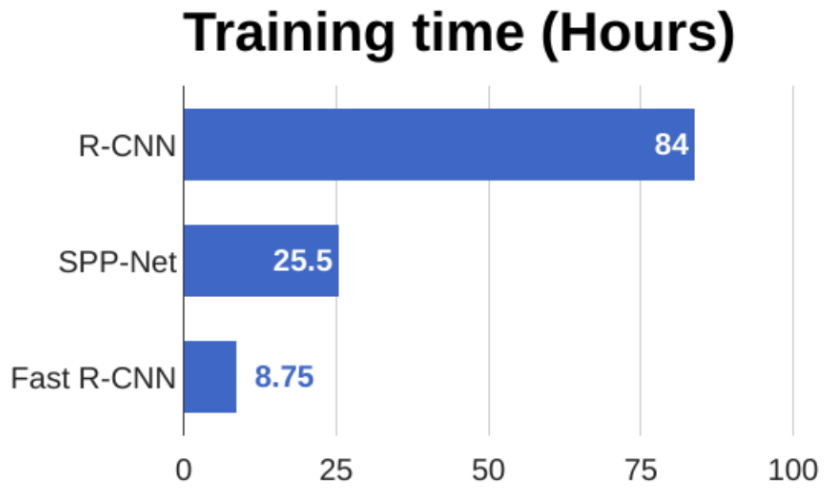


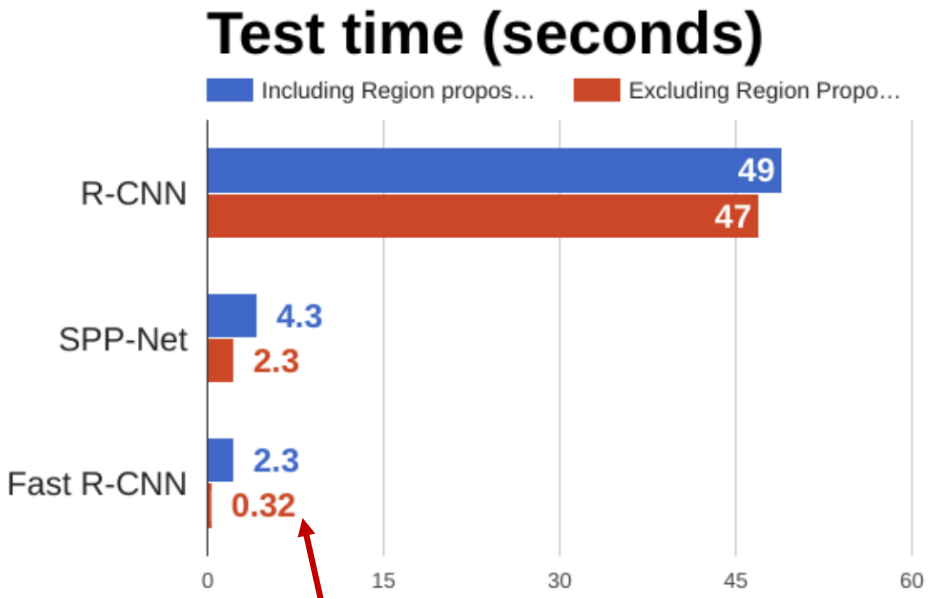


Problem: Runtime dominated by region proposals!

Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. He et al, “Spatial pyramid pooling in deep convolutional networks for visual recognition”, ECCV 2014
Girshick, “Fast R-CNN”, ICCV 2015

Fast R-CNN vs “Slow” R-CNN





Recall: Region proposals computed by heuristic ”Selective Search” algorithm on CPU -- let’s learn them with a CNN instead!

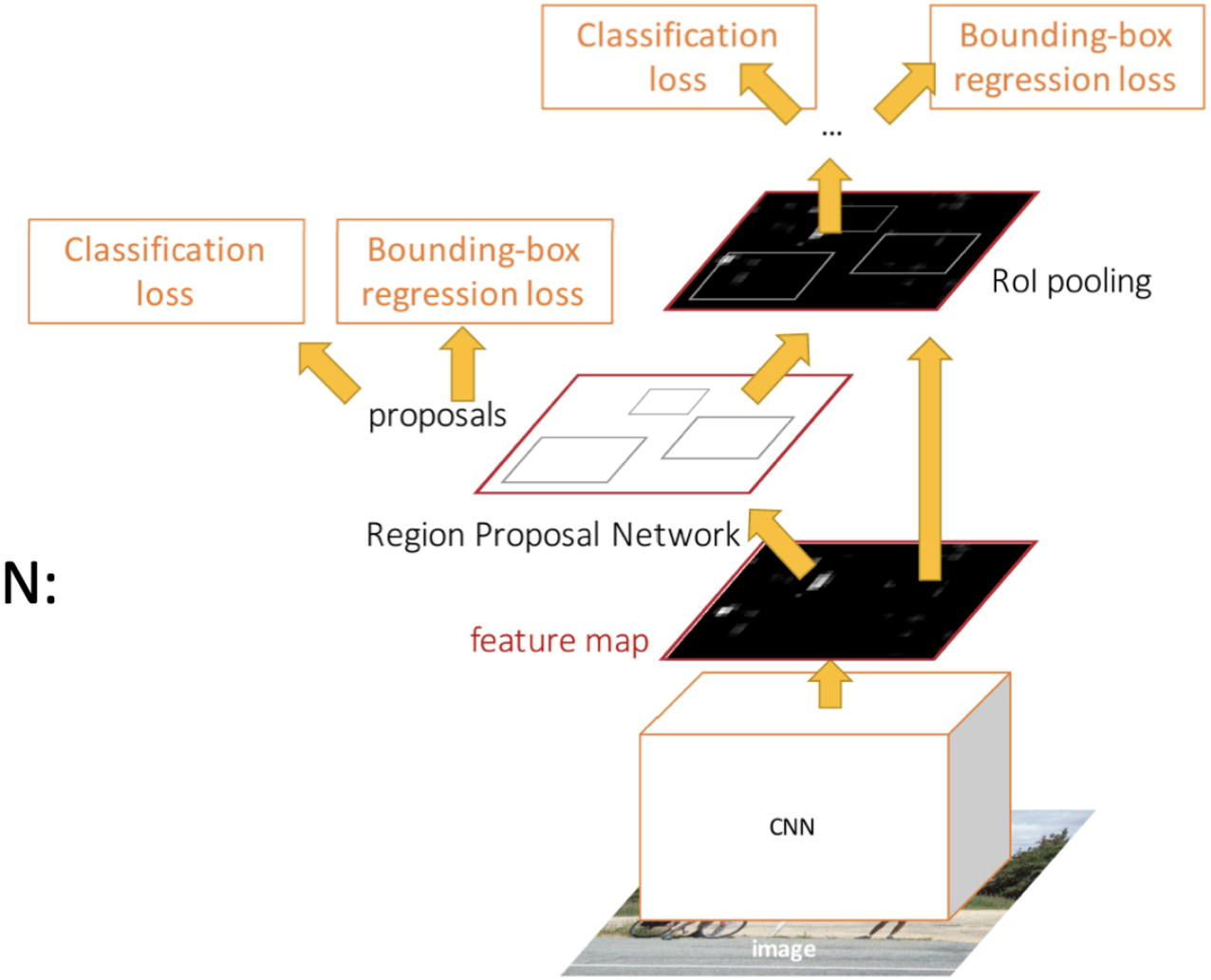
Problem: Runtime dominated by region proposals!

Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014. He et al, “Spatial pyramid pooling in deep convolutional networks for visual recognition”, ECCV 2014
Girshick, “Fast R-CNN”, ICCV 2015

Faster R-CNN: Learnable Region Proposals

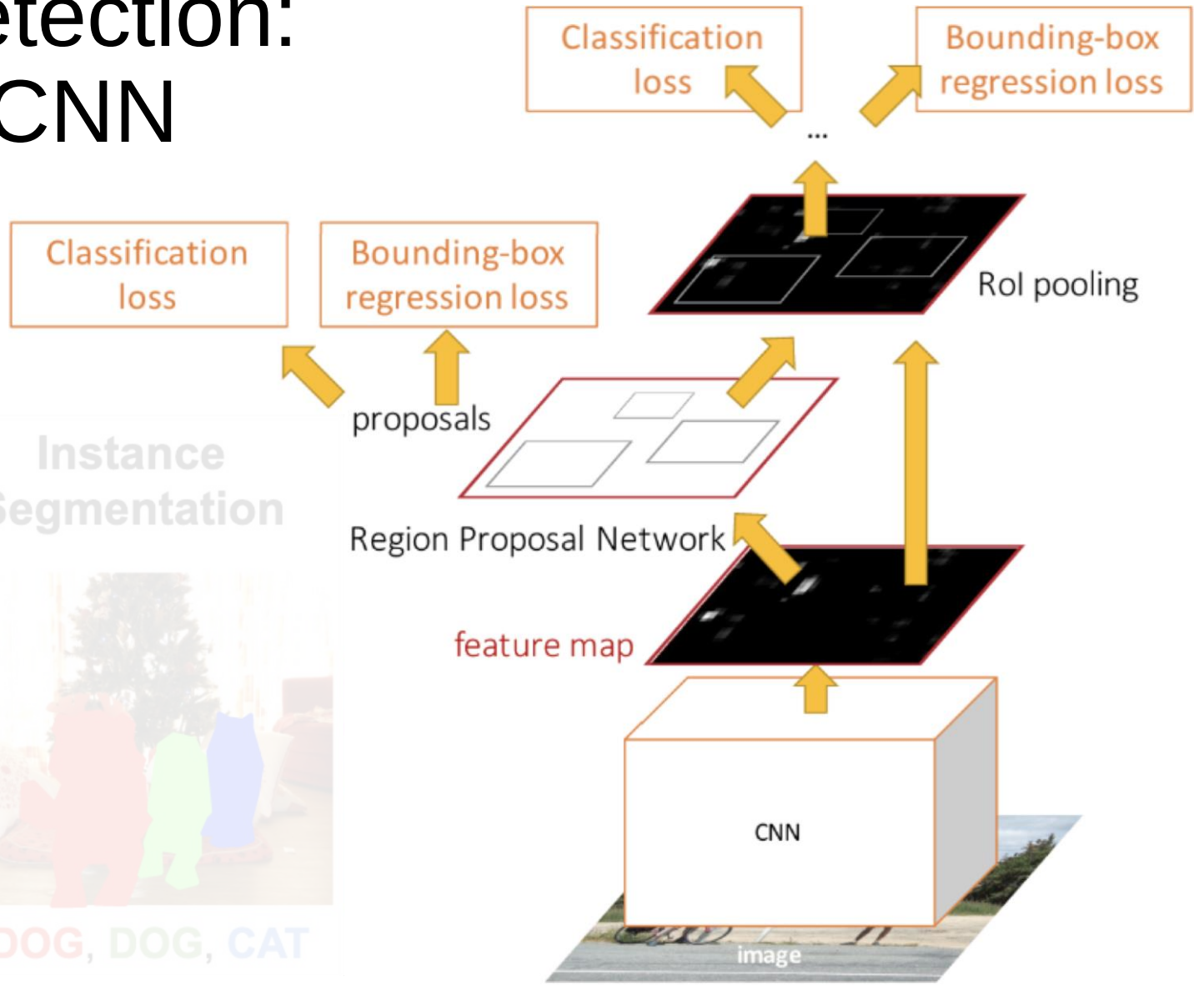
Insert **Region Proposal Network (RPN)** to predict proposals from features

Otherwise same as Fast R-CNN: Crop features for each proposal, classify each one



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015 Figure copyright 2015, Ross Girshick

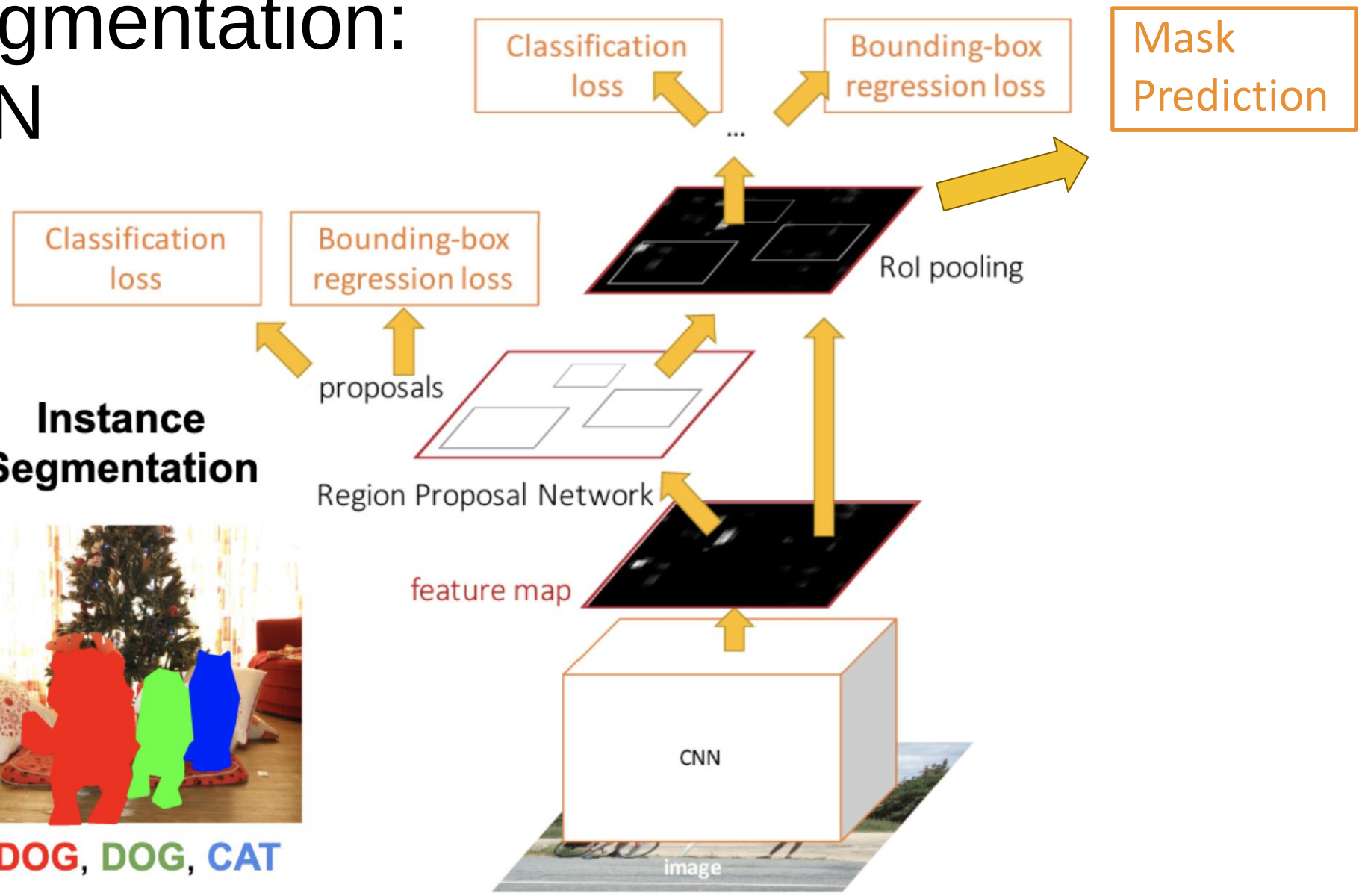
Object Detection: Faster R-CNN



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NeurIPS 2015

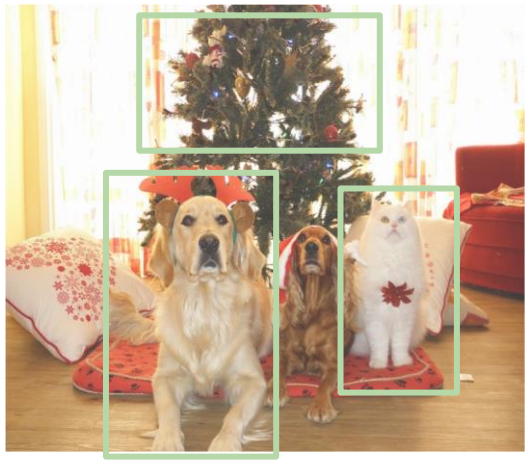
Mask R-CNN

Instance Segmentation: Mask R-CNN

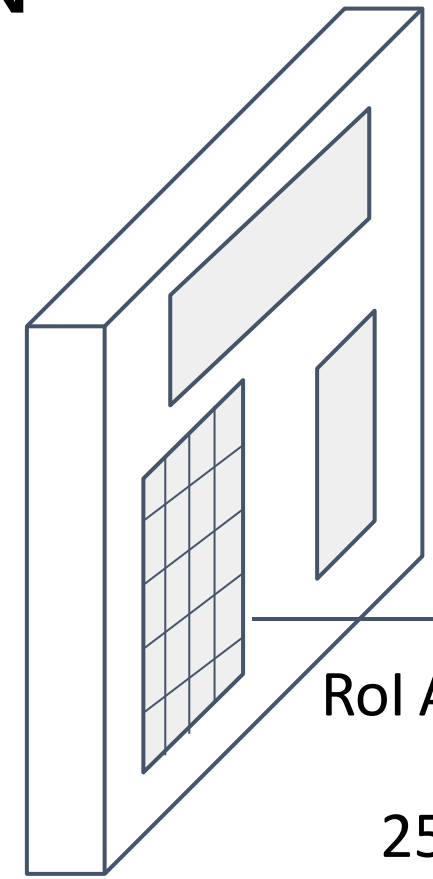


He et al, "Mask R-CNN", ICCV 2017

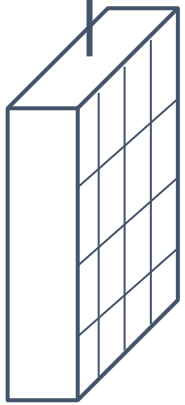
Mask R-CNN



CNN
+RPN

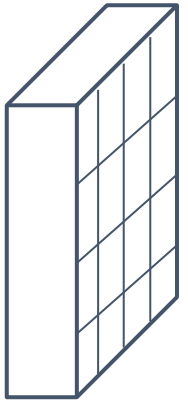


RoI Align



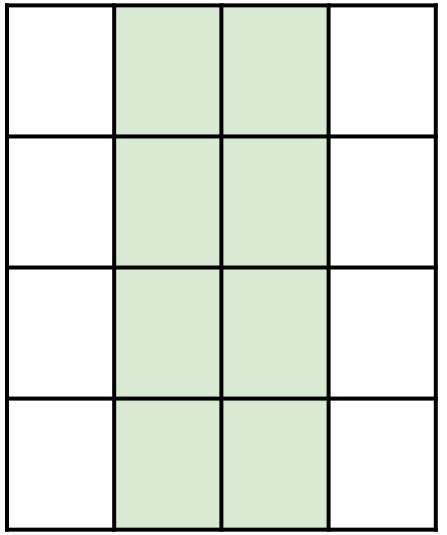
256 x 14 x 14

Conv



256 x 14 x 14

Conv

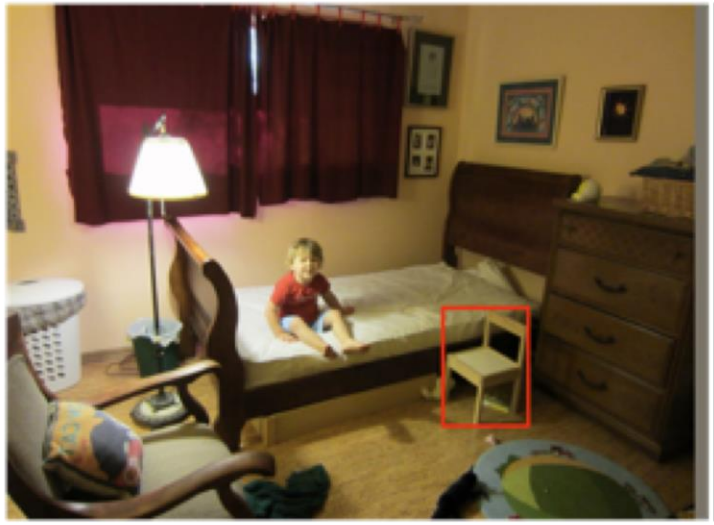


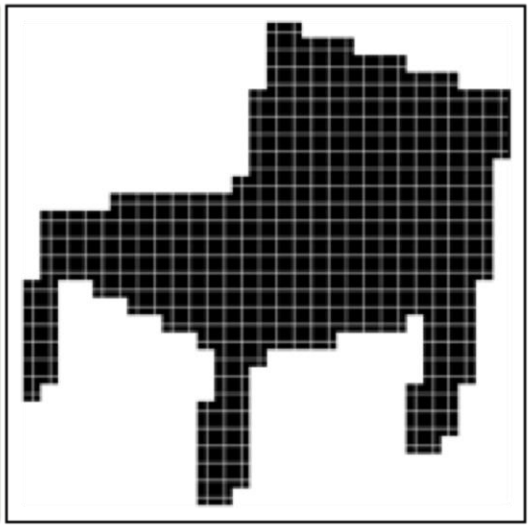
Predict a mask for each of C classes:
C x 28 x 28

Classification Scores: C
Box coordinates (per class):
4 * C

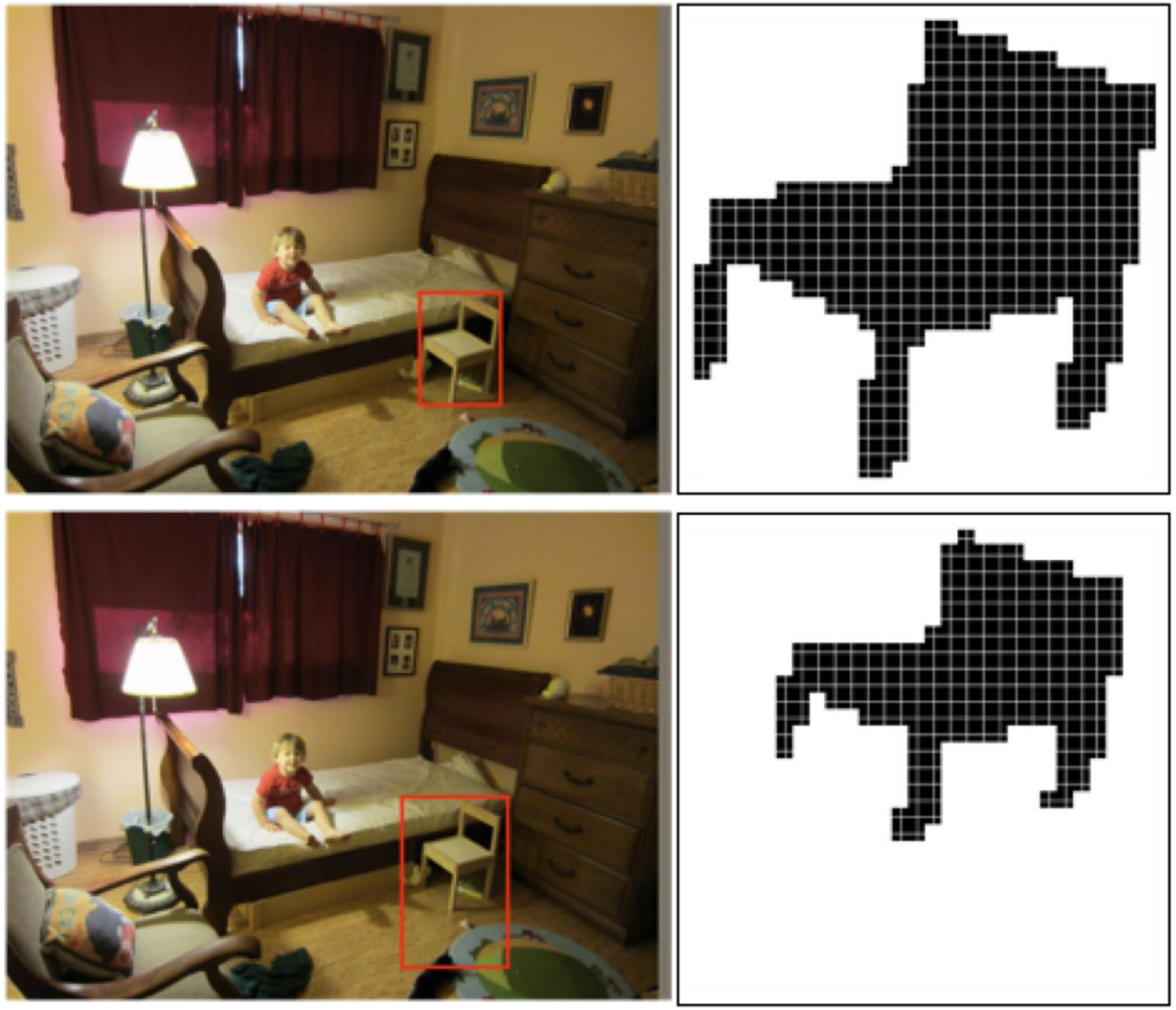
He et al, “Mask R-CNN”, ICCV 2017

Mask R-CNN: Example Training Targets

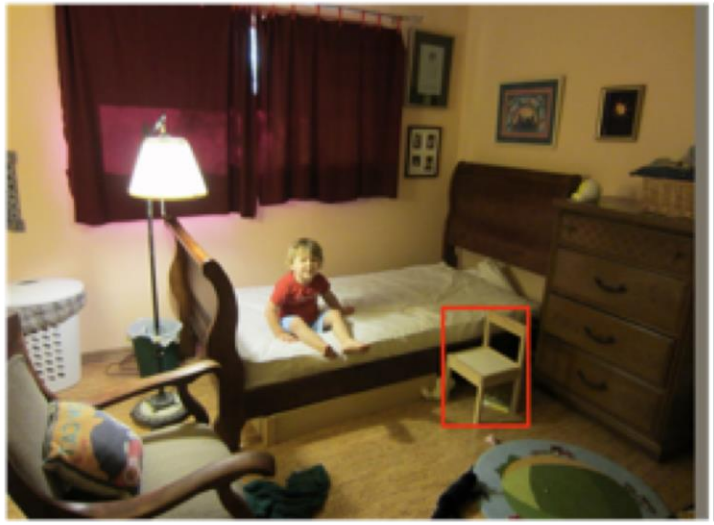
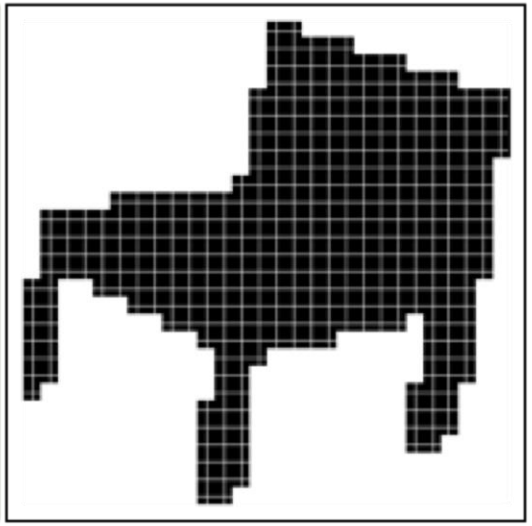
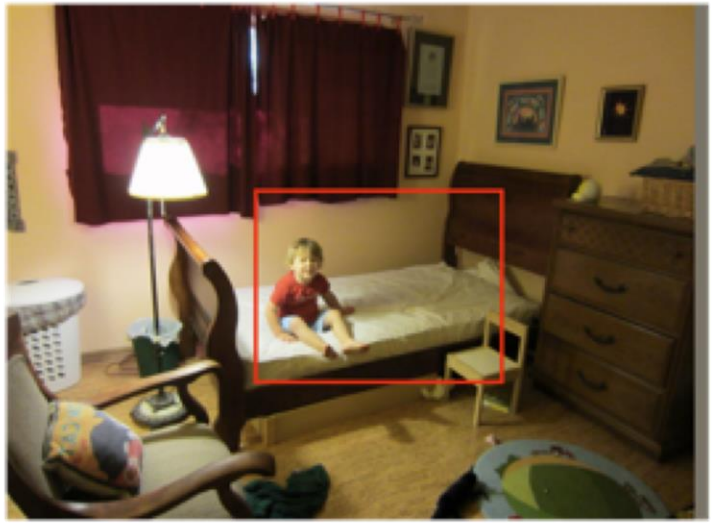
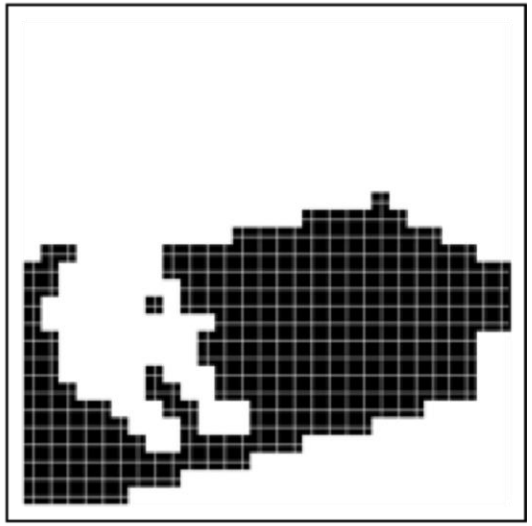
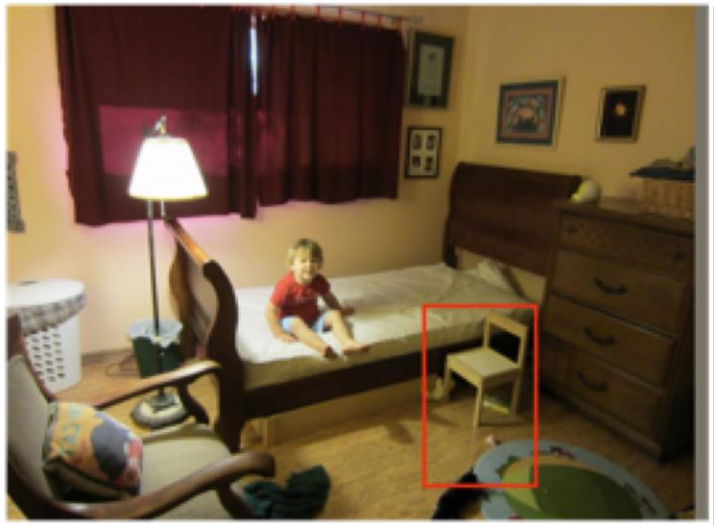
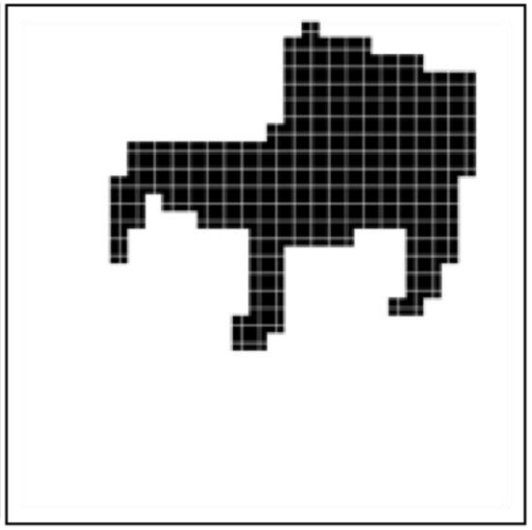




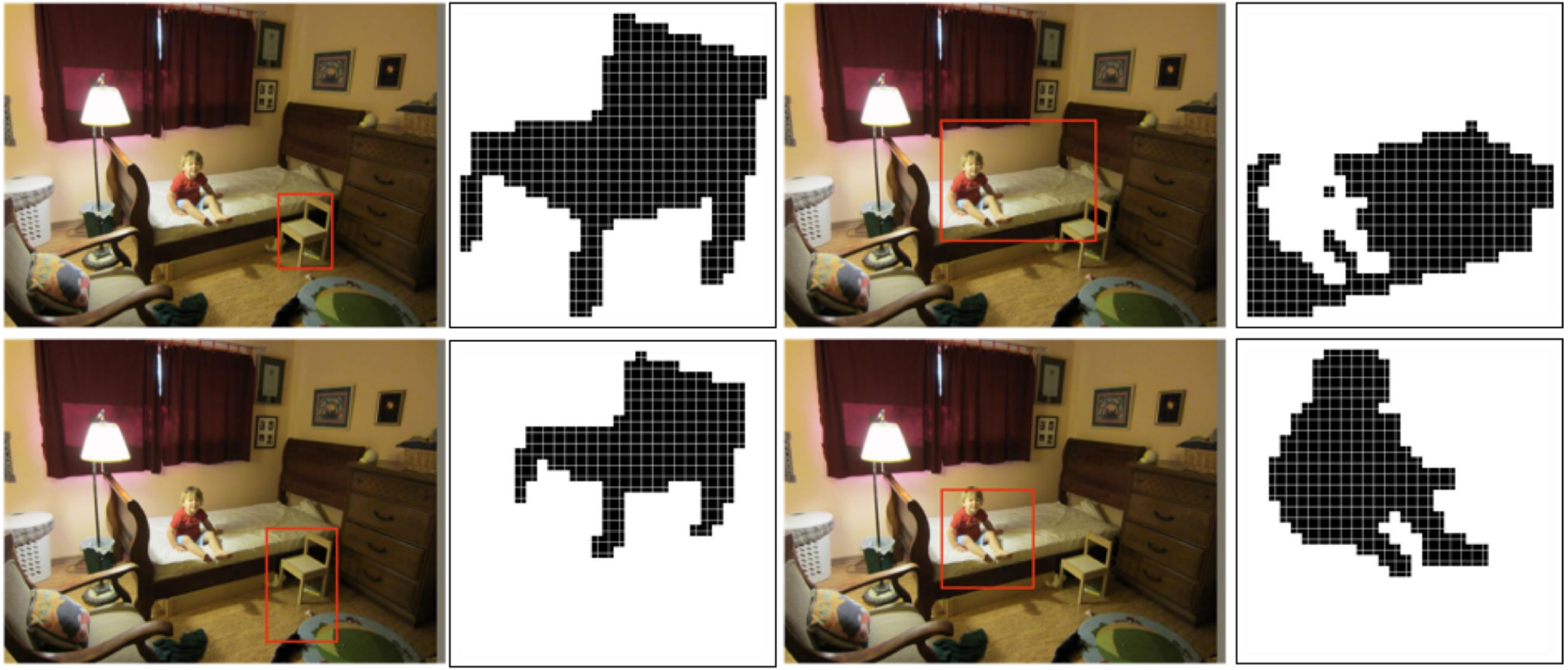
Mask R-CNN: Example Training Targets



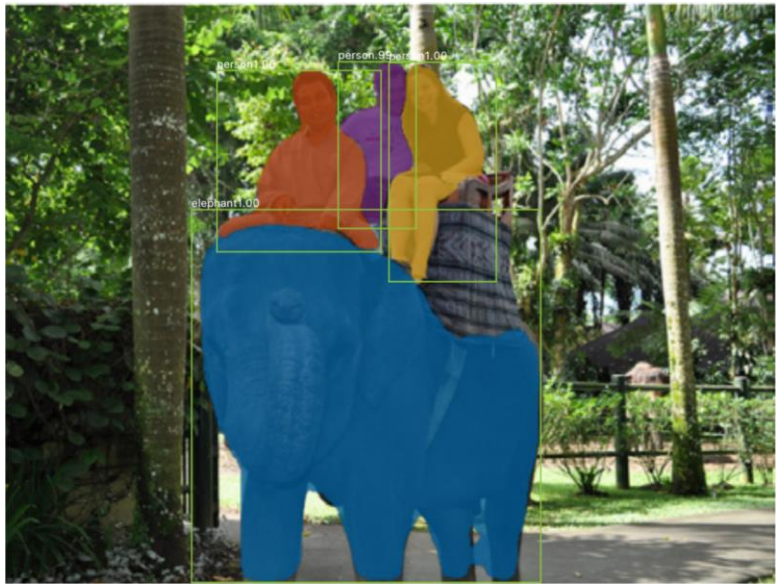
Mask R-CNN: Example Training Targets

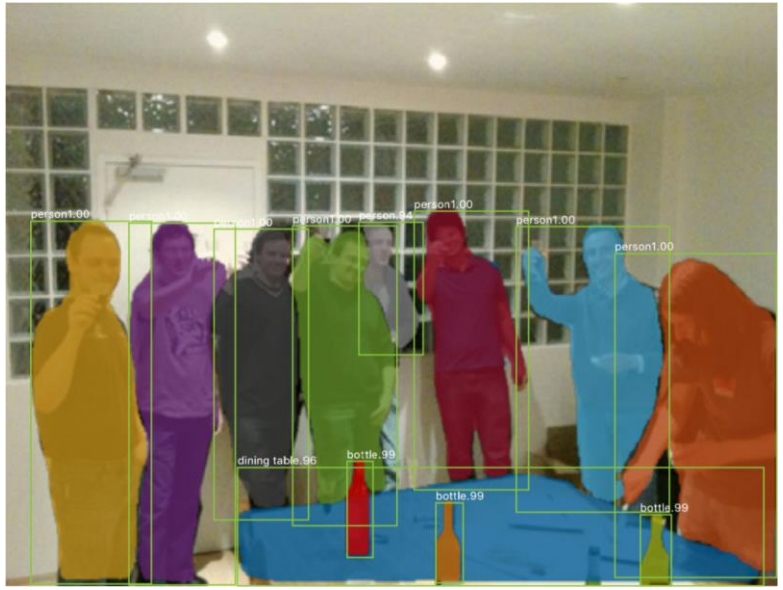







Mask R-CNN: Example Training Targets



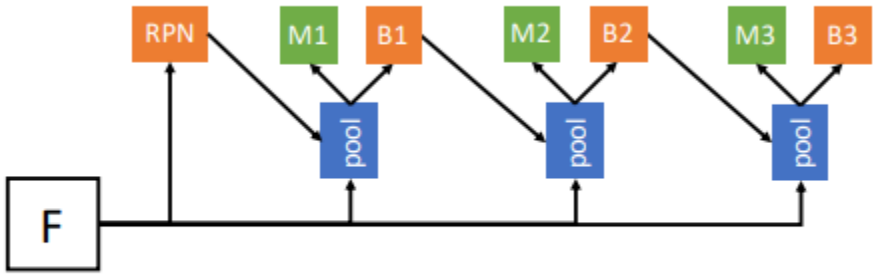
Mask R-CNN: Very Good Results!



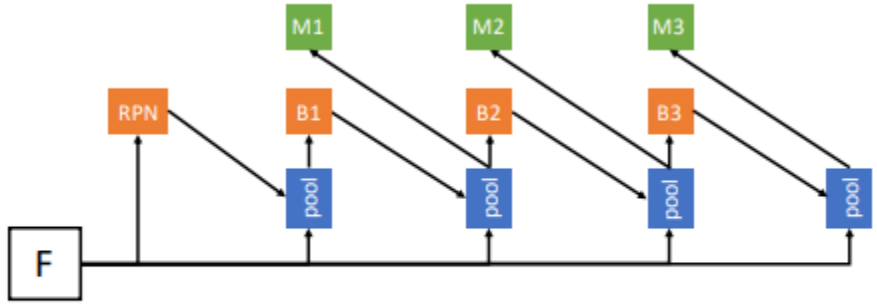


Accurate Classification and Localization

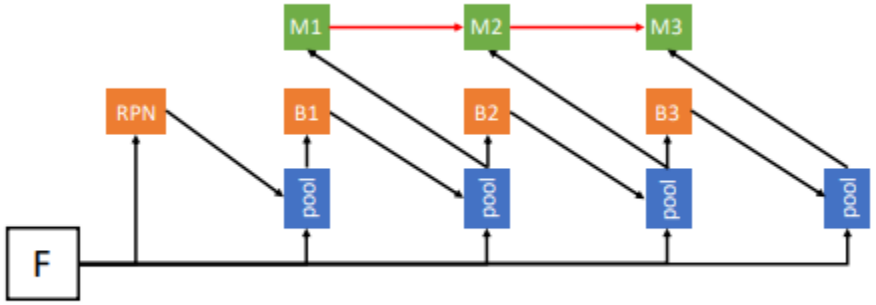
Hybrid Task Cascade



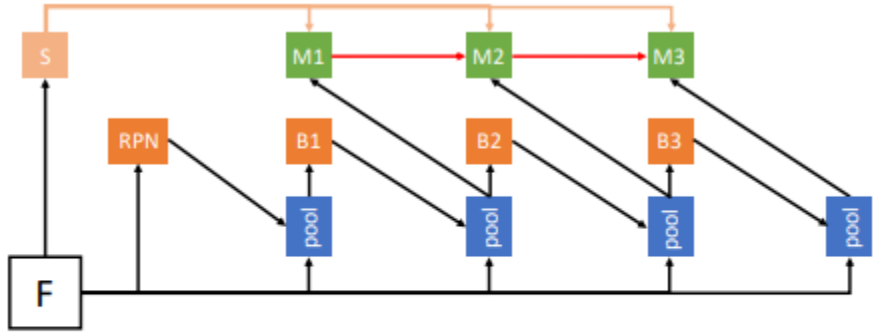
(a) Cascade Mask R-CNN



(b) Interleaved execution



(c) Mask information flow

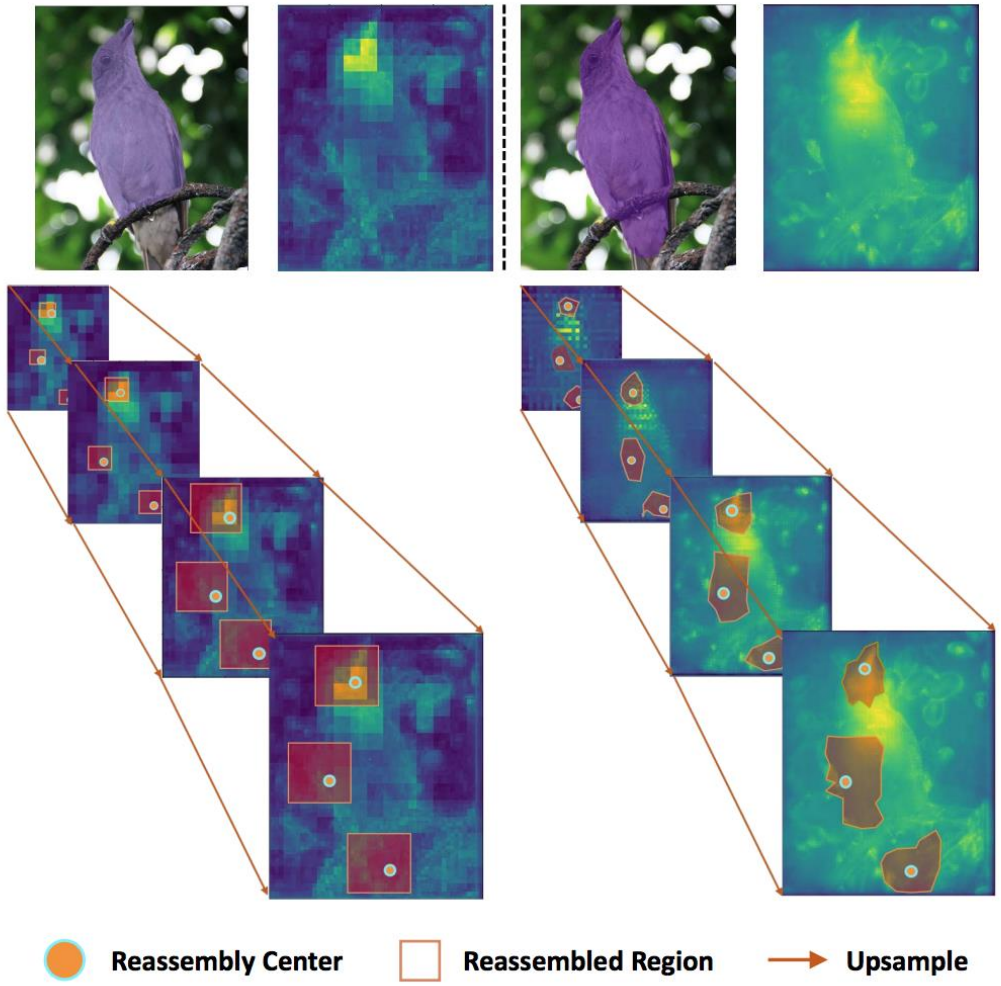


(d) Hybrid Task Cascade (semantic feature fusion with box branches is not shown on the figure for neat presentation.)

- Information fusion between bbox and mask prediction

Chen et al, “Hybrid Task Cascade for Instance Segmentation”, CVPR 2019

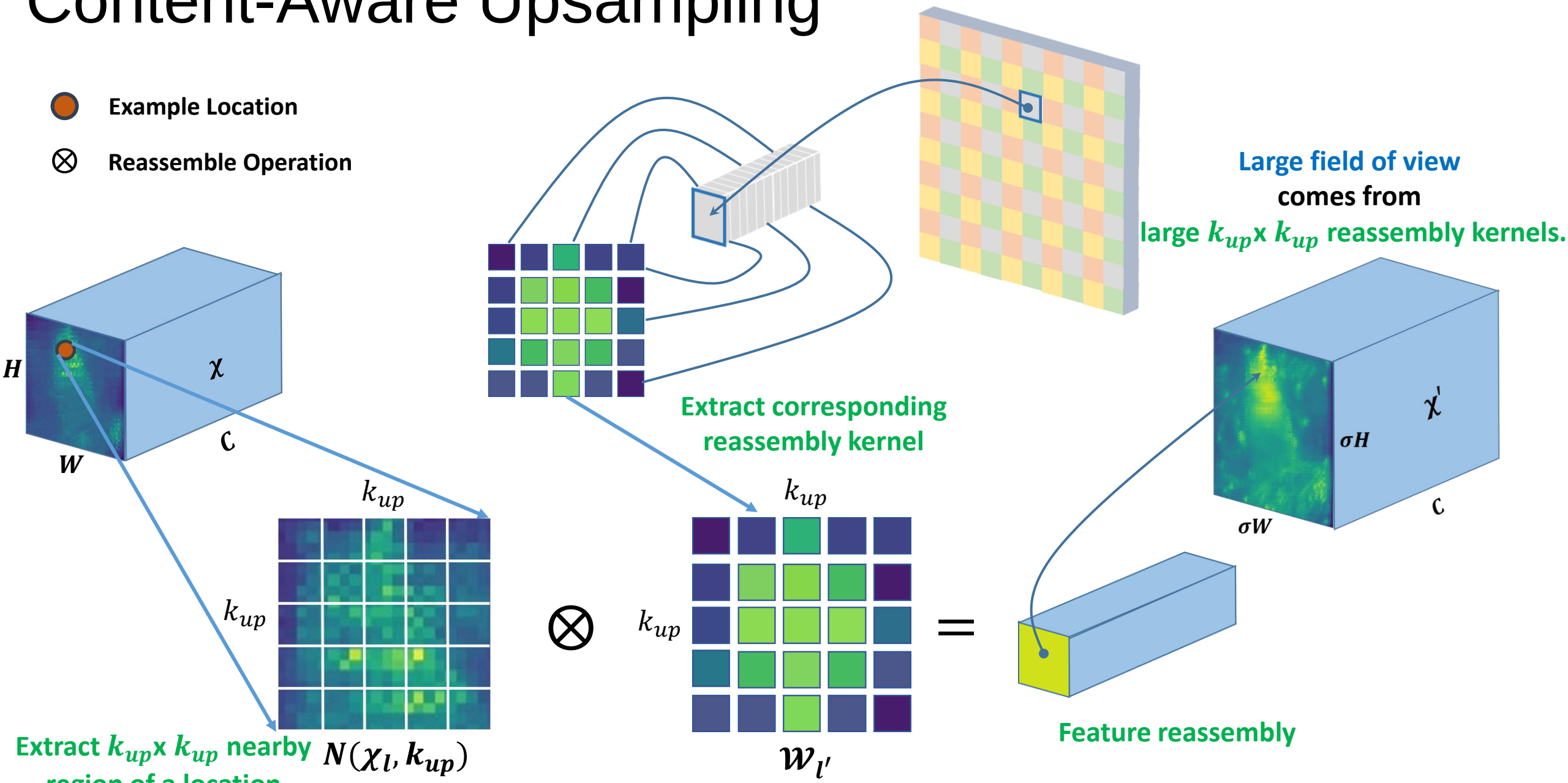
Content-Aware Upsampling



- Feature upsampling using adaptive kernels that incorporate contexts

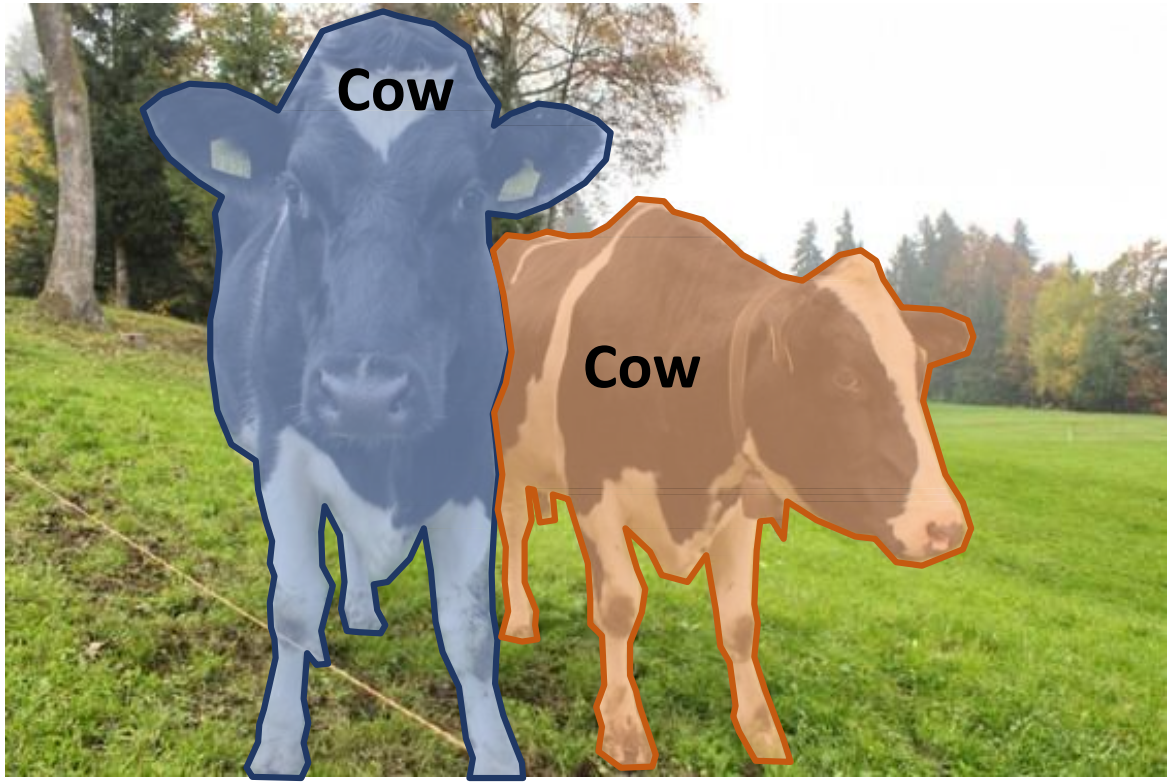
Wang et al, “CARAFE: Content-Aware ReAssembly of FEatures”, ICCV 2019

Content-Aware Upsampling

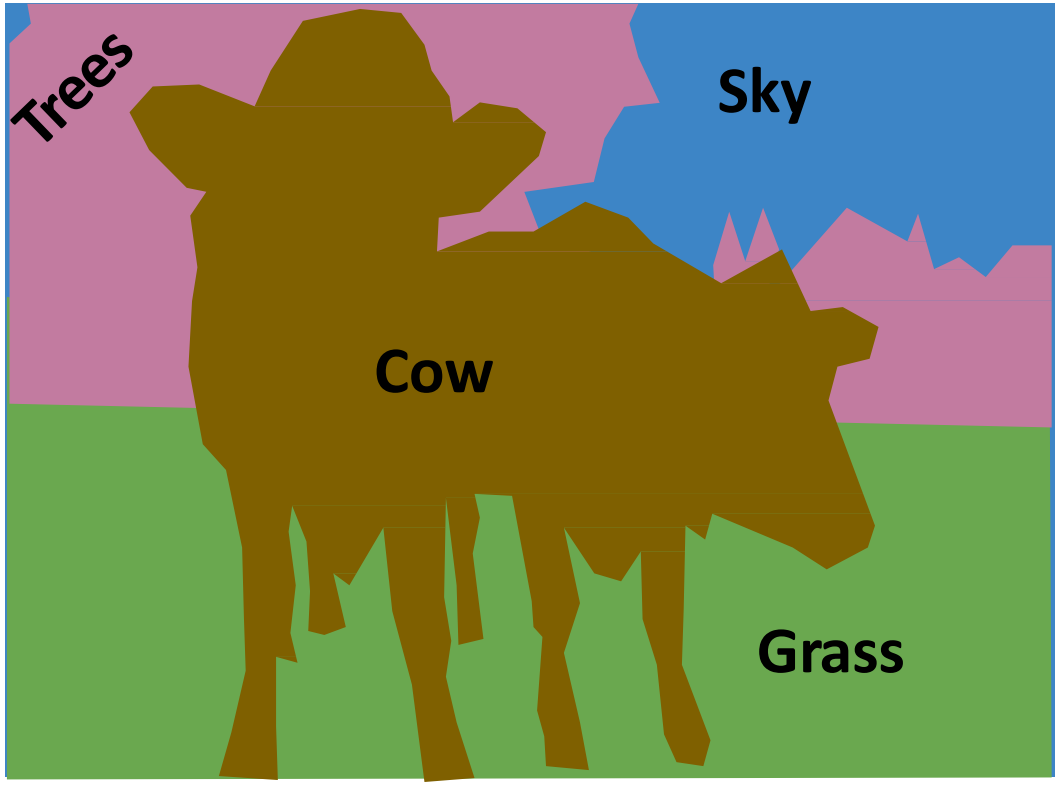


Beyond Instance Segmentation

Instance Segmentation: Separate object instances, but only things



Semantic Segmentation: Identify both things and stuff, but doesn't separate instances

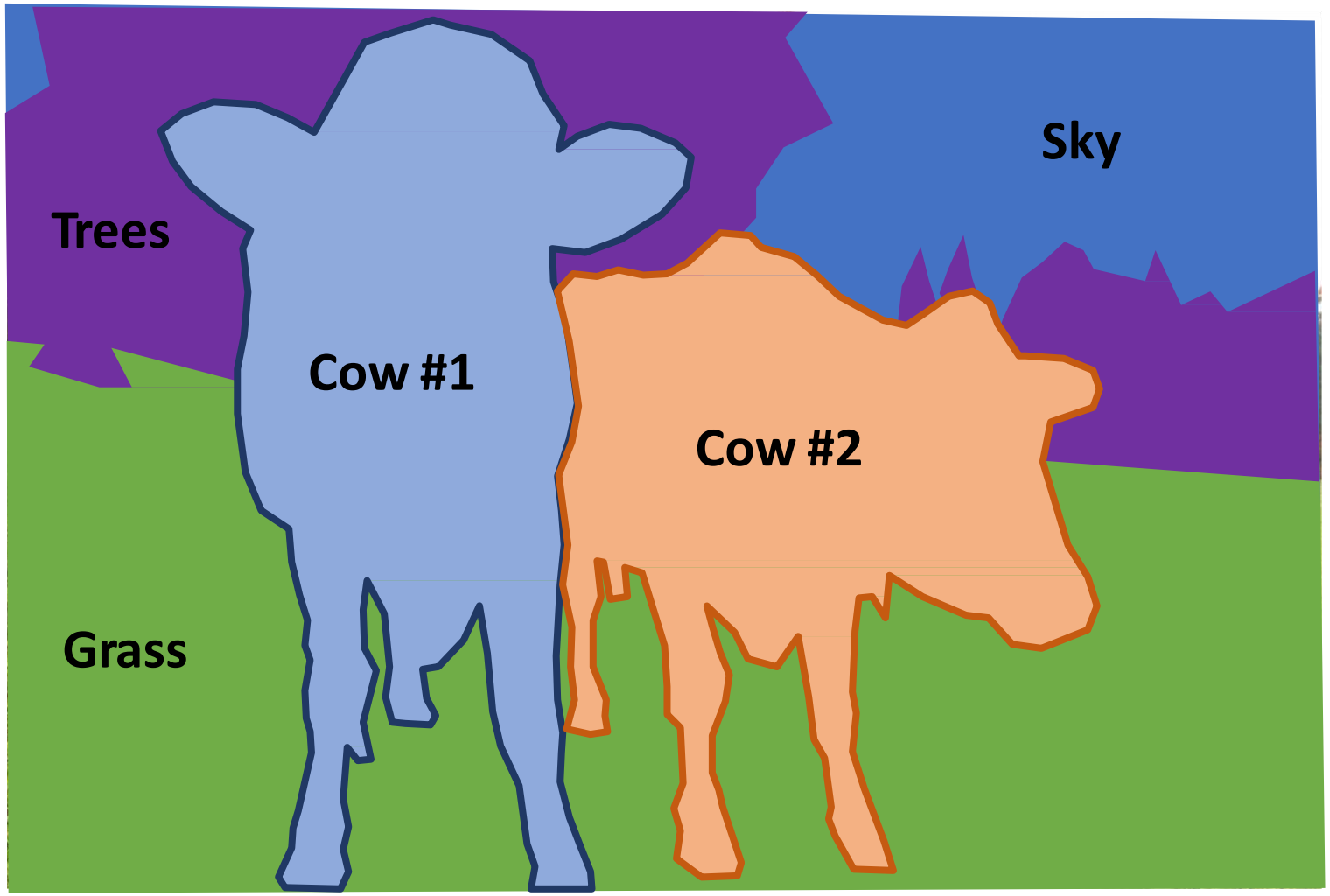


Panoptic Segmentation

Panoptic Segmentation

Label all pixels in the image (both things and stuff)

For “thing” categories also separate into instances



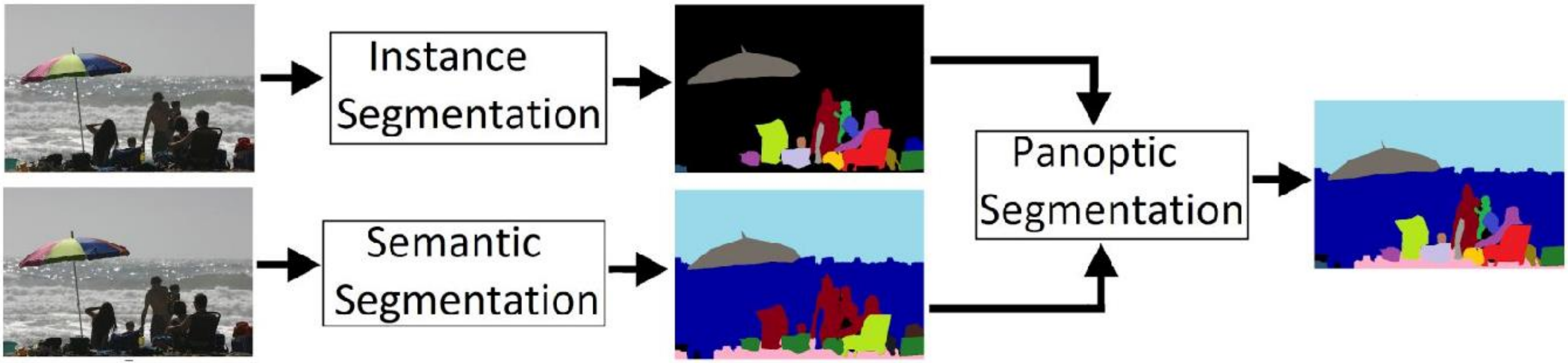
Kirillov et al, “Panoptic Segmentation”, CVPR 2019

Kirillov et al, “Panoptic Feature Pyramid Networks”, CVPR 2019

Panoptic Segmentation

Kirillov et al, “Panoptic Feature Pyramid Networks”, CVPR 2019

Panoptic Segmentation



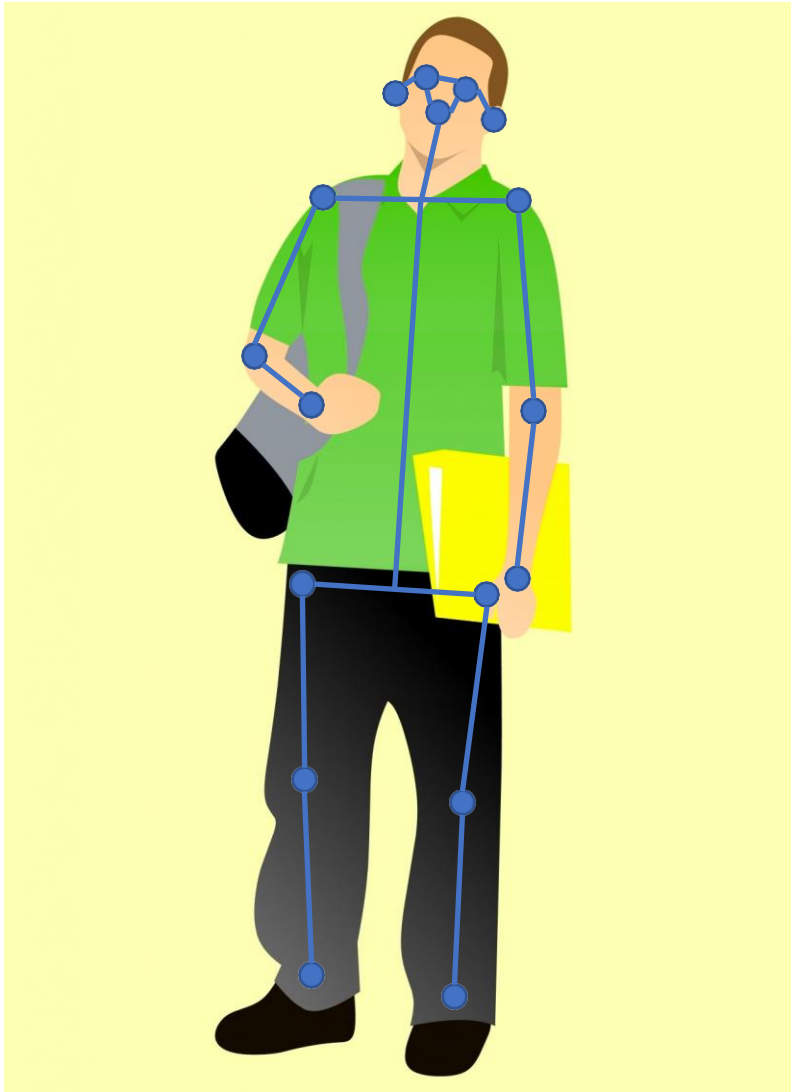
Joint Mask + X Prediction

Beyond Instance Segmentation: Human Keypoints

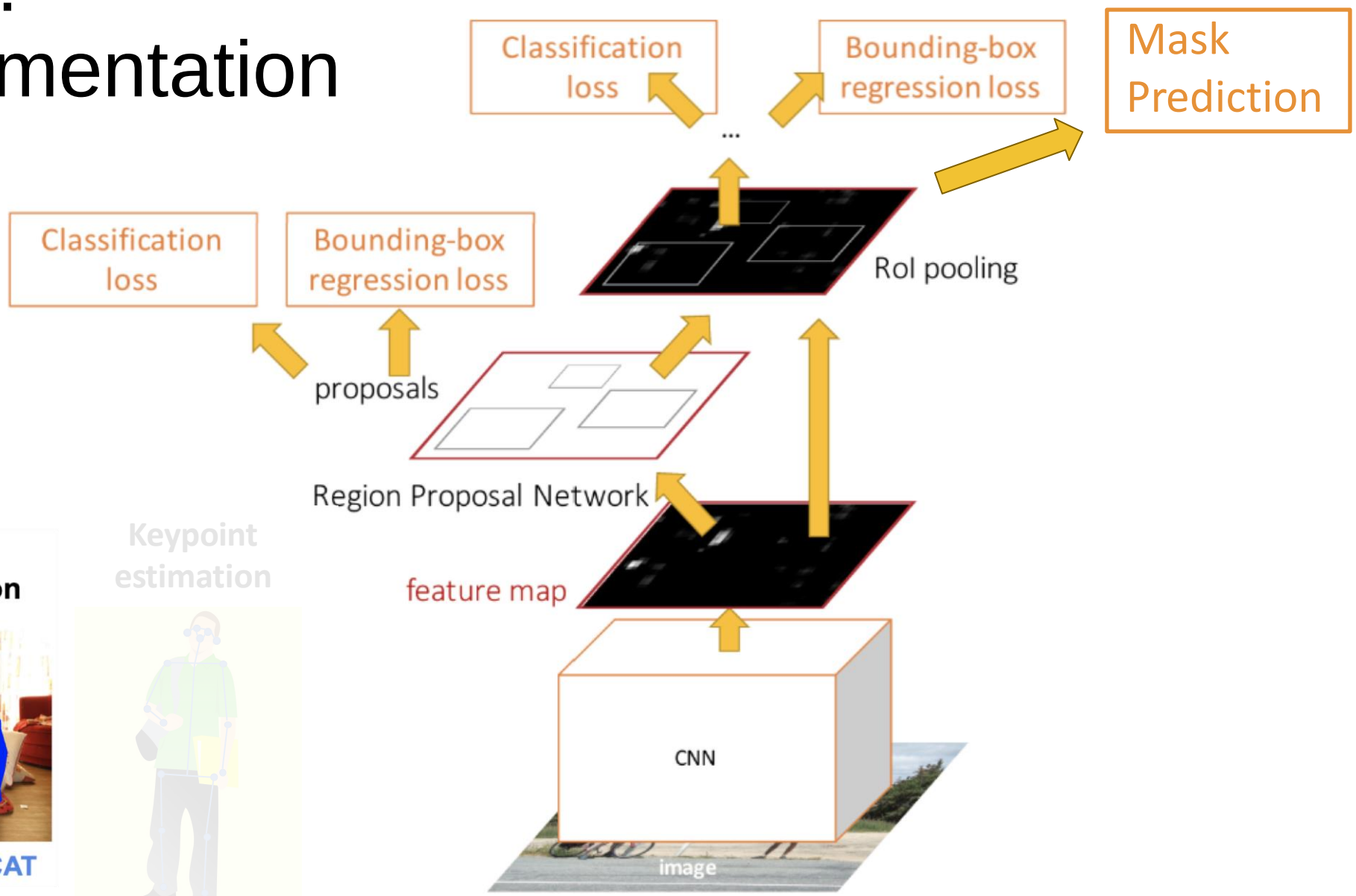
Represent the pose of a human
by locating a set of **keypoints**

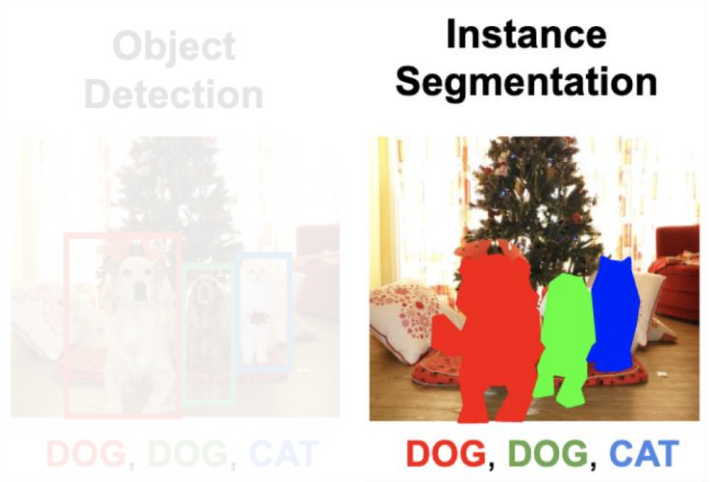
e.g. 17 keypoints:

- Nose
- Left / Right eye
- Left / Right ear
- Left / Right shoulder
- Left / Right elbow
- Left / Right wrist
- Left / Right hip
- Left / Right knee
- Left / Right ankle



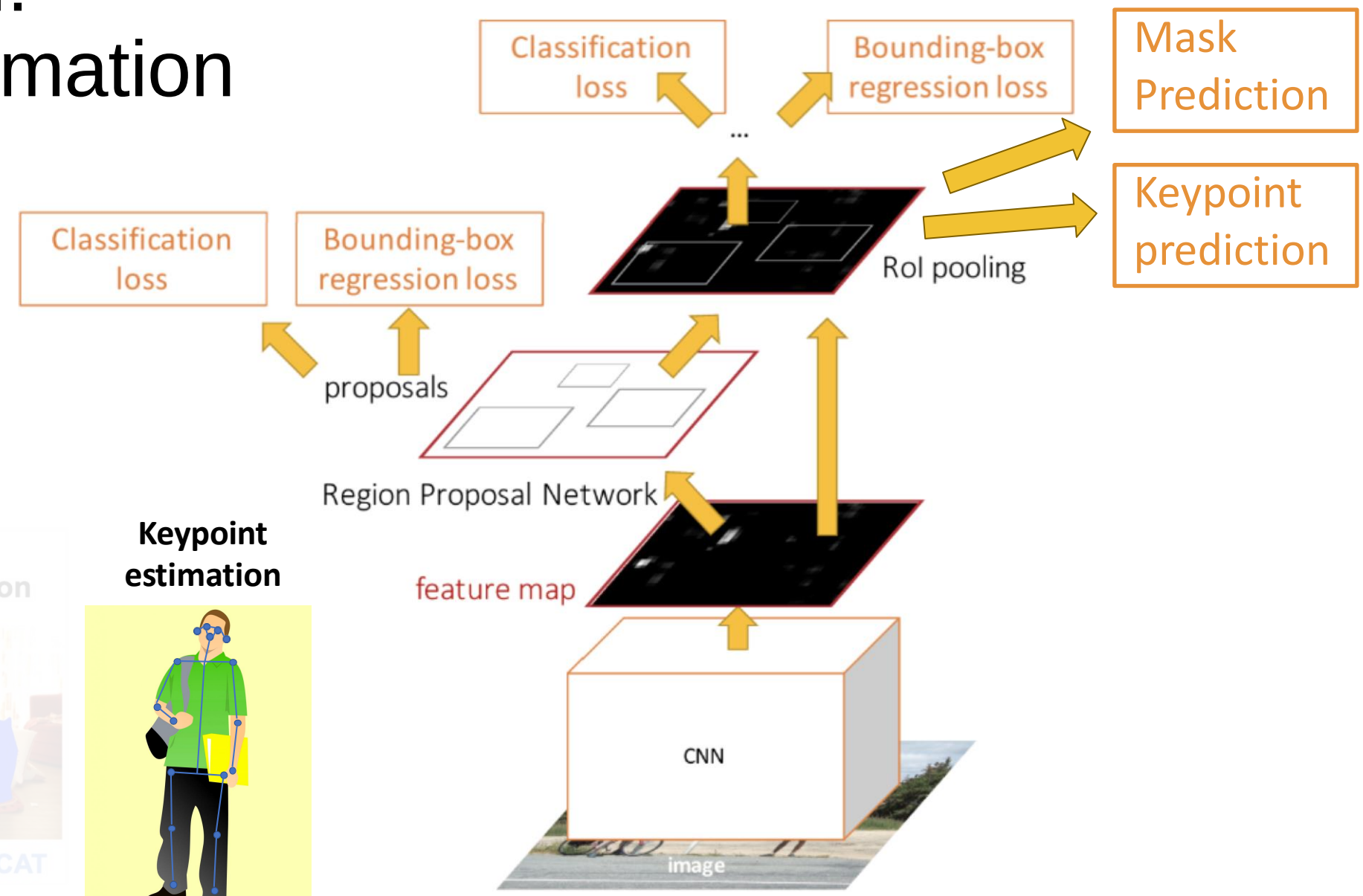
Mask R-CNN: Instance Segmentation





He et al, “Mask R-CNN”, ICCV 2017

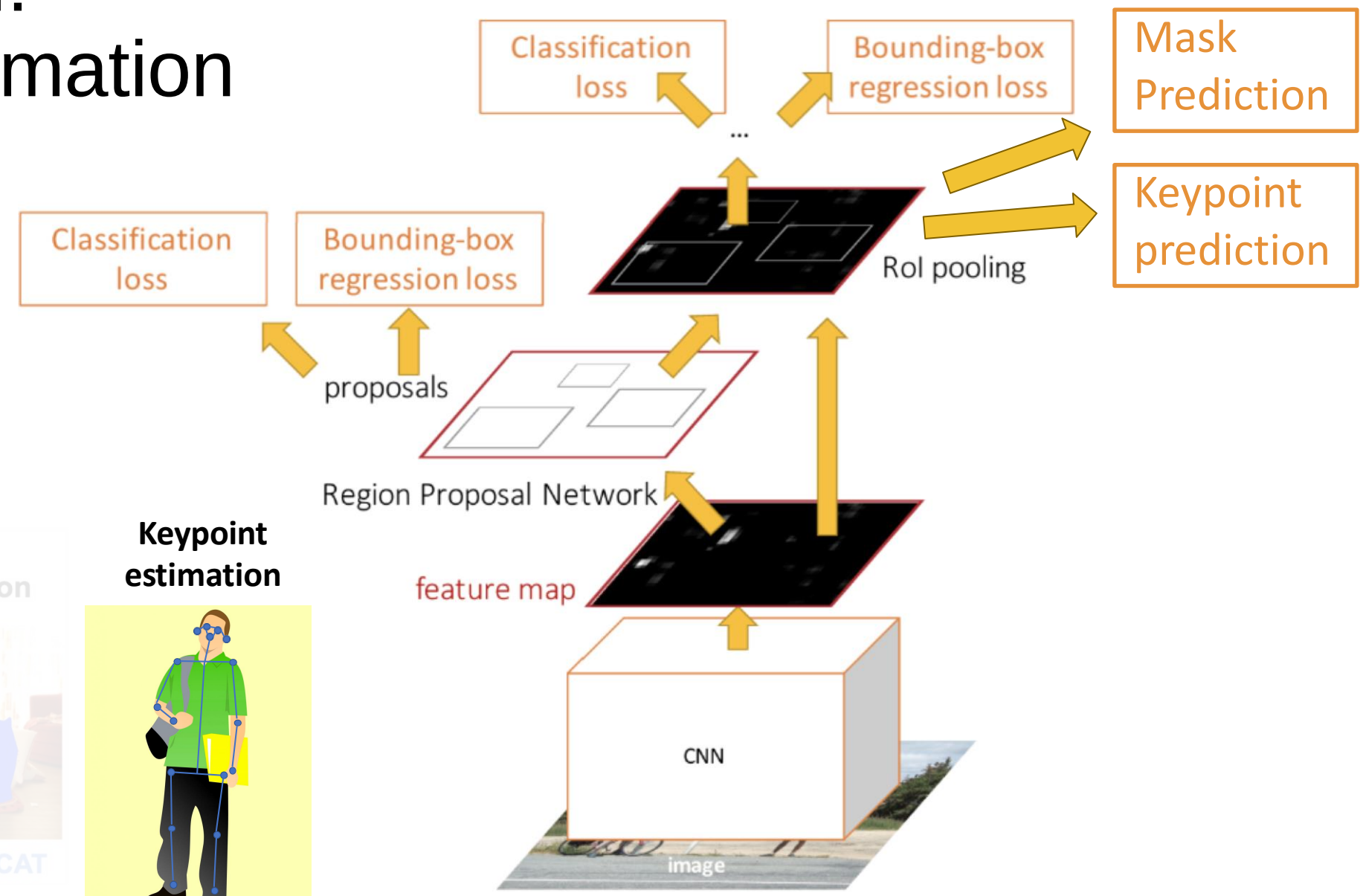
Mask R-CNN: Keypoint Estimation



Keypoint estimation

He et al, "Mask R-CNN", ICCV 2017

Mask R-CNN: Keypoint Estimation



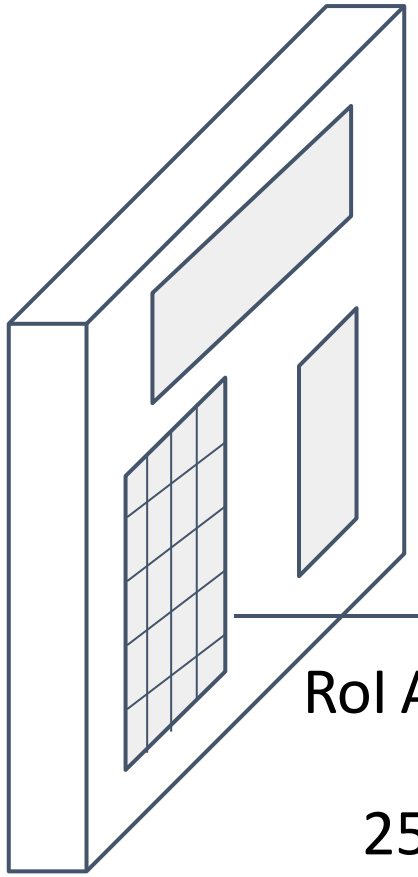
Keypoint estimation

He et al, "Mask R-CNN", ICCV 2017

Mask R-CNN: Keypoint Estimation

Classification Scores: C
Box coordinates (per class): 4 * C
Segmentation mask: C x 28 x 28

CNN
+RPN

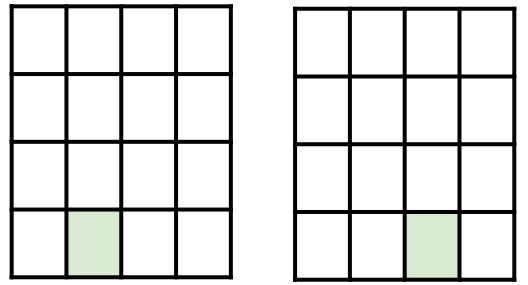


RoI Align

256 x 14 x 14

Conv…

One mask for each of
the K different keypoints



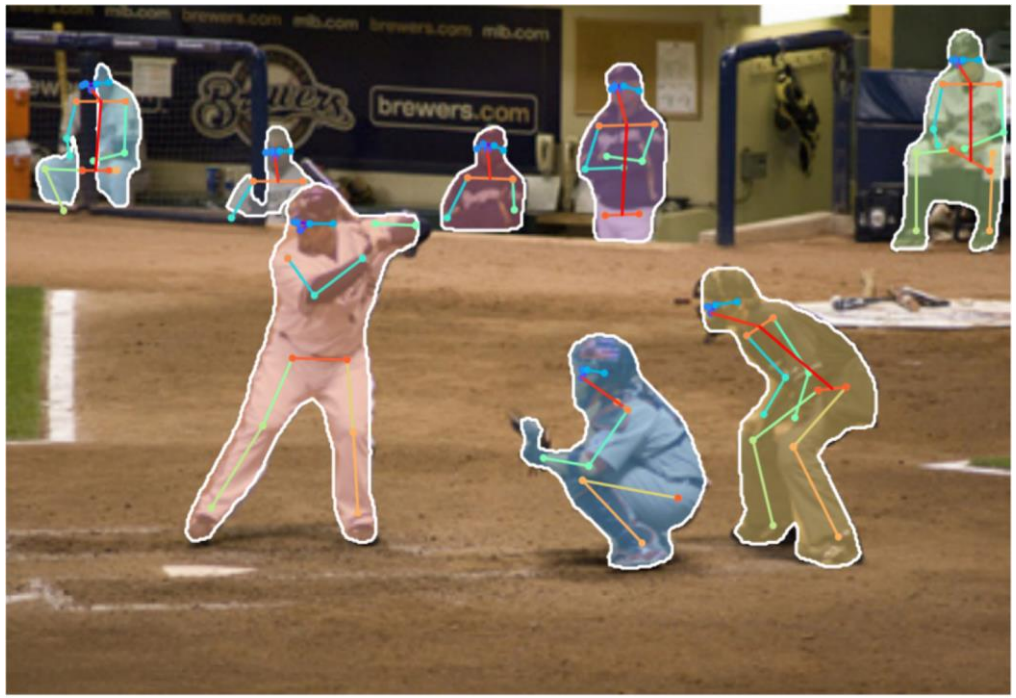
…

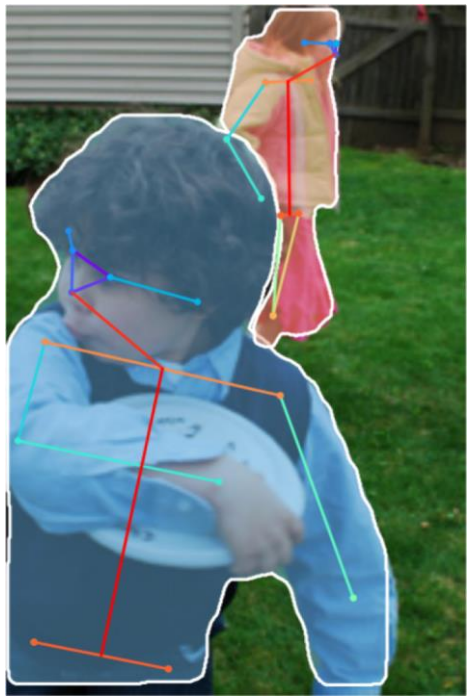
Keypoint masks:
K x 56 x 56

Ground-truth has one “pixel” turned on per keypoint. Train with softmax loss

He et al, “Mask R-CNN”, ICCV 2017

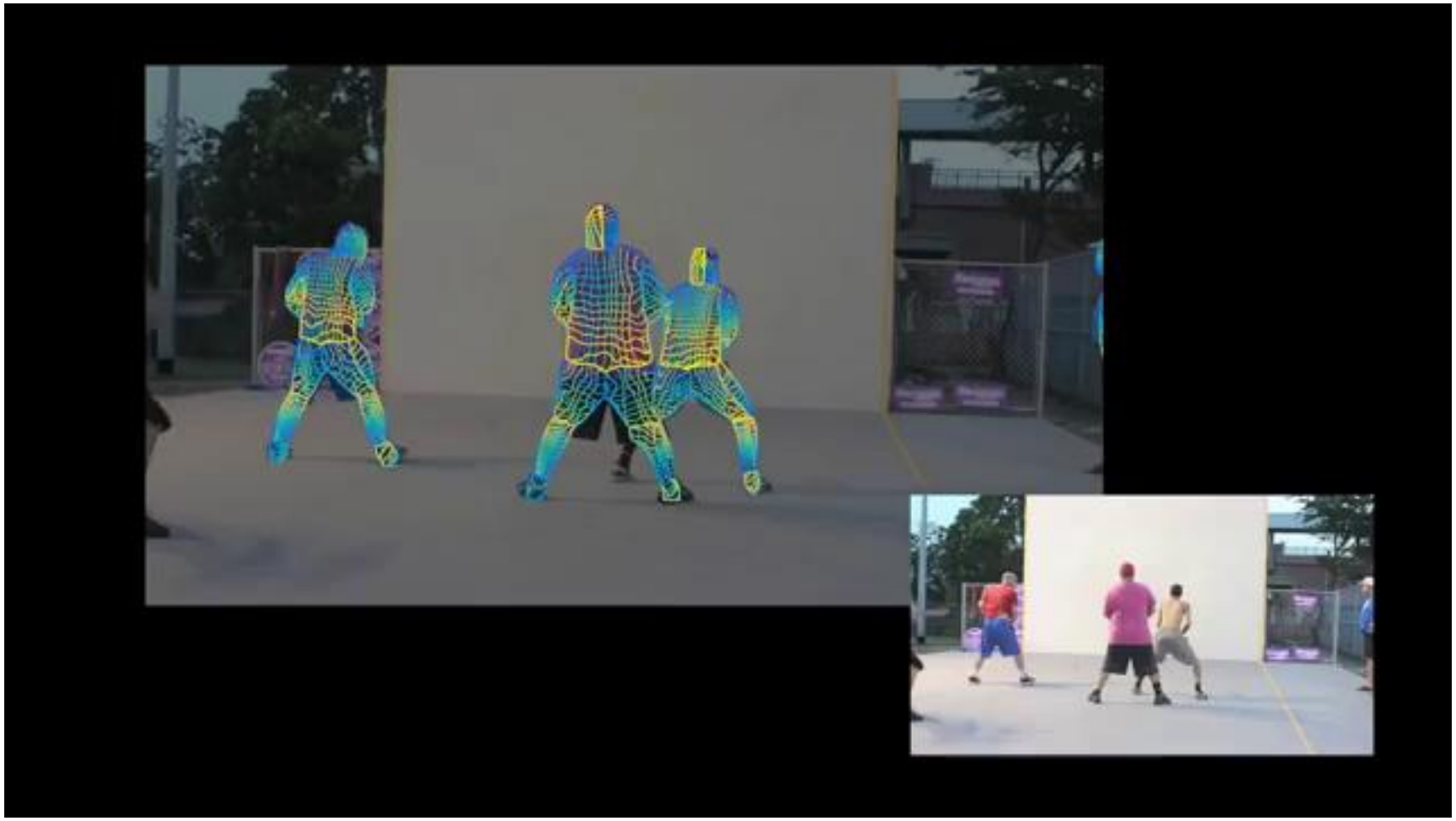
Joint Instance Segmentation and Pose Estimation





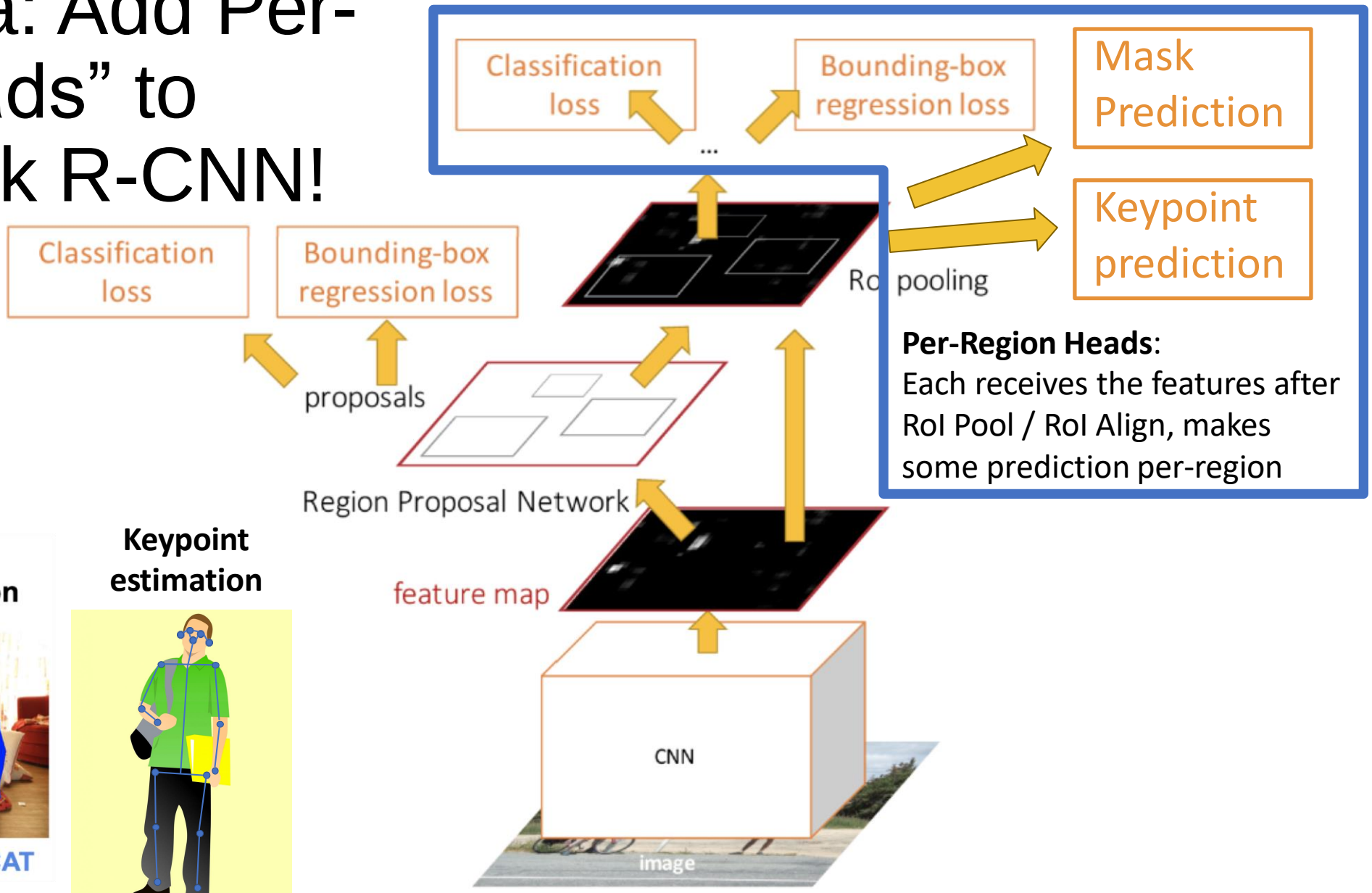
He et al, “Mask R-CNN”, ICCV 2017

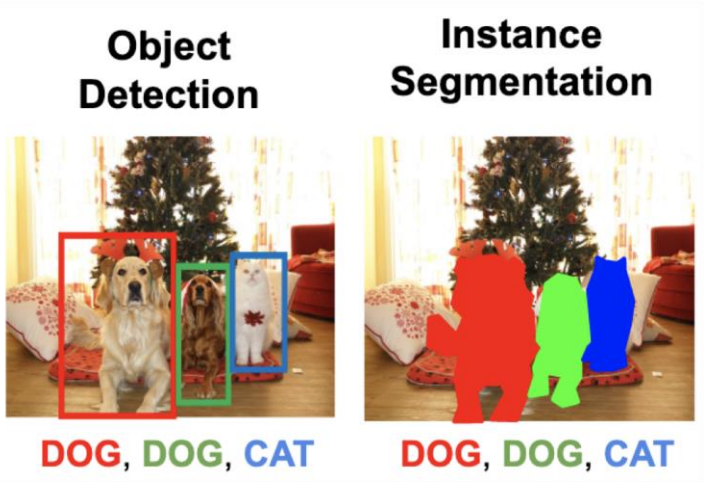
Joint Instance Segmentation and Pose Estimation



Guler et al, “DensePose: Dense Human Pose Estimation in the Wild”, CVPR 2018

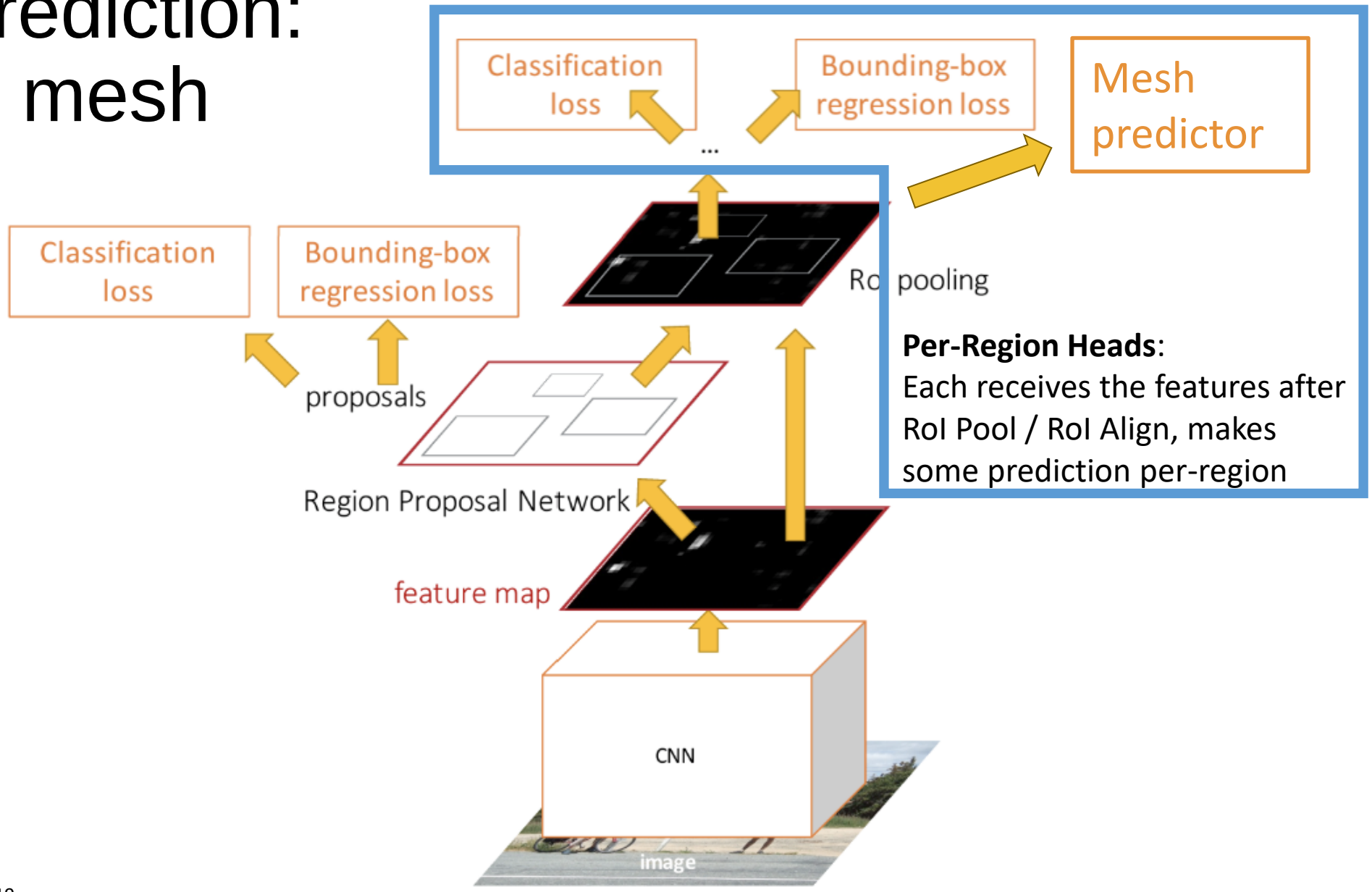
General Idea: Add Per-Region "Heads" to Faster / Mask R-CNN!





He et al, "Mask R-CNN", ICCV 2017

3D Shape Prediction: Predict a 3D mesh per region!



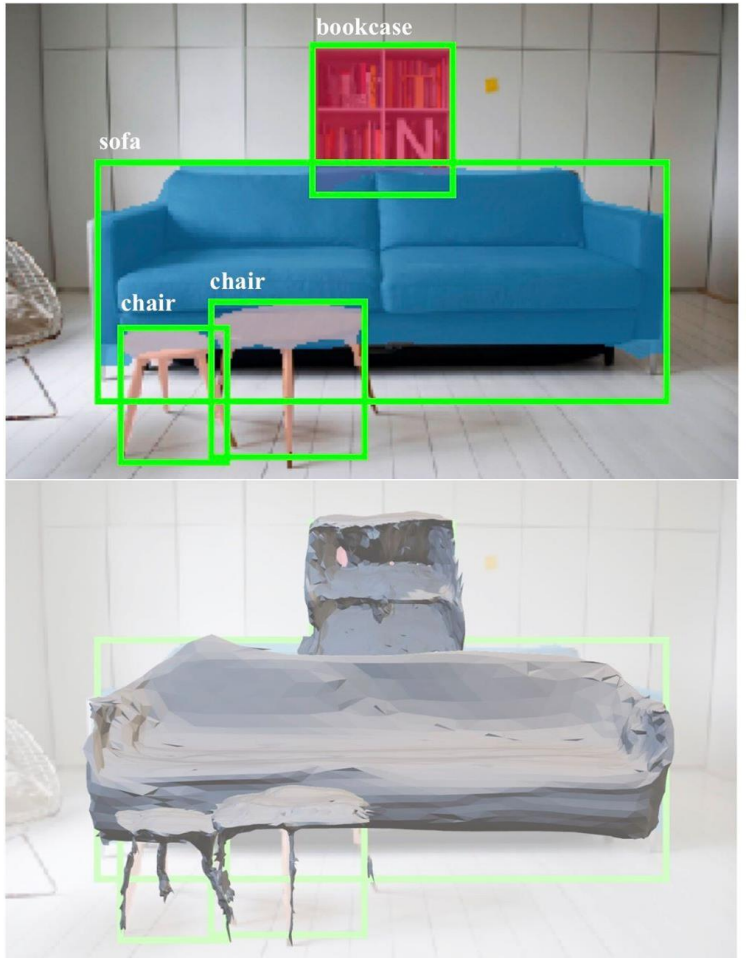
Per-Region Heads:
Each receives the features after RoI Pool / RoI Align, makes some prediction per-region

Gkioxari, Malik, and Johnson, “Mesh R-CNN”, ICCV 2019

3D Shape Prediction: Mask R-CNN + Mesh Head

Mask R-CNN:
2D Image -> 2D shapes

Mesh R-CNN:
2D Image -> **3D** shapes



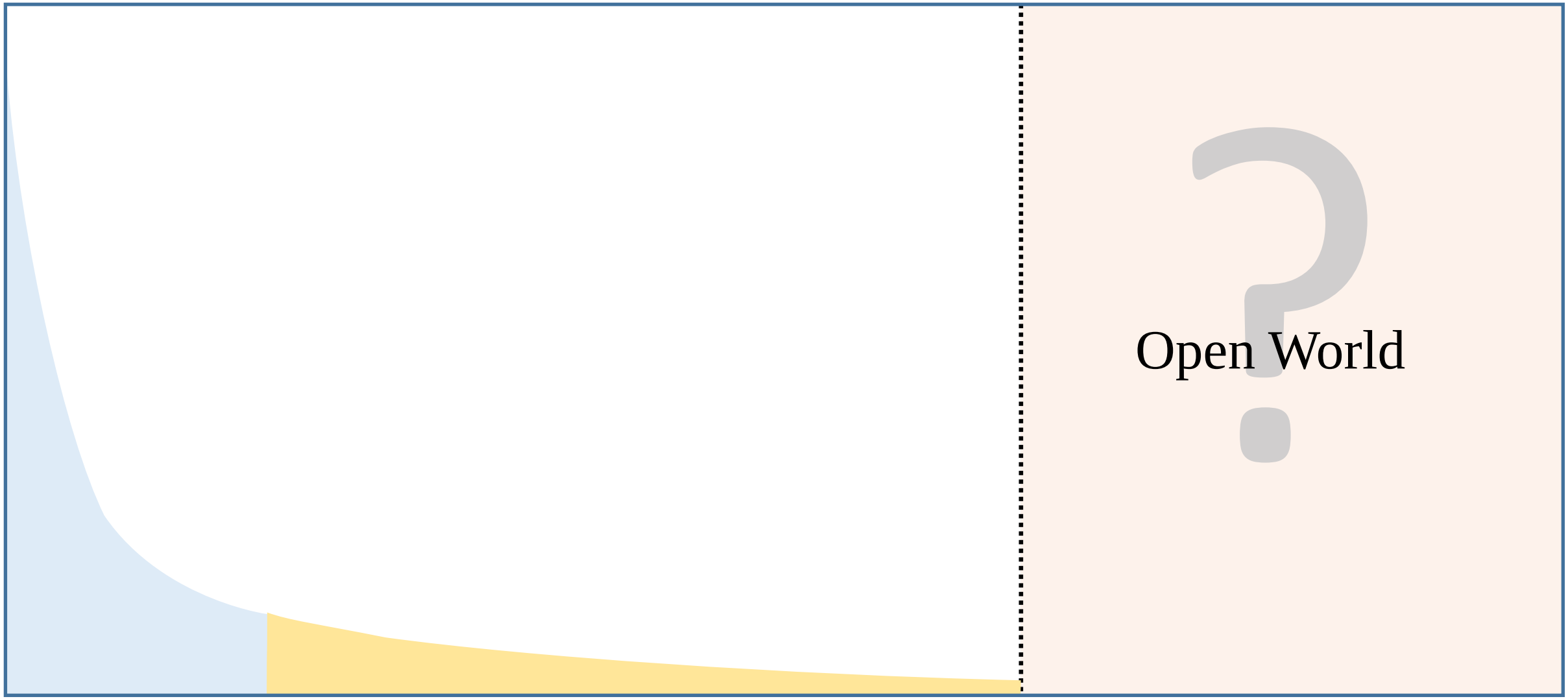
He, Gkioxari, Dollár, and Girshick, “Mask R-CNN”, ICCV 2017

Gkioxari, Malik, and Johnson, “Mesh R-CNN”, ICCV 2019

Open Problems

Problem I:
The Devils are in the Tail

Open Long-Tailed Recognition



Liu et al, “Large-Scale Long-Tailed Recognition in an Open World”, CVPR 2019

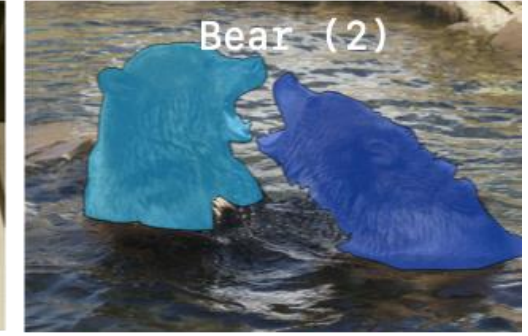
Gupta et al, “LVIS: A Dataset for Large Vocabulary Instance Segmentation”, CVPR 2019

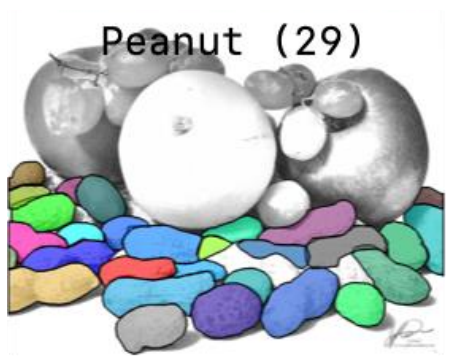

Shoulder bag (3)


Motor Scooter (4)


Table (1)


Hairbrush (3)


Bear (2)


Peanut (29)


Bed (2)

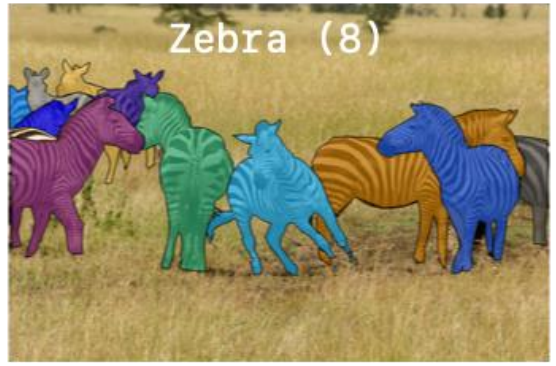

Printer (2)


Pineapple (12)


Banana (80)


Pool Table (1)


Beer Bottle (3)

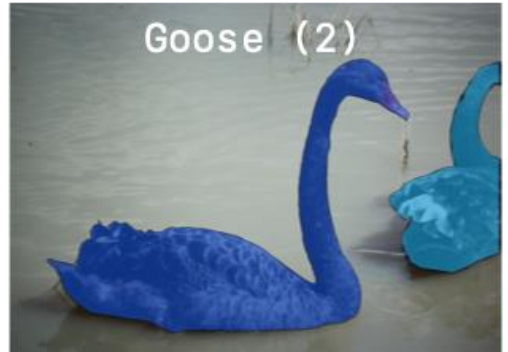

Zebra (8)

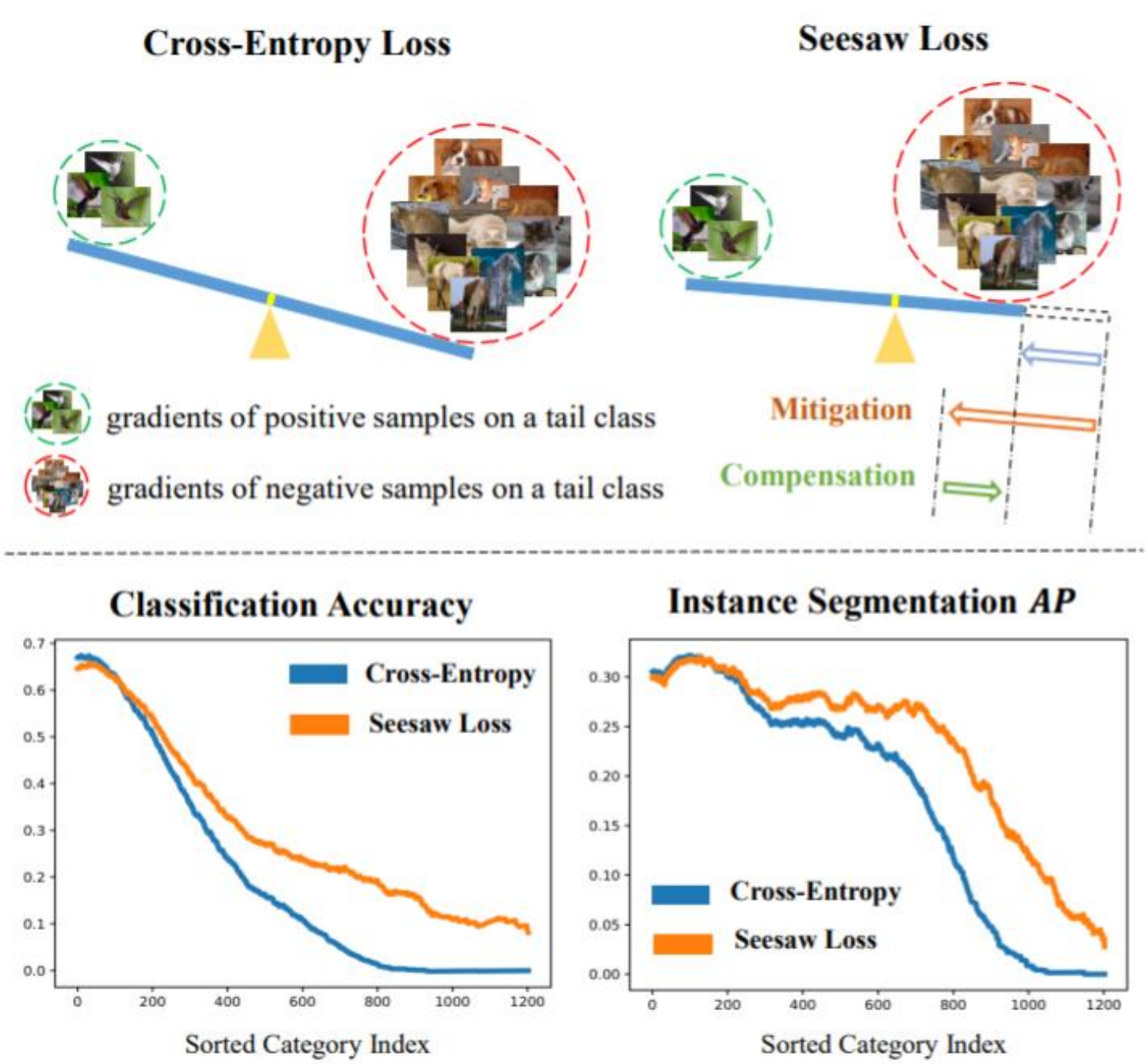

Teacup (12)


Donut (195)


Umbrella (24)

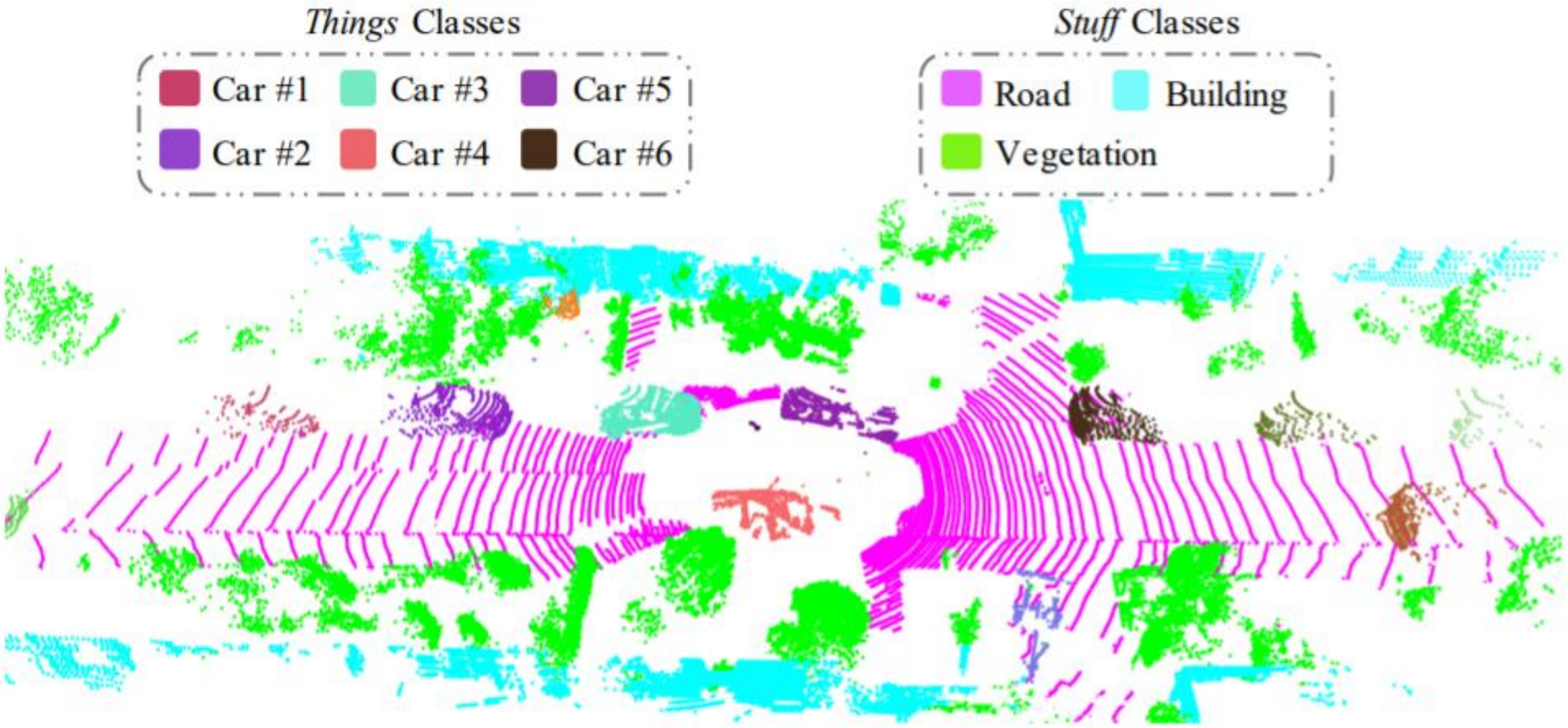

Hand Towel (2)


Goose (2)



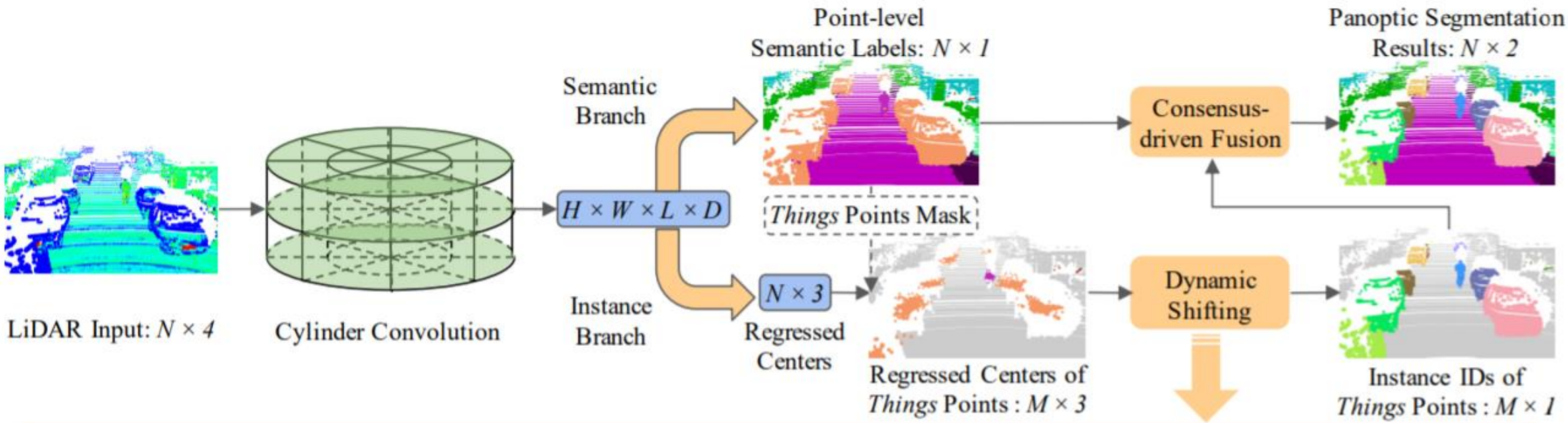
Wang et al, “Seesaw Loss for Long-Tailed Instance Segmentation”, CVPR 2021

Problem II:
The Blessing of Dimensionality



(a) LiDAR-based Panoptic Segmentation

Hong et al, “LiDAR-based Panoptic Segmentation via Dynamic Shifting Network”, CVPR 2021


LiDAR Input: $N \times 4$
Cylinder Convolution
$H \times W \times L \times D$
Semantic Branch
Instance Branch
Point-level Semantic Labels: $N \times 1$
Things Points Mask
$N \times 3$
Regressed Centers
Regressed Centers of *Things* Points : $M \times 3$
Consensus-driven Fusion
Dynamic Shifting
Panoptic Segmentation Results: $N \times 2$
Instance IDs of *Things* Points : $M \times 1$

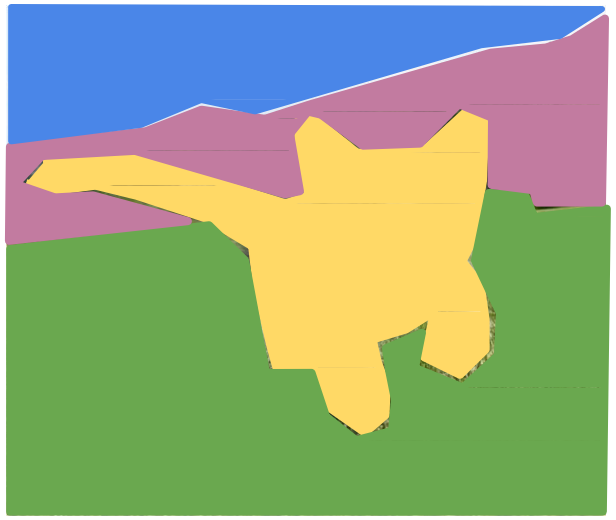
Summary: Many Structured Prediction Tasks!

Classification

CAT

No spatial extent

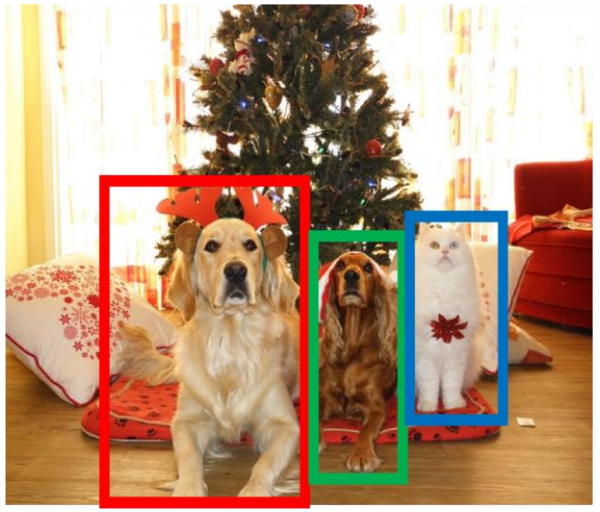
Semantic Segmentation



GRASS, **CAT**, **TREE**, **SKY**

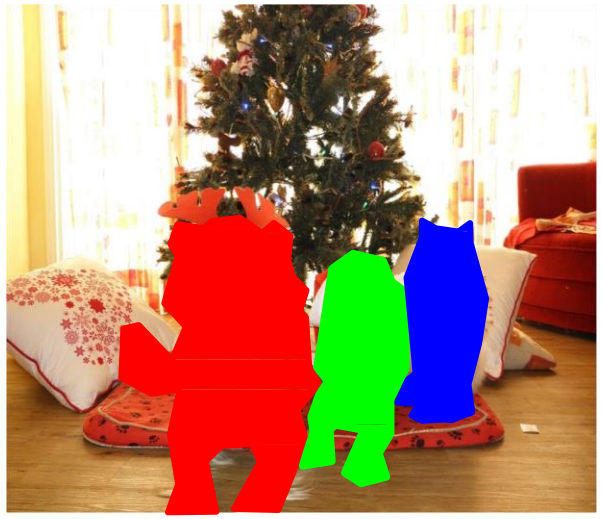
No objects, just pixels

Object Detection



DOG, **DOG**, **CAT**

Instance Segmentation



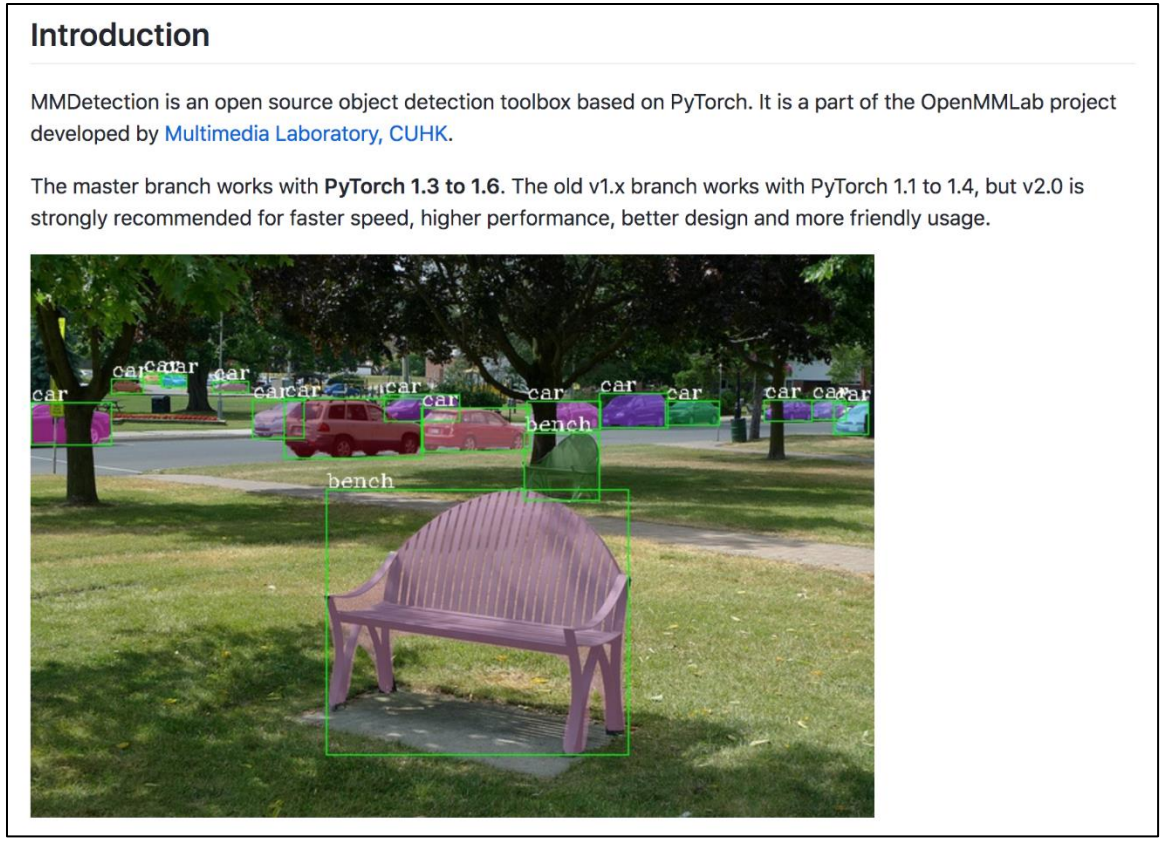
DOG, **DOG**, **CAT**

Multiple Objects

Deep Structured Prediction: Resources

ICCV Tutorial on Instance-Level Visual Recognition:
https://instancetutorial.github.io/

MMDetection:
https://github.com/open-mmlab/mmdetection



Next Time: Transformers